

accumulatedMutations_MEF

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Contents

1	Env	2
2	Libraries	2
3	Def function	4
4	Load Data	7
5	Observed Accumulated Mutation Num	7
6	Imputed Accumulated Mutation Num	9
7	Accumulated Mutation Num normalized by cell number	13
8	MPGgenotypecheck	15
9	Calculating mutations of each cell and cells of each mutation	17
10	calMulti	18
11	scMutScore Model Validation	27
12	compareClonesScore	47
13	compareClonesScore figures	52
14	save figures	67
15	check randomly clone scMutScore	68
16	tracing allClones	72
17	randomlySelectMut	75

18 lineageTracingExample	79
19 lineageTracingExample with scMutScore	81
20 ModelFigure	87
21 sessionInfo	91

1 Env

```
rm(list = ls())
setwd("/Users/liqun/Desktop/scMutrace-tutorial/ReproducibilityAnalysis/Figure5/")
```

2 Libraries

```
library(FSA)
## ## FSA v0.10.0. See citation('FSA') if used in publication.
## ## Run fishR() for related website and fishR('IFAR') for related book.
library(ineq)
library(vegan)
## Loading required package: permute
library(dplyr)
##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
library(tidyr)
library(ggpubr)
## Loading required package: ggplot2
## Registered S3 methods overwritten by 'car':
##   method      from
##   hist.boot    FSA
##   confint.boot FSA
library(Seurat)
## Loading required package: SeuratObject
## Loading required package: sp
## 'SeuratObject' was built under R 4.5.0 but the current version is
## 4.5.1; it is recommended that you reinstall 'SeuratObject' as the ABI
## for R may have changed
##
## Attaching package: 'SeuratObject'
## The following objects are masked from 'package:base':
##
```

```

##      intersect, t
library(igraph)
##
## Attaching package: 'igraph'
## The following object is masked from 'package:Seurat':
##
##      components
## The following object is masked from 'package:tidyr':
##
##      crossing
## The following objects are masked from 'package:dplyr':
##
##      as_data_frame, groups, union
## The following object is masked from 'package:vegan':
##
##      diversity
## The following object is masked from 'package:permute':
##
##      permute
## The following objects are masked from 'package:stats':
##
##      decompose, spectrum
## The following object is masked from 'package:base':
##
##      union
library(ggraph)
##
## Attaching package: 'ggraph'
## The following object is masked from 'package:sp':
##
##      geometry
library(webshot)
library(ggplot2)
library(ggpmisc)
## Loading required package: ggpp
## Registered S3 methods overwritten by 'ggpp':
##      method                  from
##      heightDetails.titleGrob ggplot2
##      widthDetails.titleGrob  ggplot2
##
## Attaching package: 'ggpp'
## The following objects are masked from 'package:ggpubr':
##
##      as_npc, as_npcx, as_npcy
## The following object is masked from 'package:ggplot2':
##
##      annotate
library(stringr)
library(corrplot)
## corrplot 0.95 loaded
library(pheatmap)
library(reshape2)
##

```

```

## Attaching package: 'reshape2'
## The following object is masked from 'package:tidyr':
##
##      smiths
library(tidygraph)
##
## Attaching package: 'tidygraph'
## The following object is masked from 'package:igraph':
##
##      groups
## The following object is masked from 'package:stats':
##
##      filter
library(networkD3)
##
## Attaching package: 'networkD3'
## The following object is masked from 'package:Seurat':
##
##      JS
## The following object is masked from 'package:SeuratObject':
##
##      JS
library(tidyverse)
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats 1.0.0      v readr 2.1.5
## v lubridate 1.9.4    v tibble 3.3.0
## v purrr 1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x lubridate::%--%()      masks igraph::%--%()
## x ggpp::annotate()      masks ggplot2::annotate()
## x tibble::as_data_frame() masks igraph::as_data_frame(), dplyr::as_data_frame()
## x purrr::compose()      masks igraph::compose()
## x igraph::crossing()     masks tidyr::crossing()
## x tidygraph::filter()   masks dplyr::filter(), stats::filter()
## x dplyr::lag()          masks stats::lag()
## x purrr::simplify()     masks igraph::simplify()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
library(patchwork)

```

3 Def function

```

retriveCellsByLocation <- function(CSM, locList){
  CSM$CellTag <- rownames(CSM)
  CSM_tmp <- CSM[, c("CellTag",locList)]
  data_MutPerCell_CSM_tmp <- data.frame(
    CellTag = rownames(CSM_tmp),
    MutNum = rowSums(CSM_tmp == "Mut"),
    WTNum = rowSums(CSM_tmp == "WT")
  )
  cells_Total <- data_MutPerCell_CSM_tmp[rowSums(data_MutPerCell_CSM_tmp[,c("MutNum", "WTNum")]) > 0, ]
  cells_WT <- data_MutPerCell_CSM_tmp[data_MutPerCell_CSM_tmp[, "WTNum"] > 0,]$CellTag
}

```

```

cells_Mut <- data_MutPerCell_CSM_tmp[data_MutPerCell_CSM_tmp[, "MutNum"] > 0,]$CellTag
cells <- list(Total = cells_Total, Mut = cells_Mut, WT = cells_WT)
return(cells)
}

retriveLocsByCells <- function(CSM, cellList){
  CSM$CellTag <- rownames(CSM)
  CSM_tmp <- CSM[CSM$CellTag %in% cellList,]
  data_CellPerMut_CSM_tmp <- data.frame(
    Location = colnames(CSM_tmp),
    CellMutNum = colSums(CSM_tmp == "Mut"),
    CellWTNum = colSums(CSM_tmp == "WT")
  )
  pos_Total <- data_CellPerMut_CSM_tmp[rowSums(data_CellPerMut_CSM_tmp[, c("CellMutNum", "CellWTNum")]) > 0,]$Location
  pos_Mut <- data_CellPerMut_CSM_tmp[data_CellPerMut_CSM_tmp[, "CellMutNum"] > 0,]$Location
  pos_WT <- data_CellPerMut_CSM_tmp[data_CellPerMut_CSM_tmp[, "CellWTNum"] > 0,]$Location
  pos <- list(Total = pos_Total, Mut = pos_Mut, WT = pos_WT)
  return(pos)
}

cal_MultiSpe <- function(matrixData) {
  res <- data.frame(
    Shannon = vegan::diversity(matrixData, index = "shannon"),
    Simpson = vegan::diversity(matrixData, index = "simpson"),
    Gini = apply(matrixData, 1, function(x) ineq(x + 0.00001, type = "Gini")),
    sumMut = rowSums(matrixData),
    Tau = NA
  )
  # Tau = sum(1 - (x / max(x))) / (n - 1)
  res$Tau <- apply(matrixData, 1, function(x) {
    if (all(x == 0)) return(NA)
    x <- x / max(x)
    sum(1 - x) / (length(x) - 1)
  })
  rownames(res) <- rownames(matrixData)
  return(res)
}

cal_scMutScore <- function(data_cell_mut, data_mut_time, data_meta){
  # data_cell_mut [cell * muts]
  # data_mut_time [muts * tissues/celltype/days]
  # data_meta [Cellbarcode, CellMutTag(tissues/celltype/days)]
  # data_cell_time_matrix
  replacement_map <- c("NoCall" = 0, "WT" = 0, "Mut" = 1)

  data_cell_mut_num <- as.data.frame(lapply(data_cell_mut, function(x) {
    ifelse(x %in% names(replacement_map), replacement_map[x], x)
  })))

  rownames(data_cell_mut_num) <- rownames(data_cell_mut)

```

```

print(dim(data_cell_mut_num))
print(dim(data_mut_time))

data_cell_time_matrix <- as.matrix(data_cell_mut_num) %*% as.matrix(data_mut_time)
rownames(data_cell_time_matrix) <- rownames(data_cell_mut)
colnames(data_cell_time_matrix) <- colnames(data_mut_time)

mutts_total <- unique(rownames(data_mut_time))

# get different tissues/time/celltype names
col_names_data <- colnames(data_cell_time_matrix)

col_length <- length(col_names_data)

col_coefficient_fix <- setNames(
  rep(1/col_length, col_length),
  col_names_data
)

data_cell_time_matrix <- data_cell_time_matrix[,col_names_data]

# Gini
res_index <- cal_MultiSpe(data_cell_time_matrix)

res_diversity <- as.data.frame((data_cell_time_matrix > 1) * rep(col_coefficient_fix[colnames(data_cell_time_matrix)],
res_diversity$Diversity <- rowSums(res_diversity)

res_diversity$Diversity[is.na(res_diversity$Diversity)] <- 0.00001

res_index$Gini[is.na(res_index$Gini)] <- 0.00001
res_index$Tau[is.na(res_index$Tau)] <- 0.00001

res_index$GiniTau <- res_index$Gini * res_index$Tau
res_index$scMutScore <- 2 / (
  (1 / (0.1 * res_index$GiniTau + 0.00001)) +
  (1 / (0.9 * res_diversity$Diversity + 0.00001))
)

res_index$Diversity <- res_diversity$Diversity

res_index$sumMut_norm <- (res_index$sumMut - min(res_index$sumMut)) /
  (max(res_index$sumMut) - min(res_index$sumMut))

res_index$GiniTau_norm <- (res_index$GiniTau - min(res_index$GiniTau)) /
  (max(res_index$GiniTau) - min(res_index$GiniTau))

res_index$scMutScoreQun <- res_index$sumMut_norm

res_index$scMutScore <- scales::rescale(res_index$scMutScore, to = c(0, 1))
res_index$scMutScoreQun <- scales::rescale(res_index$scMutScoreQun, to = c(0, 1))

res <- list(diversity = res_diversity, score = res_index, matrixD = data_cell_time_matrix)
return(res)

```

```

}

propagate_mutations <- function(data, time_points) {
  for (i in 1:nrow(data)) {
    mut_time <- which(data[i, time_points] == "Mut")
    if (length(mut_time) > 0) {
      first_mut <- min(mut_time)
      data[i, time_points[first_mut:length(time_points)]] <- "Mut"
    }
  }
  return(data)
}

```

4 Load Data

```

MEF_CSM <- readRDS("../Data/MEF/MEF_CSM.rds")
MEF_CSM_hf1 <- MEF_CSM$hf1
MEF_CSM_hf2 <- MEF_CSM$hf2

MEF_SSM <- readRDS("../Data/MEF/MEF_SSM.rds")
MEF_SSM_hf1 <- MEF_SSM$hf1
MEF_SSM_hf2 <- MEF_SSM$hf2

MEF_Meta <- readRDS("../Data/MEF/MEF_Meta.rds")
MEF_Meta_hf1 <- MEF_Meta$hf1
MEF_Meta_hf2 <- MEF_Meta$hf2

# retrieved from MPG database, see https://github.com/QunATCG/scMutrace-tutorial/tree/main/AdvancedFunc
MPG_data <- read.csv("../Data/MEF/MPG_combined.csv")

```

5 Observed Accumulated Mutation Num

```

Reps_sequencing <- c("Rep1_all", "Rep2_all")
Days_sequencing <- c("d6", "d9", "d12", "d15", "d21", "d28")

col_hf1_names <- c("hf1.d0", "hf1.d3", "hf1.d6", "hf1.d9", "hf1.d12", "hf1.d15", "hf1.d21", "hf1.d28")
col_hf2_names <- c("hf2.d0", "hf2.d3", "hf2.d6", "hf2.d9", "hf2.d12", "hf2.d15", "hf2.d21", "hf2.d28")

data_accMutNum_res <- as.data.frame(matrix(nrow = length(Days_sequencing), ncol = length(Reps_sequencing),
colnames(data_accMutNum_res) <- Reps_sequencing
rownames(data_accMutNum_res) <- Days_sequencing

for (i in rownames(data_accMutNum_res)){
  data_accMutNum_res[i, "Rep1_all"] <- length(MEF_SSM_hf1[MEF_SSM_hf1[, paste0("hf1.", i)] == "Mut",]$mut)
  data_accMutNum_res[i, "Rep2_all"] <- length(MEF_SSM_hf2[MEF_SSM_hf2[, paste0("hf2.", i)] == "Mut",]$mut)
}

data_accMutNum_res$Days <- rownames(data_accMutNum_res)

```

```

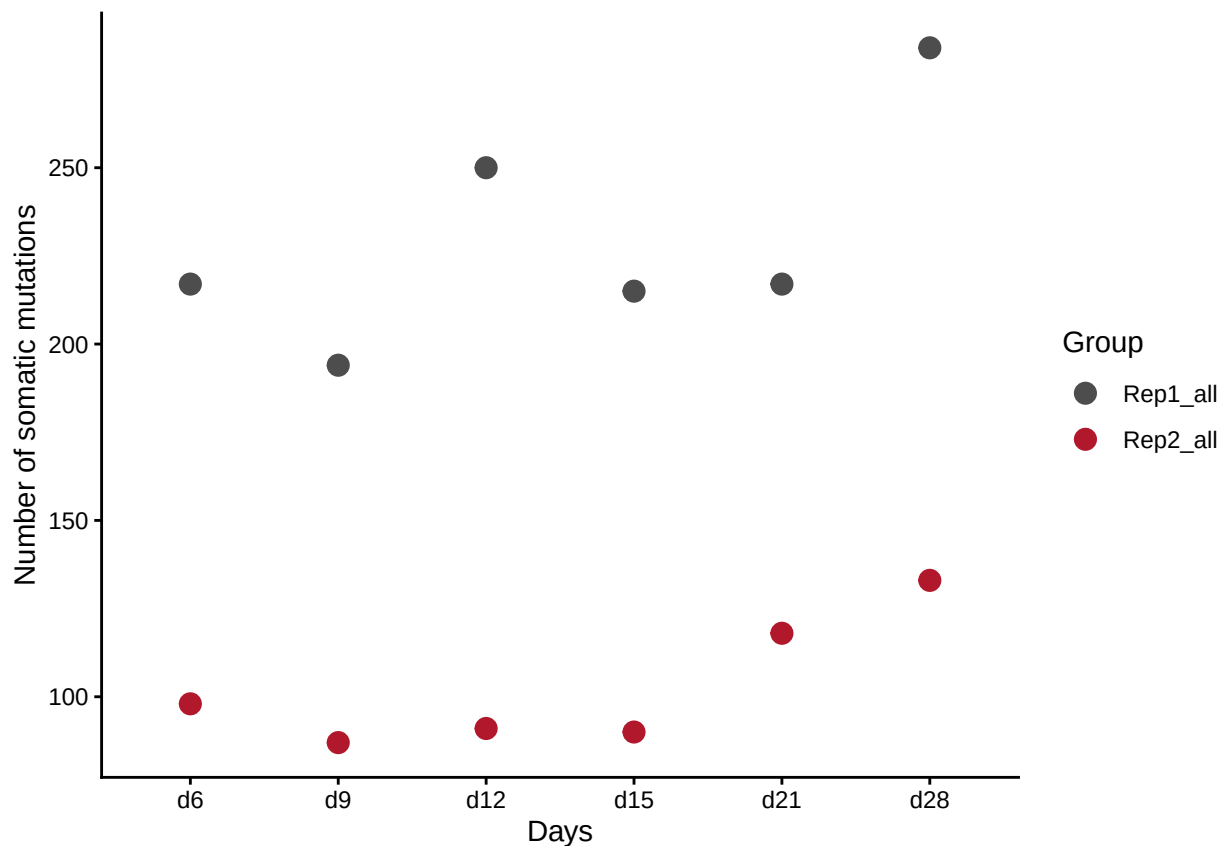
data_accMutNum_res_plot <- melt(data_accMutNum_res)
## Using Days as id variables
colnames(data_accMutNum_res_plot) <- c("Days", "Group", "MutNum")

data_accMutNum_res_plot$Days <- factor(data_accMutNum_res_plot$Days, levels = c("d6", "d9", "d12", "d15", "d21", "d28"))

p_reps_somaticNum <- ggplot(data_accMutNum_res_plot, aes(x = Days, y = MutNum, colour = Group, fill = Group)) +
  geom_point(shape = 21, size = 3, stroke = 0.8) +
  ylab("Number of somatic mutations") +
  scale_colour_manual(values = c("#4d4d4d", "#b2182b")) +
  scale_fill_manual(values = c("#4d4d4d", "#b2182b")) +
  theme_classic()

p_reps_somaticNum

```



```

replacement_map <- c("NoCall" = 0, "WT" = 0, "Mut" = 1)

MEF_SSM_hf1_hf2 <- merge(MEF_SSM_hf1[,c("mutPos", col_hf1_names)], MEF_SSM_hf2[,c("mutPos", col_hf2_names)], by = "mutPos")
MEF_SSM_hf1_hf2[is.na(MEF_SSM_hf1_hf2)] <- "NoCall"

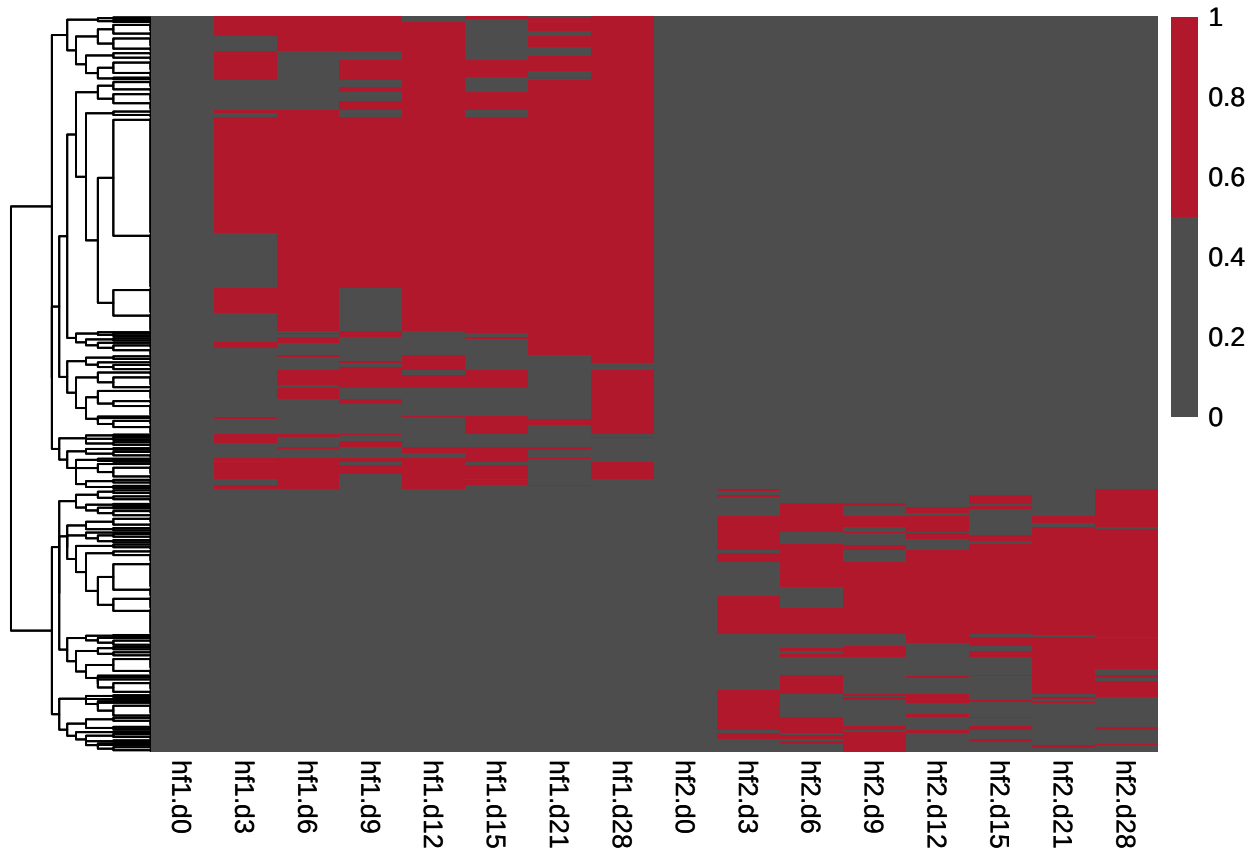
MEF_SSM_hf1_hf2_num <- as.data.frame(lapply(MEF_SSM_hf1_hf2, function(x) {
  ifelse(x %in% names(replacement_map), replacement_map[x], x)
}))

rownames(MEF_SSM_hf1_hf2_num) <- MEF_SSM_hf1_hf2$mutPos

```



```
# propagate_mutations
p_accSomaticMut_heatmap_observed <- pheatmap(MEF_SSM_hf1_hf2_num[,c(2:17)], show_rownames = F, color = c(
p_accSomaticMut_heatmap_observed
```



6 Imputed Accumulated Mutation Num

```
color_settings <- c(d0 = "#FFDC91FF", d3 = "#7876B1FF", d6 = "#4575b4",
d9 = "#91bfdb", d12 = "#e0f3f8", d15 = "#fee090",
d21 = "#fc8d59", d28 = "#BC3C29FF")

color_settings_2 <- c(Day_0 = "#FFDC91FF", Day_3 = "#7876B1FF", Day_6 = "#4575b4",
Day_9 = "#91bfdb", Day_12 = "#e0f3f8", Day_15 = "#fee090",
Day_21 = "#fc8d59", Day_28 = "#BC3C29FF")

Reps_sequencing <- c("Rep1_all", "Rep2_all")
Days_sequencing <- c("d0", "d3", "d6", "d9", "d12", "d15", "d21", "d28")

col_hf1_names <- c("hf1.d0", "hf1.d3", "hf1.d6", "hf1.d9", "hf1.d12", "hf1.d15", "hf1.d21", "hf1.d28")
col_hf2_names <- c("hf2.d0", "hf2.d3", "hf2.d6", "hf2.d9", "hf2.d12", "hf2.d15", "hf2.d21", "hf2.d28")

MEF_SSM_hf1_impute <- propagate_mutations(MEF_SSM_hf1, col_hf1_names)
MEF_SSM_hf2_impute <- propagate_mutations(MEF_SSM_hf2, col_hf2_names)
```

```

data_accMutNum_res <- as.data.frame(matrix(nrow = length(Days_sequencing), ncol = length(Reps_sequencing),
colnames(data_accMutNum_res) <- Reps_sequencing
rownames(data_accMutNum_res) <- Days_sequencing

for (i in rownames(data_accMutNum_res)){
  data_accMutNum_res[i, "Rep1_all"] <- length(MEF_SSM_hf1_impute[MEF_SSM_hf1_impute[, paste0("hf1.", i)],
  data_accMutNum_res[i, "Rep2_all"] <- length(MEF_SSM_hf2_impute[MEF_SSM_hf2_impute[, paste0("hf2.", i)],
}

data_accMutNum_res$Days <- rownames(data_accMutNum_res)
data_accMutNum_res_plot <- melt(data_accMutNum_res)
## Using Days as id variables
colnames(data_accMutNum_res_plot) <- c("Days", "Group", "MutNum")

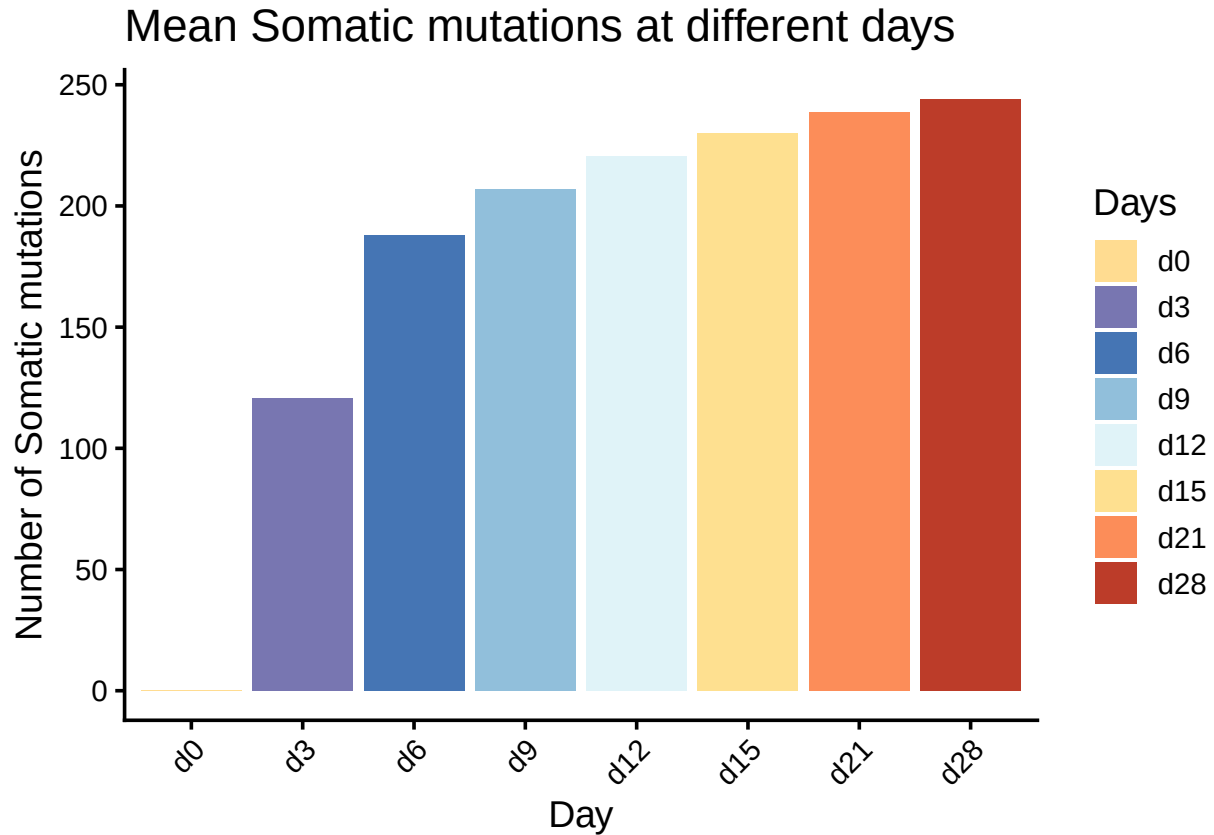
data_accMutNum_res_mean_plot <- aggregate(MutNum ~ Days, data = data_accMutNum_res_plot, FUN = mean)

data_accMutNum_res_mean_plot$Days <- factor(data_accMutNum_res_mean_plot$Days, levels = c("d0", "d3", "d7"))

p_reps_somaticNum <- ggplot(data_accMutNum_res_mean_plot, aes(x = Days, y = MutNum, fill = Days)) +
  geom_col(position = "dodge") +
  theme_minimal(base_size = 14) +
  labs(title = "Mean Somatic mutations at different days",
       x = "Day",
       y = "Number of Somatic mutations") +
  scale_fill_manual(values = color_settings) +
  theme_classic(base_size = 14) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

p_reps_somaticNum

```



```

replacement_map <- c("NoCall" = 0, "WT" = 0, "Mut" = 1)

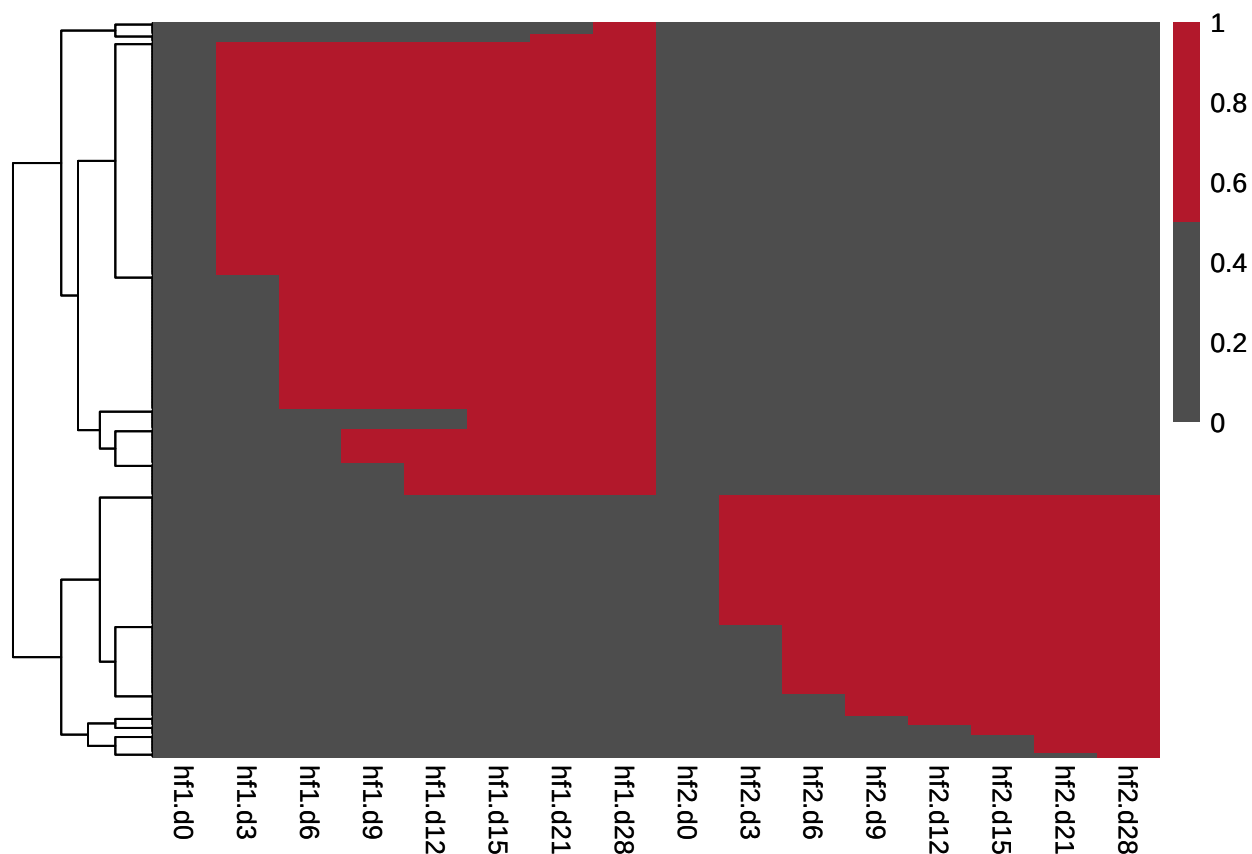
MEF_SSM_hf1_hf2_impute <- merge(MEF_SSM_hf1_impute[,c("mutPos", col_hf1_names)], MEF_SSM_hf2_impute[,c(
MEF_SSM_hf1_hf2_impute[is.na(MEF_SSM_hf1_hf2_impute)] <- "NoCall"

MEF_SSM_hf1_hf2_impute_num <- as.data.frame(lapply(MEF_SSM_hf1_hf2_impute, function(x) {
  ifelse(x %in% names(replacement_map), replacement_map[x], x)
})))

rownames(MEF_SSM_hf1_hf2_impute_num) <- MEF_SSM_hf1_hf2_impute$mutPos

# propagate_mutations
p_accSomaticMut_heatmap_impute <- pheatmap(MEF_SSM_hf1_hf2_impute_num[,c(2:17)], show_rownames = F, col
p_accSomaticMut_heatmap_impute

```



```
# cell number
hf1_cellNum <- as.data.frame(table(MEF_Meta_hf1$Day)) %>%
  mutate(Sample = "hf1")
hf2_cellNum <- as.data.frame(table(MEF_Meta_hf2$Day)) %>%
  mutate(Sample = "hf2")

hf_cellNum <- bind_rows(hf1_cellNum, hf2_cellNum)
colnames(hf_cellNum) <- c("Day", "Count", "Sample")

hf_cellNum_dayMean <- aggregate(Count ~ Day, data = hf_cellNum, FUN = mean)

days_order_row <- c("Day_0", "Day_3", "Day_6", "Day_9", "Day_12",
  "Day_15", "Day_21", "Day_28")

hf_cellNum_dayMean <- hf_cellNum_dayMean %>% filter(Day %in% days_order_row)
hf_cellNum_dayMean$Day <- factor(hf_cellNum_dayMean$Day, levels = days_order_row)

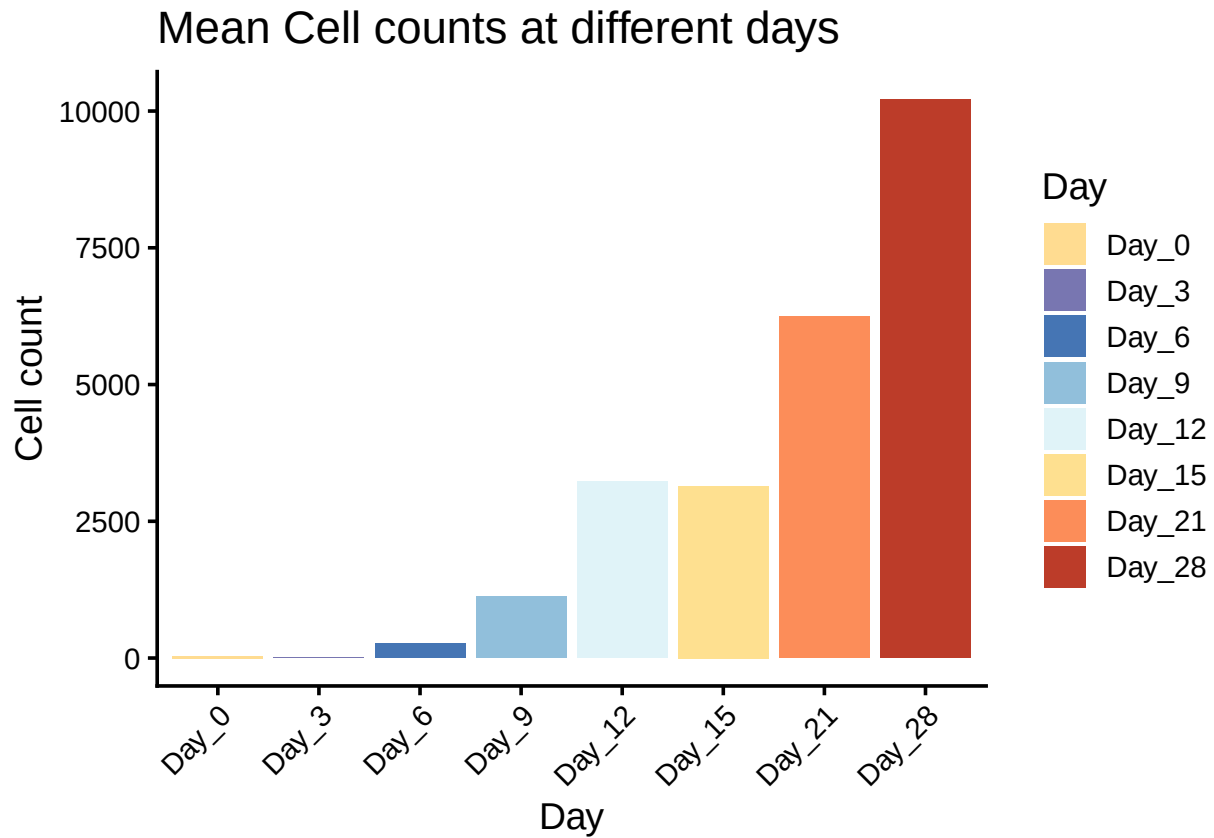
color_settings_2 <- c(Day_0 = "#FFDC91FF", Day_3 = "#7876B1FF", Day_6 = "#4575b4",
  Day_9 = "#91bffd", Day_12 = "#e0f3f8", Day_15 = "#fee090",
  Day_21 = "#fc8d59", Day_28 = "#BC3C29FF")

p_hf_cellNum <- ggplot(hf_cellNum_dayMean, aes(x = Day, y = Count, fill = Day)) +
  geom_col(position = "dodge") +
  theme_minimal(base_size = 14) +
```

```

labs(title = "Mean Cell counts at different days",
     x = "Day",
     y = "Cell count") +
scale_fill_manual(values = color_settings_2) +
theme_classic(base_size = 14) +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
p_hf_cellNum

```



7 Accumulated Mutation Num normalized by cell number

```

color_settings_randomCells <- c(hf1.d3 = "#7876B1FF", hf1.d6 = "#4575b4", hf1.d9 = "#91bfdb", hf1.d12 = "#e0f3f8",
                                hf2.d3 = "#7876B1FF", hf2.d6 = "#4575b4", hf2.d9 = "#91bfdb", hf2.d12 = "#e0f3f8")

color_settings_randomCells_day <- c(d6 = "#4575b4", d9 = "#91bfdb", d12 = "#e0f3f8", d15 = "#fee090", d21 = "#ff7f0e", d28 = "#d62728")

totalCells <- c(rownames(MEF_CSM_hf1), rownames(MEF_CSM_hf2))

selectRandomNum <- c(1000, 1314, 2000, 2186, 3000, 4000, 5000)

dataSelect <- c("hf1.d6", "hf1.d9", "hf1.d12", "hf1.d15", "hf1.d21", "hf1.d28",
                "hf2.d6", "hf2.d9", "hf2.d12", "hf2.d15", "hf2.d21", "hf2.d28")

```

```

days <- c("d6", "d9", "d12", "d15", "d21", "d28")
reps <- 20
data_random_counts <- as.data.frame(matrix(nrow = (length(selectRandomNum) * length(dataSelect) * reps),
colnames(data_random_counts) <- c("ID", "reps", "selectNum", "MutNum")

set.seed(2025)
t = 0
for (i in dataSelect){
  dataCells <- totalCells[grep(i, totalCells)]
  selectNum <- selectRandomNum
  for (j in 1:reps){
    for (k in selectNum){
      dataCells_random <- sample(dataCells, k, replace = TRUE)
      hf1_selected <- dataCells_random[dataCells_random %in% rownames(MEF_CSM_hf1)]
      hf2_selected <- dataCells_random[dataCells_random %in% rownames(MEF_CSM_hf2)]

      data_random_hf1 <- MEF_CSM_hf1[hf1_selected,]
      data_random_hf2 <- MEF_CSM_hf2[hf2_selected,]

      num_hf1 <- sum(colSums(data_random_hf1 == "Mut") > 0)
      num_hf2 <- sum(colSums(data_random_hf2 == "Mut") > 0)
      t = t + 1
      if(t %% 400 == 0){
        print(paste0("processed ", t, " iterations"))
      }
      data_random_counts[t,]$ID <- i
      data_random_counts[t,]$reps <- j
      data_random_counts[t,]$selectNum <- k
      data_random_counts[t,]$MutNum <- num_hf1 + num_hf2
    }
  }
}

## [1] "processed 400 iterations"
## [1] "processed 800 iterations"
## [1] "processed 1200 iterations"
## [1] "processed 1600 iterations"

data_random_counts <- separate(data_random_counts,col = "ID",into = c("Timecourse", "Day"), sep = "\\.")
data_random_counts$ID <- paste(data_random_counts$Timecourse, data_random_counts$Day, sep = ".")
data_random_counts$Day <- factor(data_random_counts$Day, levels = days)
data_random_counts$ID <- factor(data_random_counts$ID, levels = dataSelect)
data_random_counts_plot <- data_random_counts[data_random_counts$selectNum %in% c(1000, 2000, 3000, 4000),]

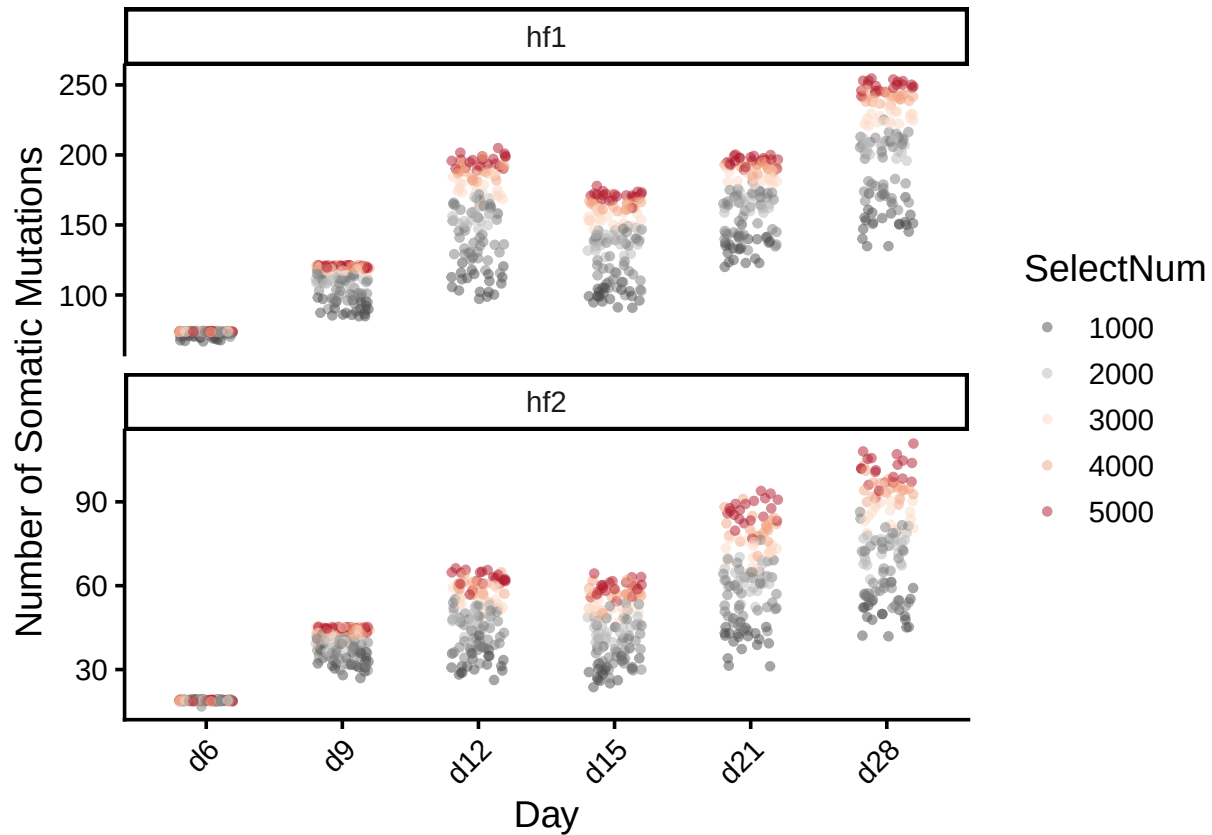
p_random_somaticNum <- ggplot(data_random_counts,
  aes(x = Day, y = MutNum, color = factor(selectNum))) +
  geom_jitter(width = 0.2, alpha = 0.5, size = 1) +
  facet_wrap(~ Timecourse, ncol = 1, scales = "free_y") +
  ylab("Number of Somatic Mutations") +
  xlab("Day") +
  labs(color = "SelectNum") +
  scale_color_manual(values = c(
    "1000" = "#4d4d4d",
    "2000" = "#bababa",

```

```

"3000" = "#fddbc7",
"4000" = "#f4a582",
"5000" = "#b2182b"
)) +
theme_classic(base_size = 14) +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
p_random_somaticNum

```



8 MPGgenotypecheck

```

# check position
MPG_data <- read.csv("../Data/MEF/MPG_combined.csv")
MEF_detected_pos <- unique(union(MEF_SSM_hf1$position_cor, MEF_SSM_hf2$position_cor))
MEF_detected_pos[MEF_detected_pos %in% MPG_data$position_cor]
## [1] "10_28334861" "10_28384990" "11_103070309" "11_59698972" "11_80397502"
## [6] "12_59094027" "12_81947565" "13_44240902" "13_59527224" "15_102984869"
## [11] "15_31472794" "15_96305641" "15_96697854" "16_20124088" "16_37897824"
## [16] "16_38799521" "17_28695383" "1_162857222" "1_178800830" "1_61723999"
## [21] "2_169768737" "2_75700504" "3_19993055" "3_22093358" "3_55288508"
## [26] "4_103363073" "4_6980700" "5_36660914" "5_66596487" "6_128290883"
## [31] "6_90475221" "7_137044144" "7_141638786" "7_16356122" "8_45733398"

```

```

## [36] "8_45807946" "8_71690586" "8_94675603" "9_124330134" "9_21429385"
## [41] "9_60385333" "10_32940591" "10_78338121" "13_119610824" "13_49547531"
## [46] "14_103312866" "17_62865029" "18_60642899" "1_137967589" "1_57915729"
## [51] "2_121793360" "5_98880003" "6_120931717" "7_13047148" "7_133890940"
## [56] "8_45786901" "8_45802144" "9_100773522" "9_102813915"

MPG_data <- MPG_data[,c(3,8:44)]
MPG_data[MPG_data == ""] <- NA

MEF_SSM_refAlt <- rbind(MEF_SSM_hf1[,c("position_cor", "ref", "alt")], MEF_SSM_hf2[,c("position_cor", "ref", "alt")])
MPG_data_refAlt <- merge(MEF_SSM_refAlt, MPG_data, by = "position_cor", all.x = T)
MPG_data_refAlt[is.na(MPG_data_refAlt)] <- 0

MPG_data_refAlt[, 4:ncol(MPG_data_refAlt)] <- apply(MPG_data_refAlt[, 4:ncol(MPG_data_refAlt)], 2, function(x) {
  refCheck <- MPG_data_refAlt[, "ref"]
  ifelse(col != 0 & col != refCheck, 1, 0)
})

MPG_data_refAlt <- MPG_data_refAlt[!duplicated(MPG_data_refAlt$position_cor),]
rownames(MPG_data_refAlt) <- MPG_data_refAlt$position_cor

MPG_data_refAlt$MutNum <- rowSums(MPG_data_refAlt[,4:40])
MPG_data_refAlt_sorted <- MPG_data_refAlt[order(-MPG_data_refAlt$MutNum), ]

colnames_order <- c("C57BL_6J", "C57BL_10J", "C57BL_6NJ", "C57BR_cdJ", "C57L_J",
  "AKR_J", "A_J", "BALB_cJ", "BTBRT_Itpr3_J", "BUB_BnJ",
  "C3H_HeH", "C3H_HeJ", "C58_J", "CAST_EiJ", "CBA_J",
  "DBA_1J", "DBA_2J", "FVB_NJ", "I_LnJ", "KK_HlJ",
  "LEWES_EiJ", "LP_J", "MOLF_EiJ",
  "NOD_ShilTJ", "NZB_B1NJ", "NZO_H1LtJ", "NZW_LacJ",
  "PWK_PhJ", "RF_J", "SEA_GnJ", "SPRET_EiJ", "ST_bJ", "WSB_EiJ",
  "ZALENDE_EiJ", "X129P2_01aHsd", "X129S1_SvImJ", "X129S5SvEv_Brd")

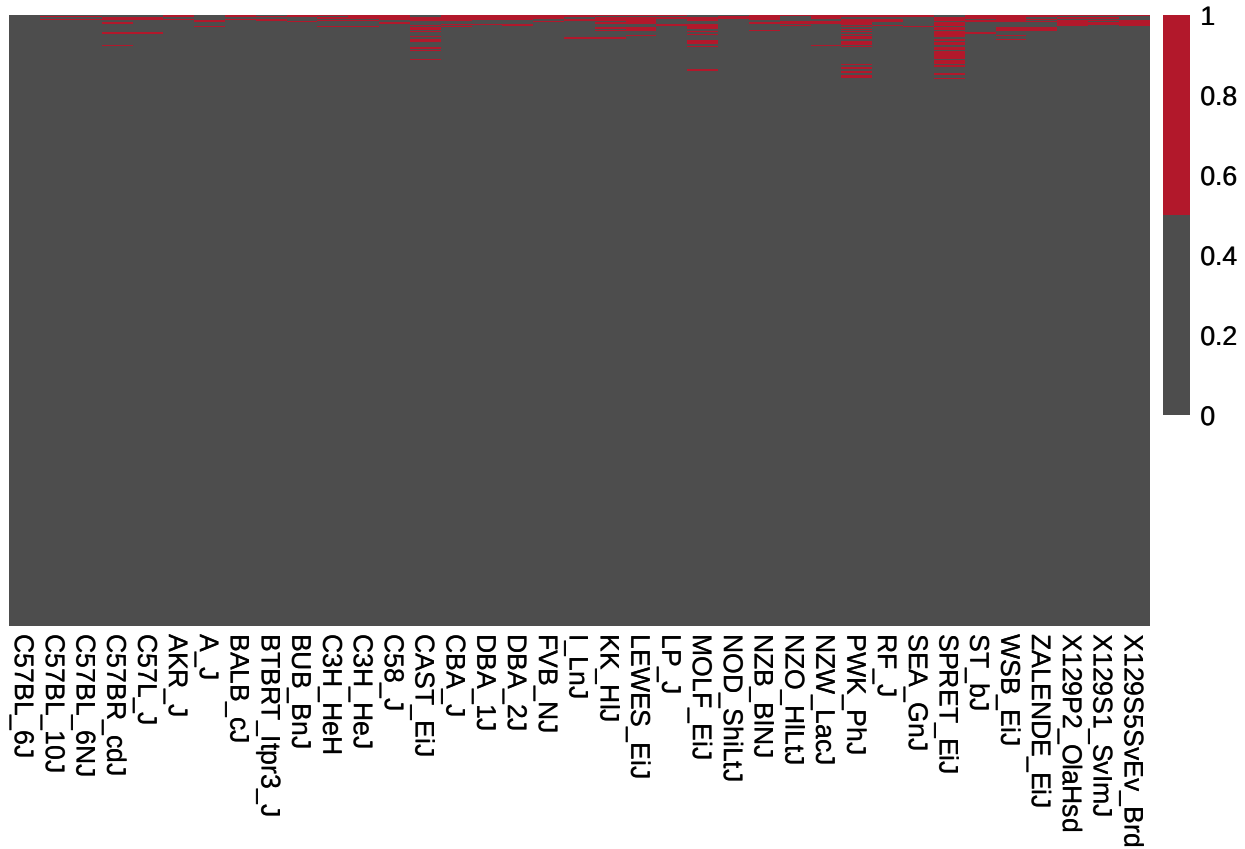
colnames_order[!colnames_order %in% colnames(MPG_data_refAlt_sorted)]
## character(0)

heatmap_data <- MPG_data_refAlt_sorted[, colnames_order]

custom_colors <- c("#4d4d4d", "#b2182b")

p_heatmapMPG <- pheatmap(heatmap_data, show_rownames = FALSE, cluster_rows = FALSE, cluster_cols = FALSE)
p_heatmapMPG

```

9 Calculating mutations of each cell and cells of each mutation

```
data_MutPerCell_hf1 <- data.frame(
  CellTag = rownames(MEF_CSM_hf1),
  MutNum_somatic = rowSums(MEF_CSM_hf1 == "Mut"),
  WTNum_somatic = rowSums(MEF_CSM_hf1 == "WT")
)

data_CellPerMut_hf1 <- data.frame(
  Location = colnames(MEF_CSM_hf1),
  CellMutNum = colSums(MEF_CSM_hf1 == "Mut"),
  CellWTNum = colSums(MEF_CSM_hf1 == "WT")
)

data_MutPerCell_hf2 <- data.frame(
  CellTag = rownames(MEF_CSM_hf2),
  MutNum_somatic = rowSums(MEF_CSM_hf2 == "Mut"),
  WTNum_somatic = rowSums(MEF_CSM_hf2 == "WT")
)

data_CellPerMut_hf2 <- data.frame(
  Location = colnames(MEF_CSM_hf2),
  CellMutNum = colSums(MEF_CSM_hf2 == "Mut"),

```

```

CellWTNum = colSums(MEF_CSM_hf2 == "WT")
)

```

10 calMulti

```

# [Cell * Time] = [Cell * Mut] * [Mut * Time]
time_types <- c("Day_6", "Day_9", "Day_12", "Day_15", "Day_21", "Day_28")
cell_num = 1
mut_num = 1

# _hf1
cells_candidate_hf1 <- data_MutPerCell_hf1[data_MutPerCell_hf1$MutNum_somatic >= cell_num,]$CellTag
cells_candidate_hf1 <- cells_candidate_hf1[cells_candidate_hf1 %in% MEF_Meta_hf1$Cellbarcode]
mut_candidate_hf1 <- data_CellPerMut_hf1[data_CellPerMut_hf1$CellMutNum >= mut_num,]$Location

cell_time_num_hf1 <- sapply(col_hf1_names, function(i) sum(str_count(cells_candidate_hf1, i)))
total_cells_hf1 <- sum(cell_time_num_hf1)
cell_time_num_ratio_hf1 <- cell_time_num_hf1/total_cells_hf1 + 2

# data_Cell_Mut
data_cell_mut_raw_hf1 <- MEF_CSM_hf1[cells_candidate_hf1, mut_candidate_hf1]

# data_Mut_Time
data_mut_time_hf1 <- as.data.frame(matrix(nrow = length(mut_candidate_hf1), ncol = length(time_types)),
rownames(data_mut_time_hf1) <- mut_candidate_hf1
colnames(data_mut_time_hf1) <- time_types

for (i in 1:nrow(data_mut_time_hf1)){
  mut_tmp <- rownames(data_mut_time_hf1)[i]
  cell_tmp <- retrieveCellsByLocation(data_cell_mut_raw_hf1, mut_tmp)
  selected_meta_tags <- MEF_Meta_hf1[MEF_Meta_hf1$Cellbarcode %in% cell_tmp$Mut,]$Day
  data_mut_time_hf1[i,]$Day_6 <- sum(str_count(selected_meta_tags, "Day_6"))
  data_mut_time_hf1[i,]$Day_9 <- sum(str_count(selected_meta_tags, "Day_9"))
  data_mut_time_hf1[i,]$Day_12 <- sum(str_count(selected_meta_tags, "Day_12"))
  data_mut_time_hf1[i,]$Day_15 <- sum(str_count(selected_meta_tags, "Day_15"))
  data_mut_time_hf1[i,]$Day_21 <- sum(str_count(selected_meta_tags, "Day_21"))
  data_mut_time_hf1[i,]$Day_28 <- sum(str_count(selected_meta_tags, "Day_28"))
}

data_mut_time_hf1_nor <- sweep(data_mut_time_hf1, 2, log2(cell_time_num_ratio_hf1[3:8]), "/")

# _hf2
cells_candidate_hf2 <- data_MutPerCell_hf2[data_MutPerCell_hf2$MutNum_somatic >= cell_num,]$CellTag
cells_candidate_hf2 <- cells_candidate_hf2[cells_candidate_hf2 %in% MEF_Meta_hf2$Cellbarcode]
mut_candidate_hf2 <- data_CellPerMut_hf2[data_CellPerMut_hf2$CellMutNum >= mut_num,]$Location

cell_time_num_hf2 <- sapply(col_hf2_names, function(i) sum(str_count(cells_candidate_hf2, i)))
total_cells_hf2 <- sum(cell_time_num_hf2)
cell_time_num_ratio_hf2 <- cell_time_num_hf2/total_cells_hf2 + 2

# data_Cell_Mut

```

```

data_cell_mut_raw_hf2 <- MEF_CSM_hf2[cells_candidate_hf2, muts_candidate_hf2]

# data_Mut_Time
data_mut_time_hf2 <- as.data.frame(matrix(nrow = length(muts_candidate_hf2), ncol = length(time_types)))
rownames(data_mut_time_hf2) <- muts_candidate_hf2
colnames(data_mut_time_hf2) <- time_types

for (i in 1:nrow(data_mut_time_hf2)){
  mut_tmp <- rownames(data_mut_time_hf2)[i]
  cell_tmp <- retrieveCellsByLocation(data_cell_mut_raw_hf2, mut_tmp)
  selected_meta_tags <- MEF_Meta_hf2[MEF_Meta_hf2$Cellbarcode %in% cell_tmp$Mut,]$Day
  data_mut_time_hf2[i,]$Day_6 <- sum(str_count(selected_meta_tags, "Day_6"))
  data_mut_time_hf2[i,]$Day_9 <- sum(str_count(selected_meta_tags, "Day_9"))
  data_mut_time_hf2[i,]$Day_12 <- sum(str_count(selected_meta_tags, "Day_12"))
  data_mut_time_hf2[i,]$Day_15 <- sum(str_count(selected_meta_tags, "Day_15"))
  data_mut_time_hf2[i,]$Day_21 <- sum(str_count(selected_meta_tags, "Day_21"))
  data_mut_time_hf2[i,]$Day_28 <- sum(str_count(selected_meta_tags, "Day_28"))
}

data_mut_time_hf2_nor <- sweep(data_mut_time_hf2, 2, log2(cell_time_num_ratio_hf2[3:8]), "/")

res_index_hf1 <- cal_scMutScore(data_cell_mut_raw_hf1, data_mut_time_hf1_nor, MEF_Meta_hf1)
## [1] 4371 297
## [1] 297 6
res_index_hf2 <- cal_scMutScore(data_cell_mut_raw_hf2, data_mut_time_hf2_nor, MEF_Meta_hf2)
## [1] 2627 156
## [1] 156 6
res_index <- rbind(res_index_hf1$score, res_index_hf2$score)

res_index$cb <- rownames(res_index)
res_index <- separate(res_index, col = "cb", into = c("celltype", "cell"), sep = "_")
res_index$Tag <- res_index$celltype
res_index <- separate(res_index, col = "celltype", into = c("rep", "Day"), sep = "\\.")

res_index$Tag <- factor(res_index$Tag, levels = c("hf1.d0", "hf1.d3", "hf1.d6", "hf1.d9", "hf1.d12", "hf1.d15", "hf1.d21", "hf1.d28",
"hf2.d0", "hf2.d3", "hf2.d6", "hf2.d9", "hf2.d12", "hf2.d15", "hf2.d21", "hf2.d28"))

res_index$Day <- factor(res_index$Day, levels = c("d0", "d3", "d6", "d9", "d12", "d15", "d21", "d28"))

res_index <- res_index[res_index$Diversity > 0, ]

color_settings <- c(d0 = "#FFDC91FF", d3 = "#7876B1FF", d6 = "#4575b4",
d9 = "#91bfdb", d12 = "#e0f3f8", d15 = "#fee090",
d21 = "#fc8d59", d28 = "#BC3C29FF")

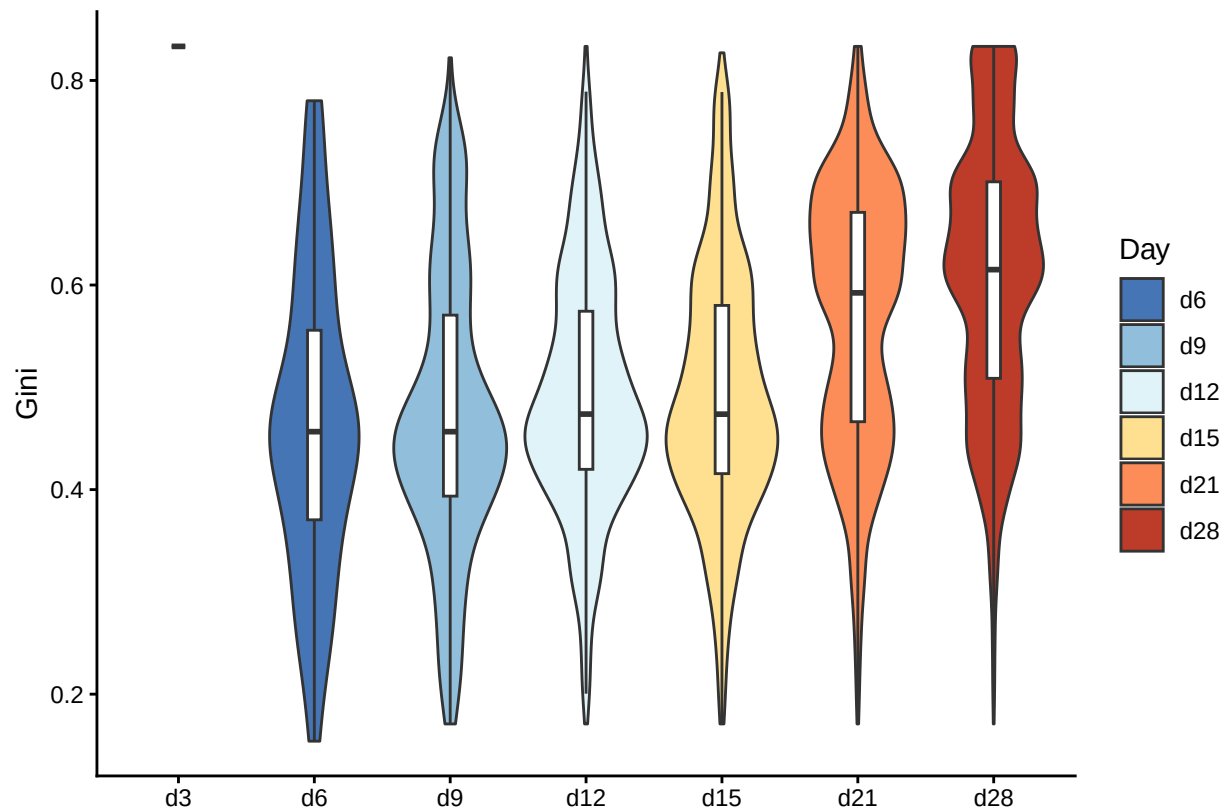
color_settings_2 <- c(Day_0 = "#FFDC91FF", Day_3 = "#7876B1FF", Day_6 = "#4575b4",
Day_9 = "#91bfdb", Day_12 = "#e0f3f8", Day_15 = "#fee090",
Day_21 = "#fc8d59", Day_28 = "#BC3C29FF")

res_index <- res_index[!is.na(res_index$scMutScore),]

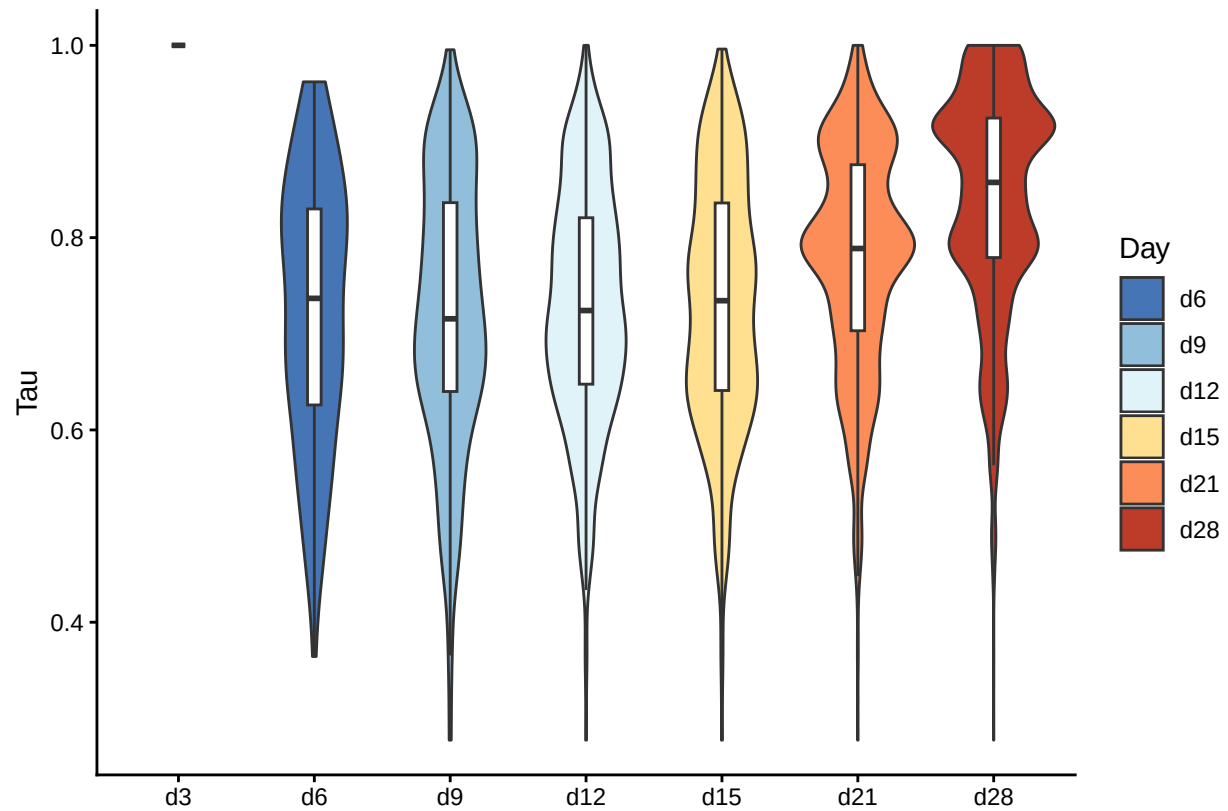
ggplot(data = res_index, mapping = aes(x = Day, y = Gini, fill = Day)) +
  geom_violin(trim = TRUE) +

```

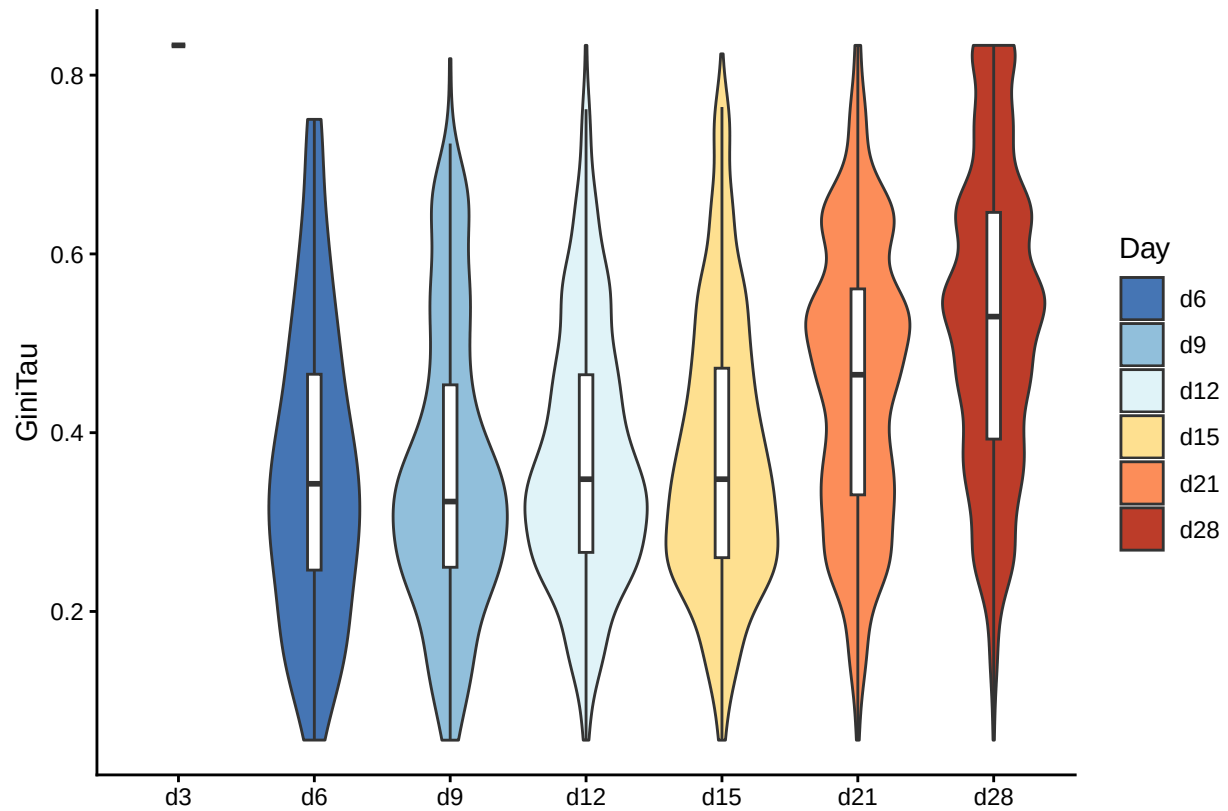
```
geom_boxplot(width = 0.1, fill = "white", outliers = FALSE) +
labs(x = "", y = "Gini") +
scale_fill_manual(values = color_settings) +
theme_classic()
## Warning: Groups with fewer than two datapoints have been dropped.
## i Set `drop = FALSE` to consider such groups for position adjustment purposes.
```



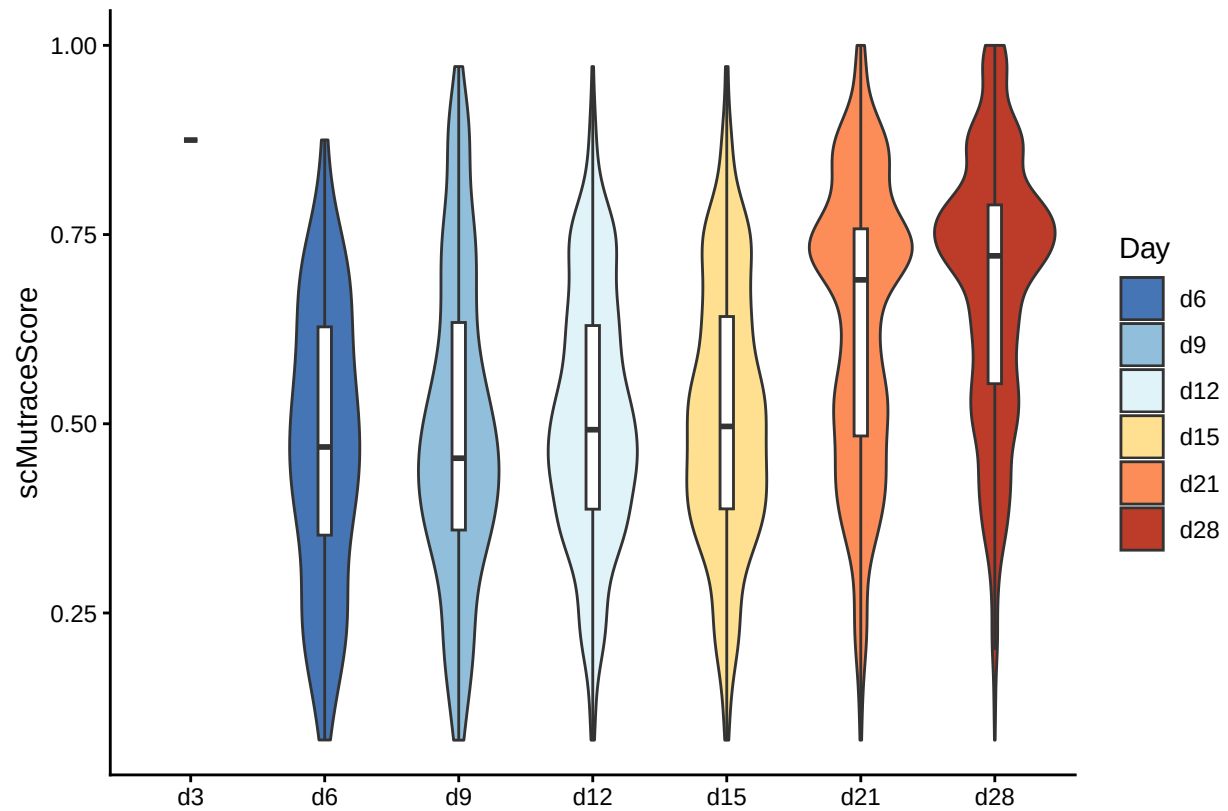
```
ggplot(data = res_index, mapping = aes(x = Day, y = Tau, fill = Day)) +
geom_violin(trim = TRUE) +
geom_boxplot(width = 0.1, fill = "white", outliers = FALSE) +
labs(x = "", y = "Tau") +
scale_fill_manual(values = color_settings) +
theme_classic()
## Warning: Groups with fewer than two datapoints have been dropped.
## i Set `drop = FALSE` to consider such groups for position adjustment purposes.
```



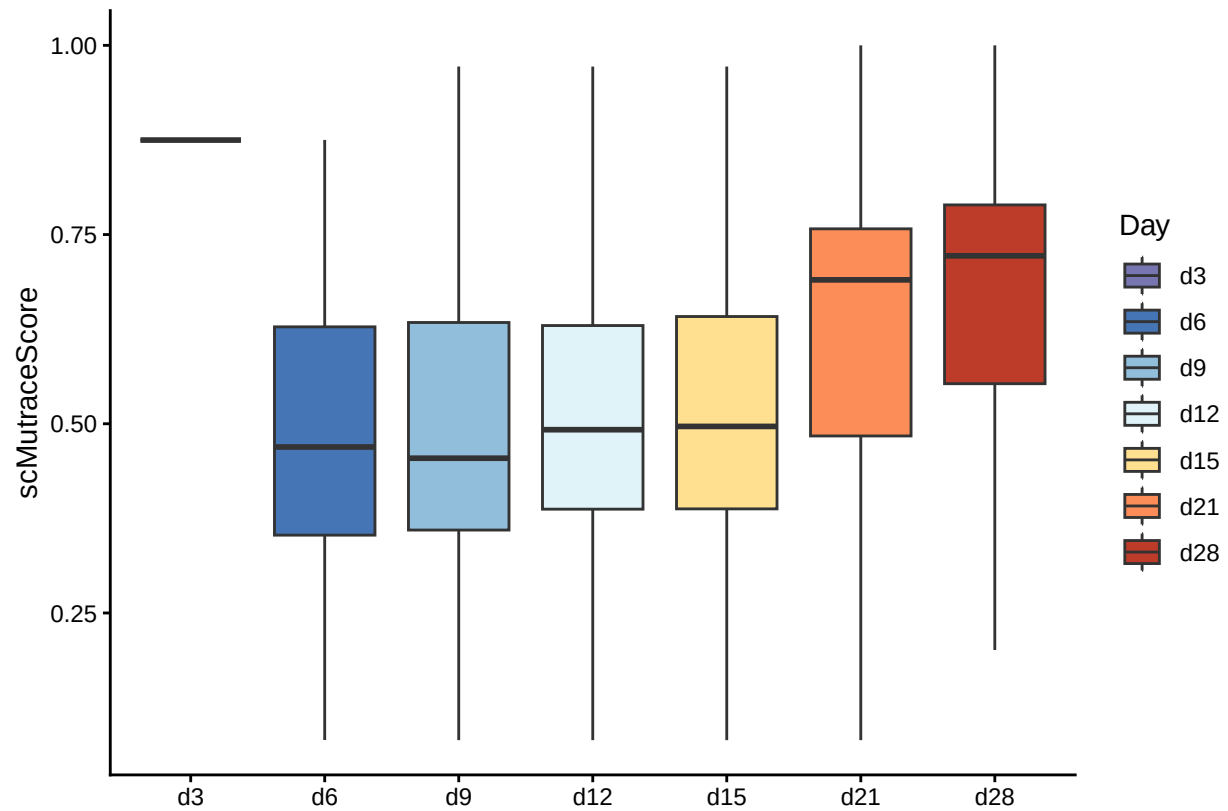
```
ggplot(data = res_index, mapping = aes(x = Day, y = GiniTau, fill = Day)) +
  geom_violin(trim = TRUE) +
  geom_boxplot(width = 0.1, fill = "white", outliers = FALSE) +
  labs(x = "", y = "GiniTau") +
  scale_fill_manual(values = color_settings) +
  theme_classic()
## Warning: Groups with fewer than two datapoints have been dropped.
## i Set `drop = FALSE` to consider such groups for position adjustment purposes.
```



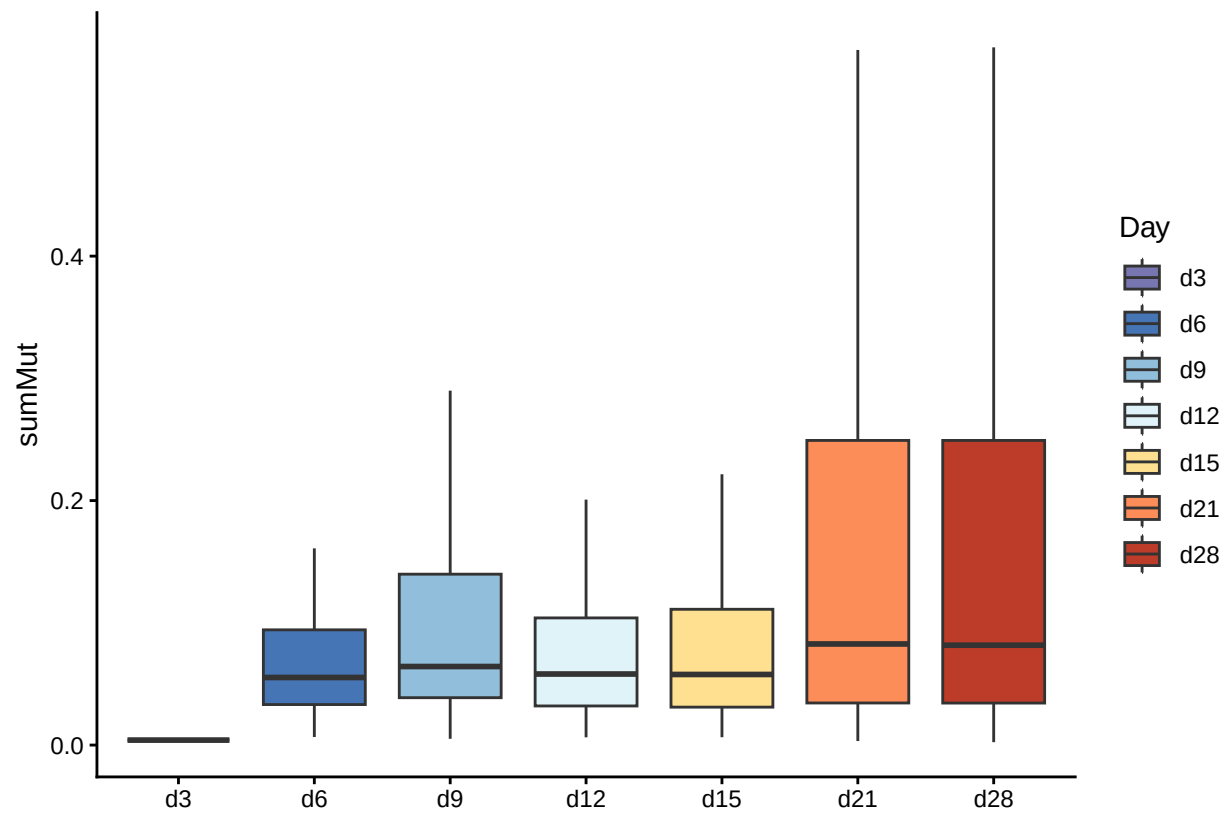
```
ggplot(data = res_index, mapping = aes(x = Day, y = scMutScore, fill = Day)) +
  geom_violin(trim = TRUE) +
  geom_boxplot(width = 0.1, fill = "white", outliers = FALSE) +
  labs(x = "", y = "scMutraceScore") +
  scale_fill_manual(values = color_settings) +
  theme_classic()
## Warning: Groups with fewer than two datapoints have been dropped.
## i Set `drop = FALSE` to consider such groups for position adjustment purposes.
```



```
ggplot(data = res_index, mapping = aes(x = Day, y = scMutScore, fill = Day)) +
  #geom_violin(trim = TRUE) +
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "scMutraceScore") +
  scale_fill_manual(values = color_settings) +
  theme_classic()
```

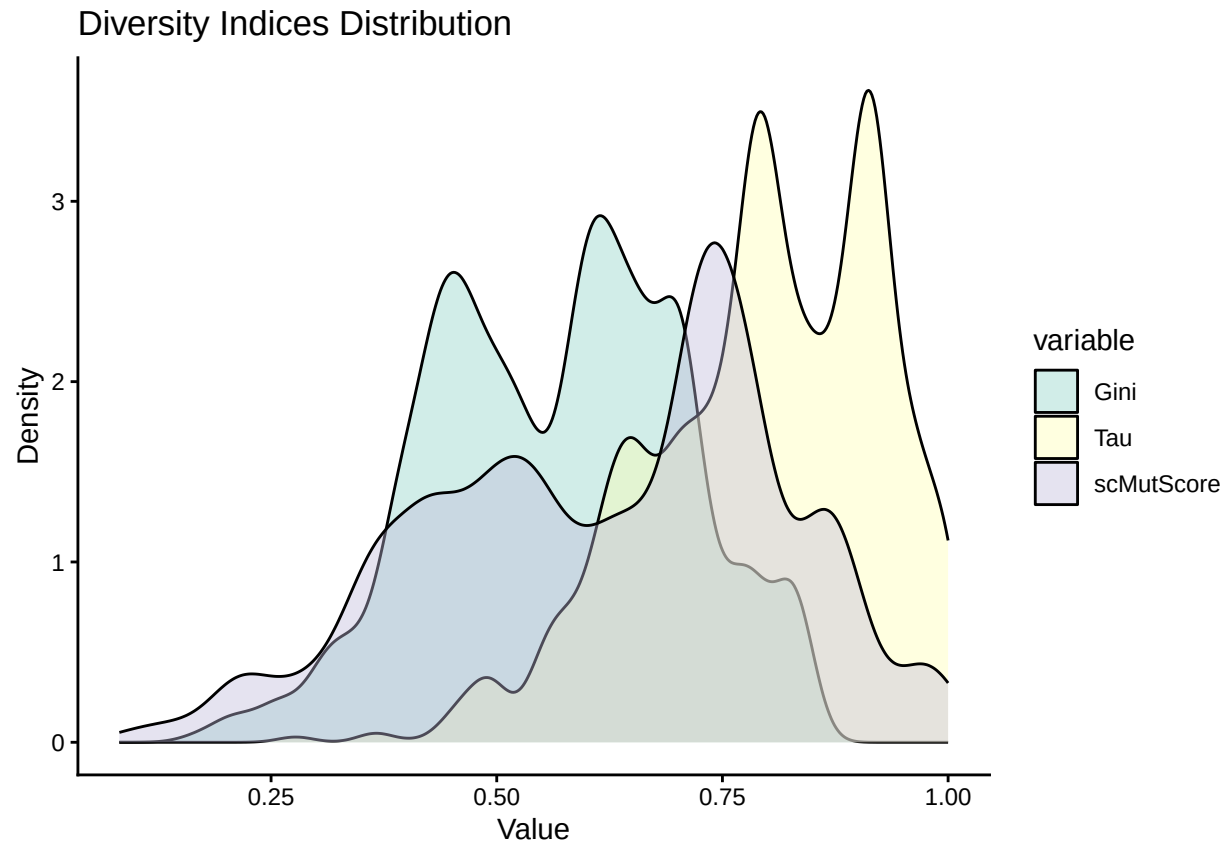


```
ggplot(data = res_index, mapping = aes(x = Day, y = scMutScoreQun, fill = Day)) +
  #geom_violin(trim = TRUE) +
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "sumMut") +
  scale_fill_manual(values = color_settings) +
  theme_classic()
```

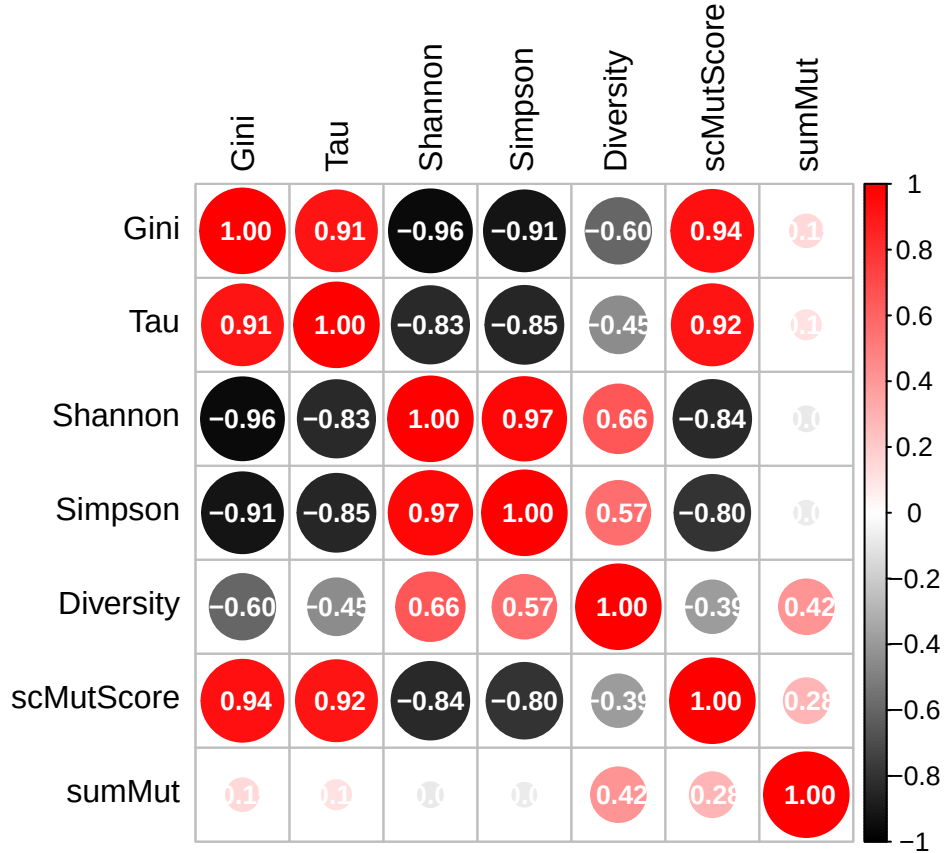



```
res_index_long <- melt(res_index[,c("Gini", "Tau", "scMutScore")])
## No id variables; using all as measure variables

ggplot(res_index_long, aes(x=value, fill=variable)) +
  geom_density(alpha=0.4) +
  theme_classic() +
  labs(x="Value", y="Density", title="Diversity Indices Distribution") +
  scale_fill_brewer(palette="Set3")
```



```
corrplot(cor(res_index[,c("Gini", "Tau", "Shannon", "Simpson", "Diversity", "scMutScore", "sumMut")], us
method="circle", col = colorRampPalette(c("black", "white", "red"))(200),
addCoef.col = "white", tl.col = "black", number.cex = 0.8)
```



11 scMutScore Model Validation

```
set.seed(2025)

# 1. init
data_check_CM_hf1 <- data_cell_mut_raw_hf1
data_check_Meta_hf1 <- MEF_Meta_hf1[MEF_Meta_hf1$Cellbarcode %in% rownames(data_check_CM_hf1),]
col_check_names_hf1 <- c("hf1.d0", "hf1.d3", "hf1.d6", "hf1.d9", "hf1.d12", "hf1.d15", "hf1.d21", "hf1.d28")
time_types_hf1 <- c("Day_6", "Day_9", "Day_12", "Day_15", "Day_21", "Day_28")

data_check_CM_hf2 <- data_cell_mut_raw_hf2
data_check_Meta_hf2 <- MEF_Meta_hf2[MEF_Meta_hf2$Cellbarcode %in% rownames(data_check_CM_hf2),]
col_check_names_hf2 <- c("hf2.d0", "hf2.d3", "hf2.d6", "hf2.d9", "hf2.d12", "hf2.d15", "hf2.d21", "hf2.d28")
time_types_hf2 <- c("Day_6", "Day_9", "Day_12", "Day_15", "Day_21", "Day_28")

col_check_names <- c(col_check_names_hf1, col_check_names_hf2)
cell_num <- 1
mut_num <- 1
mut_loss_ratio <- 0.5
cell_loss_ratio <- 0.5
iteration_times <- 100

# list for save data
```

```

data_scMutModel_list_hf1 <- vector("list", iteration_times)
data_scMutModel_list_hf2 <- vector("list", iteration_times)

num_cells_hf1 <- nrow(data_check_CM_hf1)
num_mutations_hf1 <- ncol(data_check_CM_hf1)
cells_all_hf1 <- rownames(data_check_CM_hf1)
mut_s_all_hf1 <- colnames(data_check_CM_hf1)

num_cells_hf2 <- nrow(data_check_CM_hf2)
num_mutations_hf2 <- ncol(data_check_CM_hf2)
cells_all_hf2 <- rownames(data_check_CM_hf2)
mut_s_all_hf2 <- colnames(data_check_CM_hf2)

# est cell number effects
# for hf1
data_effect_hf1 <- data_random_counts %>% dplyr::filter(selectNum == 2186 & Timecourse == "hf1")
data_effect_hf1_size <- data_effect_hf1 %>%
  group_by(Timecourse, Day) %>%
  summarise(mean_MutNum = mean(MutNum, na.rm = TRUE))
## `summarise()` has grouped output by 'Timecourse'. You can override using the
## `.groups` argument.

mean_mut_by_day_hf1 <- data_effect_hf1 %>%
  dplyr::group_by(Day) %>%
  dplyr::summarise(mean_MutNum = mean(MutNum, na.rm = TRUE)) %>%
  dplyr::arrange(match(Day, c("d6", "d9", "d12", "d15", "d21", "d28")))

data_effect_hf2 <- data_random_counts %>% dplyr::filter(selectNum == 1314 & Timecourse == "hf2")
data_effect_hf2_size <- data_effect_hf2 %>%
  group_by(Timecourse, Day) %>%
  summarise(mean_MutNum = mean(MutNum, na.rm = TRUE))
## `summarise()` has grouped output by 'Timecourse'. You can override using the
## `.groups` argument.

mean_mut_by_day_hf2 <- data_effect_hf2 %>%
  dplyr::group_by(Day) %>%
  dplyr::summarise(mean_MutNum = mean(MutNum, na.rm = TRUE)) %>%
  dplyr::arrange(match(Day, c("d6", "d9", "d12", "d15", "d21", "d28")))

# 2. loop
for (iter in seq_len(iteration_times)) {

  # 2.1 randomly loss info
  loss_cell_indices <- sample(seq_len(num_cells_hf1), ceiling(cell_loss_ratio * num_cells_hf1))
  loss_mut_indices <- sample(seq_len(num_mutations_hf1), ceiling(mut_loss_ratio * num_mutations_hf1))

  C_partial_missing <- data_check_CM_hf1
  # new data
  if(length(loss_cell_indices) > 0 && length(loss_mut_indices) > 0){
    C_partial_missing[loss_cell_indices, loss_mut_indices] <- "NoCall"
  }

  # 2.2 count cell/mut number

```

```

MutPerCell <- rowSums(C_partial_missing == "Mut")
CellPerMut <- colSums(C_partial_missing == "Mut")

cells_candidate <- cells_all_hf1[MutPerCell >= cell_num & cells_all_hf1 %in% data_check_Meta_hf1$Cellbarcode]
mut_candidate <- muts_all_hf1[CellPerMut >= mut_num]

message(paste0("Iteration: ", iter, " loss_cell_indices: ", length(loss_cell_indices), " loss_mut_indices: ", length(loss_mut_indices), "\n"))
message(paste0("Iteration: ", iter, " cells_candidate: ", length(cells_candidate), " muts_candidate: ", length(mut_candidate), "\n"))

data_cell_mut_raw_check <- C_partial_missing[cells_candidate, mut_candidate, drop = TRUE]

# 2.3 Mut × Time
cell_time_num <- sapply(col_check_names_hf1, function(i) sum(str_count(cells_candidate, i)))
total_cells <- sum(cell_time_num)
cell_time_ratio <- cell_time_num / total_cells + 2 # +2

# init Mut × Time
M_partial_missing <- matrix(0, nrow = length(mut_candidate), ncol = length(time_types_hf1),
                             dimnames = list(mut_candidate, time_types_hf1))

# Calculate the number of each mutation at each time
for (t in seq_along(time_types_hf1)) {
  time_label <- time_types_hf1[t]
  # Find out in which cells each mutation occurs
  mut_cell_list <- apply(data_cell_mut_raw_check, 2, function(x) which(x == "Mut"))
  # Count the number of times each mutation occurs at that point in time
  for (mut in mut_candidate) {
    cells_with_mut <- rownames(data_cell_mut_raw_check)[data_cell_mut_raw_check[, mut] == "Mut"]
    selected_days <- data_check_Meta_hf1$Day[data_check_Meta_hf1$Cellbarcode %in% cells_with_mut]
    M_partial_missing[mut, t] <- sum(str_count(selected_days, time_label))
  }
}

# 2.4 normalized Mut × Time
M_partial_missing_norm <- sweep(M_partial_missing, 2, log2(cell_time_ratio[3:8]), "/")
#M_partial_missing_norm <- sweep(M_partial_missing, 2, mean_mut_by_day_hf1$mean_MutNum, "/")

# 2.5 scMutScore
res_index_tmp <- cal_scMutScore(data_cell_mut_raw_check, M_partial_missing_norm, data_check_Meta_hf1)
res_index_tmp$iteration <- paste0("iter", iter)
res_index_tmp <- res_index_tmp[res_index_tmp$Diversity > 0, ]
res_index_tmp$cb <- rownames(res_index_tmp)
res_index_tmp$iteration <- paste0("iter", iter)
rownames(res_index_tmp) <- NULL
data_scMutModel_list_hf1[[iter]] <- res_index_tmp
}

## Iteration: 1 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 1 cells_candidate: 3452 muts_candidate: 287
## [1] 3452 287
## [1] 287 6
## Iteration: 2 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 2 cells_candidate: 3307 muts_candidate: 293
## [1] 3307 293

```

```

## [1] 293 6
## Iteration: 3 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 3 cells_candidate: 3402 muts_candidate: 287
## [1] 3402 287
## [1] 287 6
## Iteration: 4 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 4 cells_candidate: 3330 muts_candidate: 290
## [1] 3330 290
## [1] 290 6
## Iteration: 5 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 5 cells_candidate: 3451 muts_candidate: 292
## [1] 3451 292
## [1] 292 6
## Iteration: 6 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 6 cells_candidate: 3324 muts_candidate: 296
## [1] 3324 296
## [1] 296 6
## Iteration: 7 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 7 cells_candidate: 3345 muts_candidate: 291
## [1] 3345 291
## [1] 291 6
## Iteration: 8 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 8 cells_candidate: 3516 muts_candidate: 290
## [1] 3516 290
## [1] 290 6
## Iteration: 9 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 9 cells_candidate: 3324 muts_candidate: 292
## [1] 3324 292
## [1] 292 6
## Iteration: 10 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 10 cells_candidate: 3449 muts_candidate: 291
## [1] 3449 291
## [1] 291 6
## Iteration: 11 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 11 cells_candidate: 3451 muts_candidate: 288
## [1] 3451 288
## [1] 288 6
## Iteration: 12 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 12 cells_candidate: 3446 muts_candidate: 291
## [1] 3446 291
## [1] 291 6
## Iteration: 13 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 13 cells_candidate: 3444 muts_candidate: 293
## [1] 3444 293
## [1] 293 6
## Iteration: 14 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 14 cells_candidate: 3402 muts_candidate: 293
## [1] 3402 293
## [1] 293 6
## Iteration: 15 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 15 cells_candidate: 3291 muts_candidate: 288
## [1] 3291 288
## [1] 288 6

```

```

## Iteration: 16 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 16 cells_candidate: 3393 muts_candidate: 293
## [1] 3393 293
## [1] 293 6
## Iteration: 17 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 17 cells_candidate: 3225 muts_candidate: 294
## [1] 3225 294
## [1] 294 6
## Iteration: 18 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 18 cells_candidate: 3418 muts_candidate: 290
## [1] 3418 290
## [1] 290 6
## Iteration: 19 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 19 cells_candidate: 3449 muts_candidate: 296
## [1] 3449 296
## [1] 296 6
## Iteration: 20 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 20 cells_candidate: 3221 muts_candidate: 290
## [1] 3221 290
## [1] 290 6
## Iteration: 21 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 21 cells_candidate: 3246 muts_candidate: 293
## [1] 3246 293
## [1] 293 6
## Iteration: 22 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 22 cells_candidate: 3334 muts_candidate: 292
## [1] 3334 292
## [1] 292 6
## Iteration: 23 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 23 cells_candidate: 3167 muts_candidate: 295
## [1] 3167 295
## [1] 295 6
## Iteration: 24 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 24 cells_candidate: 3211 muts_candidate: 292
## [1] 3211 292
## [1] 292 6
## Iteration: 25 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 25 cells_candidate: 3502 muts_candidate: 291
## [1] 3502 291
## [1] 291 6
## Iteration: 26 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 26 cells_candidate: 3241 muts_candidate: 292
## [1] 3241 292
## [1] 292 6
## Iteration: 27 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 27 cells_candidate: 3384 muts_candidate: 292
## [1] 3384 292
## [1] 292 6
## Iteration: 28 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 28 cells_candidate: 3368 muts_candidate: 291
## [1] 3368 291
## [1] 291 6
## Iteration: 29 loss_cell_indices: 2186 loss_mut_indices: 149

```

```

## Iteration: 29 cells_candidate: 3437 muts_candidate: 293
## [1] 3437 293
## [1] 293 6
## Iteration: 30 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 30 cells_candidate: 3338 muts_candidate: 294
## [1] 3338 294
## [1] 294 6
## Iteration: 31 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 31 cells_candidate: 3394 muts_candidate: 293
## [1] 3394 293
## [1] 293 6
## Iteration: 32 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 32 cells_candidate: 3410 muts_candidate: 294
## [1] 3410 294
## [1] 294 6
## Iteration: 33 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 33 cells_candidate: 3371 muts_candidate: 290
## [1] 3371 290
## [1] 290 6
## Iteration: 34 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 34 cells_candidate: 3504 muts_candidate: 294
## [1] 3504 294
## [1] 294 6
## Iteration: 35 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 35 cells_candidate: 3368 muts_candidate: 290
## [1] 3368 290
## [1] 290 6
## Iteration: 36 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 36 cells_candidate: 3378 muts_candidate: 295
## [1] 3378 295
## [1] 295 6
## Iteration: 37 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 37 cells_candidate: 3323 muts_candidate: 294
## [1] 3323 294
## [1] 294 6
## Iteration: 38 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 38 cells_candidate: 3321 muts_candidate: 291
## [1] 3321 291
## [1] 291 6
## Iteration: 39 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 39 cells_candidate: 3420 muts_candidate: 290
## [1] 3420 290
## [1] 290 6
## Iteration: 40 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 40 cells_candidate: 3215 muts_candidate: 294
## [1] 3215 294
## [1] 294 6
## Iteration: 41 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 41 cells_candidate: 3474 muts_candidate: 291
## [1] 3474 291
## [1] 291 6
## Iteration: 42 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 42 cells_candidate: 3525 muts_candidate: 285

```



```

## [1] 3525 285
## [1] 285 6
## Iteration: 43 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 43 cells_candidate: 3364 muts_candidate: 288
## [1] 3364 288
## [1] 288 6
## Iteration: 44 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 44 cells_candidate: 3305 muts_candidate: 294
## [1] 3305 294
## [1] 294 6
## Iteration: 45 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 45 cells_candidate: 3362 muts_candidate: 292
## [1] 3362 292
## [1] 292 6
## Iteration: 46 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 46 cells_candidate: 3323 muts_candidate: 294
## [1] 3323 294
## [1] 294 6
## Iteration: 47 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 47 cells_candidate: 3417 muts_candidate: 291
## [1] 3417 291
## [1] 291 6
## Iteration: 48 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 48 cells_candidate: 3456 muts_candidate: 291
## [1] 3456 291
## [1] 291 6
## Iteration: 49 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 49 cells_candidate: 3506 muts_candidate: 292
## [1] 3506 292
## [1] 292 6
## Iteration: 50 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 50 cells_candidate: 3379 muts_candidate: 295
## [1] 3379 295
## [1] 295 6
## Iteration: 51 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 51 cells_candidate: 3239 muts_candidate: 294
## [1] 3239 294
## [1] 294 6
## Iteration: 52 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 52 cells_candidate: 3373 muts_candidate: 291
## [1] 3373 291
## [1] 291 6
## Iteration: 53 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 53 cells_candidate: 3338 muts_candidate: 287
## [1] 3338 287
## [1] 287 6
## Iteration: 54 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 54 cells_candidate: 3232 muts_candidate: 292
## [1] 3232 292
## [1] 292 6
## Iteration: 55 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 55 cells_candidate: 3425 muts_candidate: 289
## [1] 3425 289

```

```

## [1] 289 6
## Iteration: 56 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 56 cells_candidate: 3218 muts_candidate: 289
## [1] 3218 289
## [1] 289 6
## Iteration: 57 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 57 cells_candidate: 3285 muts_candidate: 294
## [1] 3285 294
## [1] 294 6
## Iteration: 58 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 58 cells_candidate: 3260 muts_candidate: 292
## [1] 3260 292
## [1] 292 6
## Iteration: 59 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 59 cells_candidate: 3532 muts_candidate: 290
## [1] 3532 290
## [1] 290 6
## Iteration: 60 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 60 cells_candidate: 3368 muts_candidate: 292
## [1] 3368 292
## [1] 292 6
## Iteration: 61 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 61 cells_candidate: 3414 muts_candidate: 295
## [1] 3414 295
## [1] 295 6
## Iteration: 62 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 62 cells_candidate: 3468 muts_candidate: 289
## [1] 3468 289
## [1] 289 6
## Iteration: 63 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 63 cells_candidate: 3327 muts_candidate: 289
## [1] 3327 289
## [1] 289 6
## Iteration: 64 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 64 cells_candidate: 3384 muts_candidate: 288
## [1] 3384 288
## [1] 288 6
## Iteration: 65 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 65 cells_candidate: 3232 muts_candidate: 290
## [1] 3232 290
## [1] 290 6
## Iteration: 66 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 66 cells_candidate: 3428 muts_candidate: 293
## [1] 3428 293
## [1] 293 6
## Iteration: 67 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 67 cells_candidate: 3204 muts_candidate: 290
## [1] 3204 290
## [1] 290 6
## Iteration: 68 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 68 cells_candidate: 3336 muts_candidate: 290
## [1] 3336 290
## [1] 290 6

```

```

## Iteration: 69 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 69 cells_candidate: 3297 muts_candidate: 288
## [1] 3297 288
## [1] 288 6
## Iteration: 70 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 70 cells_candidate: 3453 muts_candidate: 294
## [1] 3453 294
## [1] 294 6
## Iteration: 71 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 71 cells_candidate: 3295 muts_candidate: 293
## [1] 3295 293
## [1] 293 6
## Iteration: 72 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 72 cells_candidate: 3283 muts_candidate: 291
## [1] 3283 291
## [1] 291 6
## Iteration: 73 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 73 cells_candidate: 3265 muts_candidate: 293
## [1] 3265 293
## [1] 293 6
## Iteration: 74 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 74 cells_candidate: 3445 muts_candidate: 290
## [1] 3445 290
## [1] 290 6
## Iteration: 75 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 75 cells_candidate: 3306 muts_candidate: 293
## [1] 3306 293
## [1] 293 6
## Iteration: 76 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 76 cells_candidate: 3277 muts_candidate: 287
## [1] 3277 287
## [1] 287 6
## Iteration: 77 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 77 cells_candidate: 3423 muts_candidate: 288
## [1] 3423 288
## [1] 288 6
## Iteration: 78 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 78 cells_candidate: 3331 muts_candidate: 292
## [1] 3331 292
## [1] 292 6
## Iteration: 79 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 79 cells_candidate: 3371 muts_candidate: 290
## [1] 3371 290
## [1] 290 6
## Iteration: 80 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 80 cells_candidate: 3364 muts_candidate: 289
## [1] 3364 289
## [1] 289 6
## Iteration: 81 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 81 cells_candidate: 3534 muts_candidate: 292
## [1] 3534 292
## [1] 292 6
## Iteration: 82 loss_cell_indices: 2186 loss_mut_indices: 149

```

```

## Iteration: 82 cells_candidate: 3522 muts_candidate: 290
## [1] 3522 290
## [1] 290 6
## Iteration: 83 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 83 cells_candidate: 3356 muts_candidate: 288
## [1] 3356 288
## [1] 288 6
## Iteration: 84 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 84 cells_candidate: 3363 muts_candidate: 292
## [1] 3363 292
## [1] 292 6
## Iteration: 85 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 85 cells_candidate: 3351 muts_candidate: 289
## [1] 3351 289
## [1] 289 6
## Iteration: 86 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 86 cells_candidate: 3360 muts_candidate: 290
## [1] 3360 290
## [1] 290 6
## Iteration: 87 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 87 cells_candidate: 3358 muts_candidate: 291
## [1] 3358 291
## [1] 291 6
## Iteration: 88 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 88 cells_candidate: 3482 muts_candidate: 291
## [1] 3482 291
## [1] 291 6
## Iteration: 89 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 89 cells_candidate: 3390 muts_candidate: 291
## [1] 3390 291
## [1] 291 6
## Iteration: 90 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 90 cells_candidate: 3309 muts_candidate: 289
## [1] 3309 289
## [1] 289 6
## Iteration: 91 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 91 cells_candidate: 3510 muts_candidate: 290
## [1] 3510 290
## [1] 290 6
## Iteration: 92 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 92 cells_candidate: 3434 muts_candidate: 293
## [1] 3434 293
## [1] 293 6
## Iteration: 93 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 93 cells_candidate: 3429 muts_candidate: 292
## [1] 3429 292
## [1] 292 6
## Iteration: 94 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 94 cells_candidate: 3228 muts_candidate: 289
## [1] 3228 289
## [1] 289 6
## Iteration: 95 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 95 cells_candidate: 3352 muts_candidate: 292

```

```

## [1] 3352 292
## [1] 292 6
## Iteration: 96 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 96 cells_candidate: 3267 muts_candidate: 290
## [1] 3267 290
## [1] 290 6
## Iteration: 97 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 97 cells_candidate: 3366 muts_candidate: 290
## [1] 3366 290
## [1] 290 6
## Iteration: 98 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 98 cells_candidate: 3296 muts_candidate: 289
## [1] 3296 289
## [1] 289 6
## Iteration: 99 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 99 cells_candidate: 3295 muts_candidate: 293
## [1] 3295 293
## [1] 293 6
## Iteration: 100 loss_cell_indices: 2186 loss_mut_indices: 149
## Iteration: 100 cells_candidate: 3405 muts_candidate: 292
## [1] 3405 292
## [1] 292 6

for (iter in seq_len(iteration_times)) {

  # 2.1 randomly loss info
  loss_cell_indices <- sample(seq_len(num_cells_hf2), ceiling(cell_loss_ratio * num_cells_hf2))
  loss_mut_indices <- sample(seq_len(num_mutations_hf2), ceiling(mut_loss_ratio * num_mutations_hf2))

  C_partial_missing <- data_check_CM_hf2
  # new data
  if(length(loss_cell_indices) > 0 && length(loss_mut_indices) > 0){
    C_partial_missing[loss_cell_indices, loss_mut_indices] <- "NoCall"
  }

  # 2.2 count cell/mut number
  MutPerCell <- rowSums(C_partial_missing == "Mut")
  CellPerMut <- colSums(C_partial_missing == "Mut")

  cells_candidate <- cells_all_hf2[MutPerCell >= cell_num & cells_all_hf2 %in% data_check_Meta_hf2$Cell]
  muts_candidate <- muts_all_hf2[CellPerMut >= mut_num]

  message(paste0("Iteration: ", iter, " loss_cell_indices: ", length(loss_cell_indices), " loss_mut_ind",
  message(paste0("Iteration: ", iter, " cells_candidate: ", length(cells_candidate), " muts_candidate: "

  data_cell_mut_raw_check <- C_partial_missing[cells_candidate, muts_candidate, drop = TRUE]

  # 2.3 Mut × Time
  cell_time_num <- sapply(col_check_names_hf2, function(i) sum(str_count(cells_candidate, i)))
  total_cells <- sum(cell_time_num)
  cell_time_ratio <- cell_time_num / total_cells + 2 # +2

  # init Mut × Time

```

```

M_partial_missing <- matrix(0, nrow = length(muts_candidate), ncol = length(time_types_hf2),
                             dimnames = list(muts_candidate, time_types_hf2))

# Calculate the number of each mutation at each time
for (t in seq_along(time_types_hf2)) {
  time_label <- time_types_hf2[t]
  # Find out in which cells each mutation occurs
  mut_cell_list <- apply(data_cell_mut_raw_check, 2, function(x) which(x == "Mut"))
  # Count the number of times each mutation occurs at that point in time
  for (mut in muts_candidate) {
    cells_with_mut <- rownames(data_cell_mut_raw_check)[data_cell_mut_raw_check[, mut] == "Mut"]
    selected_days <- data_check_Meta_hf2$Day[data_check_Meta_hf2$Cellbarcode %in% cells_with_mut]
    M_partial_missing[mut, t] <- sum(str_count(selected_days, time_label))
  }
}

# 2.4 normalized Mut × Time
M_partial_missing_norm <- sweep(M_partial_missing, 2, log2(cell_time_ratio[3:8]), "/")
#M_partial_missing_norm <- sweep(M_partial_missing, 2, mean_mut_by_day_hf2$mean_MutNum, "/")

# 2.5 scMutScore
res_index_tmp <- cal_scMutScore(data_cell_mut_raw_check, M_partial_missing_norm, data_check_Meta_hf2)
res_index_tmp$iteration <- paste0("iter", iter)
res_index_tmp <- res_index_tmp[res_index_tmp$Diversity > 0, ]
res_index_tmp$cb <- rownames(res_index_tmp)
res_index_tmp$iteration <- paste0("iter", iter)
rownames(res_index_tmp) <- NULL
data_scMutModel_list_hf2[[iter]] <- res_index_tmp
}

## Iteration: 1 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 1 cells_candidate: 2151 muts_candidate: 146
## [1] 2151 146
## [1] 146 6
## Iteration: 2 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 2 cells_candidate: 2101 muts_candidate: 154
## [1] 2101 154
## [1] 154 6
## Iteration: 3 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 3 cells_candidate: 1983 muts_candidate: 150
## [1] 1983 150
## [1] 150 6
## Iteration: 4 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 4 cells_candidate: 2031 muts_candidate: 149
## [1] 2031 149
## [1] 149 6
## Iteration: 5 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 5 cells_candidate: 2094 muts_candidate: 146
## [1] 2094 146
## [1] 146 6
## Iteration: 6 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 6 cells_candidate: 2128 muts_candidate: 149
## [1] 2128 149
## [1] 149 6

```

```

## Iteration: 7 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 7 cells_candidate: 2065 muts_candidate: 144
## [1] 2065 144
## [1] 144 6
## Iteration: 8 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 8 cells_candidate: 2177 muts_candidate: 150
## [1] 2177 150
## [1] 150 6
## Iteration: 9 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 9 cells_candidate: 2106 muts_candidate: 149
## [1] 2106 149
## [1] 149 6
## Iteration: 10 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 10 cells_candidate: 2159 muts_candidate: 150
## [1] 2159 150
## [1] 150 6
## Iteration: 11 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 11 cells_candidate: 2152 muts_candidate: 146
## [1] 2152 146
## [1] 146 6
## Iteration: 12 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 12 cells_candidate: 2054 muts_candidate: 151
## [1] 2054 151
## [1] 151 6
## Iteration: 13 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 13 cells_candidate: 1929 muts_candidate: 149
## [1] 1929 149
## [1] 149 6
## Iteration: 14 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 14 cells_candidate: 1935 muts_candidate: 152
## [1] 1935 152
## [1] 152 6
## Iteration: 15 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 15 cells_candidate: 2011 muts_candidate: 149
## [1] 2011 149
## [1] 149 6
## Iteration: 16 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 16 cells_candidate: 2057 muts_candidate: 152
## [1] 2057 152
## [1] 152 6
## Iteration: 17 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 17 cells_candidate: 2046 muts_candidate: 145
## [1] 2046 145
## [1] 145 6
## Iteration: 18 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 18 cells_candidate: 2089 muts_candidate: 149
## [1] 2089 149
## [1] 149 6
## Iteration: 19 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 19 cells_candidate: 1793 muts_candidate: 151
## [1] 1793 151
## [1] 151 6
## Iteration: 20 loss_cell_indices: 1314 loss_mut_indices: 78

```

```

## Iteration: 20 cells_candidate: 1968 muts_candidate: 152
## [1] 1968 152
## [1] 152 6
## Iteration: 21 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 21 cells_candidate: 2010 muts_candidate: 152
## [1] 2010 152
## [1] 152 6
## Iteration: 22 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 22 cells_candidate: 2056 muts_candidate: 142
## [1] 2056 142
## [1] 142 6
## Iteration: 23 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 23 cells_candidate: 2016 muts_candidate: 144
## [1] 2016 144
## [1] 144 6
## Iteration: 24 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 24 cells_candidate: 2042 muts_candidate: 145
## [1] 2042 145
## [1] 145 6
## Iteration: 25 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 25 cells_candidate: 1911 muts_candidate: 147
## [1] 1911 147
## [1] 147 6
## Iteration: 26 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 26 cells_candidate: 1915 muts_candidate: 148
## [1] 1915 148
## [1] 148 6
## Iteration: 27 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 27 cells_candidate: 1864 muts_candidate: 149
## [1] 1864 149
## [1] 149 6
## Iteration: 28 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 28 cells_candidate: 1857 muts_candidate: 151
## [1] 1857 151
## [1] 151 6
## Iteration: 29 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 29 cells_candidate: 2103 muts_candidate: 147
## [1] 2103 147
## [1] 147 6
## Iteration: 30 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 30 cells_candidate: 1939 muts_candidate: 147
## [1] 1939 147
## [1] 147 6
## Iteration: 31 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 31 cells_candidate: 2046 muts_candidate: 146
## [1] 2046 146
## [1] 146 6
## Iteration: 32 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 32 cells_candidate: 2093 muts_candidate: 152
## [1] 2093 152
## [1] 152 6
## Iteration: 33 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 33 cells_candidate: 2128 muts_candidate: 149

```



```

## [1] 2128 149
## [1] 149 6
## Iteration: 34 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 34 cells_candidate: 1999 muts_candidate: 151
## [1] 1999 151
## [1] 151 6
## Iteration: 35 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 35 cells_candidate: 2050 muts_candidate: 150
## [1] 2050 150
## [1] 150 6
## Iteration: 36 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 36 cells_candidate: 1918 muts_candidate: 150
## [1] 1918 150
## [1] 150 6
## Iteration: 37 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 37 cells_candidate: 1805 muts_candidate: 148
## [1] 1805 148
## [1] 148 6
## Iteration: 38 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 38 cells_candidate: 1935 muts_candidate: 147
## [1] 1935 147
## [1] 147 6
## Iteration: 39 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 39 cells_candidate: 2071 muts_candidate: 150
## [1] 2071 150
## [1] 150 6
## Iteration: 40 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 40 cells_candidate: 1997 muts_candidate: 150
## [1] 1997 150
## [1] 150 6
## Iteration: 41 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 41 cells_candidate: 2010 muts_candidate: 150
## [1] 2010 150
## [1] 150 6
## Iteration: 42 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 42 cells_candidate: 1912 muts_candidate: 147
## [1] 1912 147
## [1] 147 6
## Iteration: 43 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 43 cells_candidate: 1944 muts_candidate: 150
## [1] 1944 150
## [1] 150 6
## Iteration: 44 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 44 cells_candidate: 1868 muts_candidate: 147
## [1] 1868 147
## [1] 147 6
## Iteration: 45 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 45 cells_candidate: 1882 muts_candidate: 150
## [1] 1882 150
## [1] 150 6
## Iteration: 46 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 46 cells_candidate: 1873 muts_candidate: 153
## [1] 1873 153

```

```

## [1] 153 6
## Iteration: 47 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 47 cells_candidate: 1991 muts_candidate: 147
## [1] 1991 147
## [1] 147 6
## Iteration: 48 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 48 cells_candidate: 1942 muts_candidate: 148
## [1] 1942 148
## [1] 148 6
## Iteration: 49 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 49 cells_candidate: 1956 muts_candidate: 150
## [1] 1956 150
## [1] 150 6
## Iteration: 50 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 50 cells_candidate: 1927 muts_candidate: 150
## [1] 1927 150
## [1] 150 6
## Iteration: 51 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 51 cells_candidate: 2022 muts_candidate: 150
## [1] 2022 150
## [1] 150 6
## Iteration: 52 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 52 cells_candidate: 1897 muts_candidate: 151
## [1] 1897 151
## [1] 151 6
## Iteration: 53 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 53 cells_candidate: 2059 muts_candidate: 152
## [1] 2059 152
## [1] 152 6
## Iteration: 54 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 54 cells_candidate: 2072 muts_candidate: 148
## [1] 2072 148
## [1] 148 6
## Iteration: 55 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 55 cells_candidate: 1972 muts_candidate: 148
## [1] 1972 148
## [1] 148 6
## Iteration: 56 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 56 cells_candidate: 2046 muts_candidate: 150
## [1] 2046 150
## [1] 150 6
## Iteration: 57 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 57 cells_candidate: 1978 muts_candidate: 148
## [1] 1978 148
## [1] 148 6
## Iteration: 58 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 58 cells_candidate: 2125 muts_candidate: 148
## [1] 2125 148
## [1] 148 6
## Iteration: 59 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 59 cells_candidate: 2089 muts_candidate: 148
## [1] 2089 148
## [1] 148 6

```

```

## Iteration: 60 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 60 cells_candidate: 1921 muts_candidate: 148
## [1] 1921 148
## [1] 148 6
## Iteration: 61 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 61 cells_candidate: 1896 muts_candidate: 152
## [1] 1896 152
## [1] 152 6
## Iteration: 62 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 62 cells_candidate: 2142 muts_candidate: 152
## [1] 2142 152
## [1] 152 6
## Iteration: 63 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 63 cells_candidate: 2131 muts_candidate: 151
## [1] 2131 151
## [1] 151 6
## Iteration: 64 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 64 cells_candidate: 2096 muts_candidate: 151
## [1] 2096 151
## [1] 151 6
## Iteration: 65 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 65 cells_candidate: 2053 muts_candidate: 153
## [1] 2053 153
## [1] 153 6
## Iteration: 66 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 66 cells_candidate: 2015 muts_candidate: 147
## [1] 2015 147
## [1] 147 6
## Iteration: 67 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 67 cells_candidate: 2298 muts_candidate: 152
## [1] 2298 152
## [1] 152 6
## Iteration: 68 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 68 cells_candidate: 2203 muts_candidate: 148
## [1] 2203 148
## [1] 148 6
## Iteration: 69 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 69 cells_candidate: 2082 muts_candidate: 149
## [1] 2082 149
## [1] 149 6
## Iteration: 70 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 70 cells_candidate: 1846 muts_candidate: 152
## [1] 1846 152
## [1] 152 6
## Iteration: 71 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 71 cells_candidate: 1910 muts_candidate: 147
## [1] 1910 147
## [1] 147 6
## Iteration: 72 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 72 cells_candidate: 1994 muts_candidate: 150
## [1] 1994 150
## [1] 150 6
## Iteration: 73 loss_cell_indices: 1314 loss_mut_indices: 78

```

```

## Iteration: 73 cells_candidate: 2079 muts_candidate: 148
## [1] 2079 148
## [1] 148 6
## Iteration: 74 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 74 cells_candidate: 1868 muts_candidate: 147
## [1] 1868 147
## [1] 147 6
## Iteration: 75 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 75 cells_candidate: 2036 muts_candidate: 149
## [1] 2036 149
## [1] 149 6
## Iteration: 76 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 76 cells_candidate: 2133 muts_candidate: 148
## [1] 2133 148
## [1] 148 6
## Iteration: 77 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 77 cells_candidate: 1856 muts_candidate: 150
## [1] 1856 150
## [1] 150 6
## Iteration: 78 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 78 cells_candidate: 2114 muts_candidate: 146
## [1] 2114 146
## [1] 146 6
## Iteration: 79 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 79 cells_candidate: 1968 muts_candidate: 149
## [1] 1968 149
## [1] 149 6
## Iteration: 80 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 80 cells_candidate: 2072 muts_candidate: 152
## [1] 2072 152
## [1] 152 6
## Iteration: 81 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 81 cells_candidate: 1992 muts_candidate: 150
## [1] 1992 150
## [1] 150 6
## Iteration: 82 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 82 cells_candidate: 2213 muts_candidate: 150
## [1] 2213 150
## [1] 150 6
## Iteration: 83 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 83 cells_candidate: 1965 muts_candidate: 148
## [1] 1965 148
## [1] 148 6
## Iteration: 84 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 84 cells_candidate: 2111 muts_candidate: 149
## [1] 2111 149
## [1] 149 6
## Iteration: 85 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 85 cells_candidate: 1991 muts_candidate: 153
## [1] 1991 153
## [1] 153 6
## Iteration: 86 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 86 cells_candidate: 1979 muts_candidate: 151

```

```

## [1] 1979 151
## [1] 151 6
## Iteration: 87 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 87 cells_candidate: 2053 muts_candidate: 151
## [1] 2053 151
## [1] 151 6
## Iteration: 88 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 88 cells_candidate: 2038 muts_candidate: 149
## [1] 2038 149
## [1] 149 6
## Iteration: 89 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 89 cells_candidate: 1989 muts_candidate: 151
## [1] 1989 151
## [1] 151 6
## Iteration: 90 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 90 cells_candidate: 1858 muts_candidate: 147
## [1] 1858 147
## [1] 147 6
## Iteration: 91 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 91 cells_candidate: 1837 muts_candidate: 150
## [1] 1837 150
## [1] 150 6
## Iteration: 92 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 92 cells_candidate: 2042 muts_candidate: 148
## [1] 2042 148
## [1] 148 6
## Iteration: 93 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 93 cells_candidate: 1955 muts_candidate: 150
## [1] 1955 150
## [1] 150 6
## Iteration: 94 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 94 cells_candidate: 1955 muts_candidate: 149
## [1] 1955 149
## [1] 149 6
## Iteration: 95 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 95 cells_candidate: 2104 muts_candidate: 146
## [1] 2104 146
## [1] 146 6
## Iteration: 96 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 96 cells_candidate: 2178 muts_candidate: 150
## [1] 2178 150
## [1] 150 6
## Iteration: 97 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 97 cells_candidate: 1999 muts_candidate: 146
## [1] 1999 146
## [1] 146 6
## Iteration: 98 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 98 cells_candidate: 2126 muts_candidate: 150
## [1] 2126 150
## [1] 150 6
## Iteration: 99 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 99 cells_candidate: 1997 muts_candidate: 150
## [1] 1997 150

```

```

## [1] 150    6
## Iteration: 100 loss_cell_indices: 1314 loss_mut_indices: 78
## Iteration: 100 cells_candidate: 1954 muts_candidate: 145
## [1] 1954   145
## [1] 145    6

# 3. merge res
data_scMutModel_res_hf1 <- do.call(rbind, data_scMutModel_list_hf1)
data_scMutModel_res_hf2 <- do.call(rbind, data_scMutModel_list_hf2)
data_scMutModel_res <- rbind(data_scMutModel_res_hf1, data_scMutModel_res_hf2)

# 4. column
data_scMutModel_res <- separate(data_scMutModel_res, col = "cb", into = c("celltype", "cell"), sep = "_")
data_scMutModel_res$Tag <- data_scMutModel_res$celltype
data_scMutModel_res <- separate(data_scMutModel_res, col = "celltype", into = c("rep", "Day"), sep = "\\.")

data_scMutModel_res <- data_scMutModel_res[data_scMutModel_res$Day %in% c("d6", "d9", "d12", "d15", "d21"), ]

data_scMutModel_res$Tag <- factor(data_scMutModel_res$Tag, levels = col_check_names)
data_scMutModel_res$Day <- factor(data_scMutModel_res$Day, levels = c("d6", "d9", "d12", "d15", "d21", "d28"))

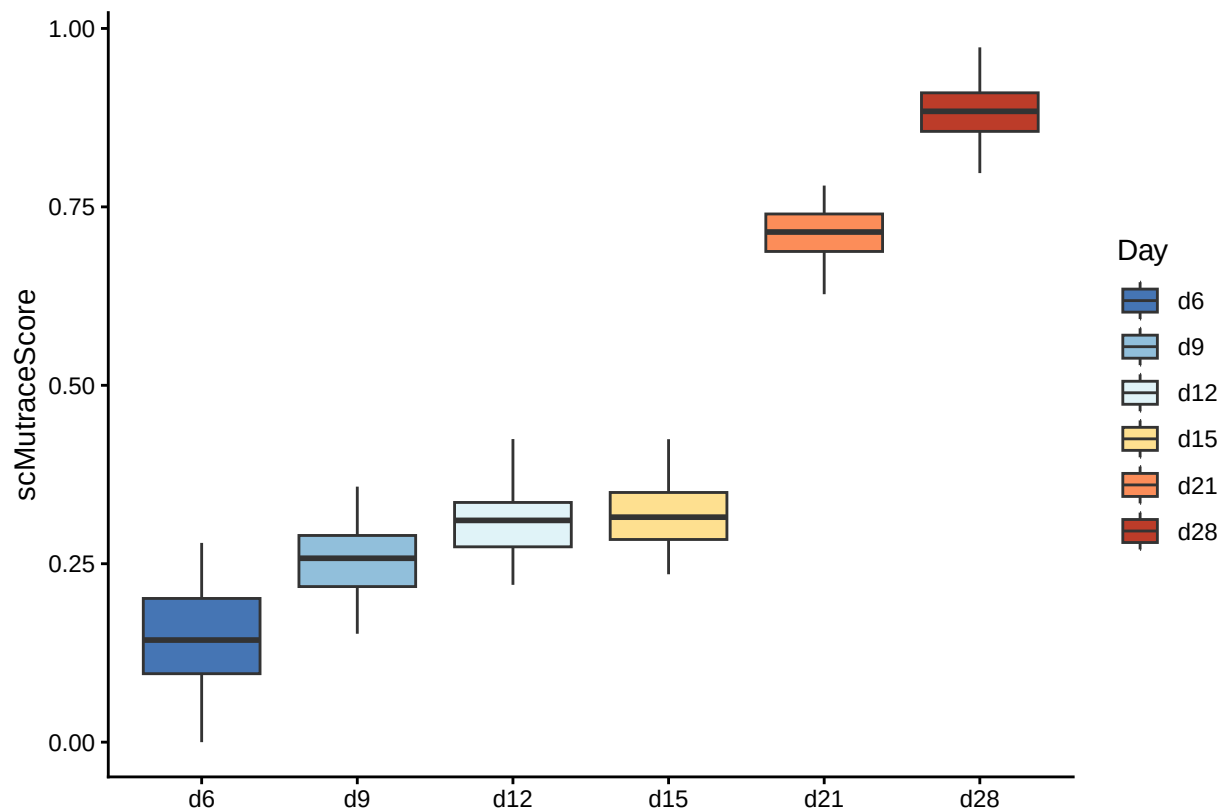
# 5. plot
data_scMutModel_res_summary <- data_scMutModel_res %>%
  dplyr::group_by(iteration, Day) %>%
  dplyr::summarise(mean_scMutScore = mean(scMutScore, na.rm = TRUE)) %>%
  dplyr::ungroup()
## `summarise()` has grouped output by 'iteration'. You can override using the
## `.groups` argument.

data_scMutModel_res_summary <- data_scMutModel_res_summary %>%
  dplyr::mutate(
    mean_scMutScore_scaled = scales::rescale(mean_scMutScore, to = c(0, 1))
  )

p_iter_scMutraceScore <- ggplot(data = data_scMutModel_res_summary, mapping = aes(x = Day, y = mean_scMutScore_scaled)) +
  #geom_violin(trim = TRUE, alpha = 0.6) +
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "scMutraceScore") +
  scale_fill_manual(values = color_settings) +
  theme_classic()

p_iter_scMutraceScore

```



12 compareClonesScore

```
# generate data for comparision
MEF_meta_hf1_hf2 <- rbind(MEF_Meta_hf1, MEF_Meta_hf2)
res_index_scMut <- data.frame(Cellbarcode = rownames(res_index), scMutScore = res_index$scMutScore, Tag

res_index_scMut_meta <- merge(res_index_scMut, MEF_meta_hf1_hf2, by = "Cellbarcode")
res_index_scMut_meta$Day <- factor(res_index_scMut_meta$Day, levels = c("Day_0", "Day_3", "Day_6", "Day_9", "Day_12", "Day_15", "Day_21", "Day_28"))
res_index_scMut_meta <- res_index_scMut_meta[res_index_scMut_meta$Day %in% c("Day_6", "Day_9", "Day_12", "Day_15", "Day_21", "Day_28"),]

colNames_data_clone_score <- c("D0_tag", "D3_tag", "D13_tag", "Day", "Mean_Day_score", "Mean_Day_pseudo")
data_clone_score <- as.data.frame(matrix(ncol = length(colNames_data_clone_score)))
colnames(data_clone_score) <- colNames_data_clone_score
d0_tags <- unique(MEF_meta_hf1_hf2$CellTagD0_48k)

count_line = 0

for (i in 1:length(d0_tags)){
  d0_tag_tmp <- d0_tags[i]
  d3_tags <- unique(MEF_meta_hf1_hf2[MEF_meta_hf1_hf2$CellTagD0_48k == d0_tag_tmp,]$CellTagD3_48k)
  days_reprogramming <- unique(MEF_meta_hf1_hf2[MEF_meta_hf1_hf2$CellTagD0_48k == d0_tag_tmp,]$Day)
  for (day in 1:length(days_reprogramming)){
    if (length(d3_tags) >= 1){
```

```

for (j in 1:length(d3_tags)){
  d3_tag_tmp <- d3_tags[j]
  d13_tags <- unique(MEF_meta_hf1_hf2[MEF_meta_hf1_hf2$CellTagD0_48k == d0_tag_tmp & MEF_meta_hf1_hf2$CellTagD3_48k == d3_tag_tmp])
  if(length(d13_tags) >= 1){
    for (k in 1:length(d13_tags)){
      d13_tag_tmp <- d13_tags[k]
      count_line = count_line + 1
      data_clone_score[count_line,]$Day = days_reprogramming[day]
      data_clone_score[count_line,]$D0_tag = d0_tag_tmp
      data_clone_score[count_line,]$D3_tag = d3_tag_tmp
      data_clone_score[count_line,]$D13_tag = d13_tag_tmp
    }
  }else{
    count_line = count_line + 1
    data_clone_score[count_line,]$Day = days_reprogramming[day]
    data_clone_score[count_line,]$D0_tag = d0_tag_tmp
    data_clone_score[count_line,]$D3_tag = d3_tag_tmp
    data_clone_score[count_line,]$D13_tag = "NotAva"
  }
}
}
}

data_clone_score$ID <- paste(data_clone_score$D0_tag, data_clone_score$D3_tag, data_clone_score$D13_tag, sep = "_")
data_clone_score <- data_clone_score[data_clone_score$ID != "NotAva_NotAva_NotAva",]

for (i in 1:nrow(data_clone_score)){

  d0_tag_tmp <- data_clone_score[i,]$D0_tag
  d3_tag_tmp <- data_clone_score[i,]$D3_tag
  d13_tag_tmp <- data_clone_score[i,]$D13_tag
  day_tmp <- data_clone_score[i,]$Day

  data_clone_score[i,]$Mean_Day_score <- "NotAva"
  data_clone_score[i,]$Mean_Day_pseudotime <- "NotAva"

  id_check <- paste(d0_tag_tmp, d3_tag_tmp, d13_tag_tmp, sep = "_")
  data_clone_score[i,]$ID <- id_check

  scMutscore_tmp <- res_index_scMut_meta[res_index_scMut_meta$cloneTimeTag == id_check & res_index_scMut_meta$cloneTimeTag != "NotAva"]
  pseudotime_tmp <- res_index_scMut_meta[res_index_scMut_meta$cloneTimeTag == id_check & res_index_scMut_meta$cloneTimeTag != "NotAva"]

  scMutscore_tmp <- as.numeric(scMutscore_tmp[scMutscore_tmp != "NotAva"])
  pseudotime_tmp <- as.numeric(pseudotime_tmp[pseudotime_tmp != "NotAva"])
}

```



```

data_clone_score[i,]$Mean_Day_score <- mean(scMutscore_tmp)
data_clone_score[i,]$Mean_Day_pseudotime <- mean(pseudotime_tmp)
}

# filter
# clonetagDay
data_clone_score_clonetagDay <- data_clone_score[data_clone_score$Mean_Day_score != "NaN" & data_clone_score$Mean_Day_pseudotime != "NaN"]
ID_counts <- table(data_clone_score_clonetagDay$ID)
valid_IDs <- names(ID_counts[ID_counts >= 2])
data_clone_score_clonetagDay <- subset(data_clone_score_clonetagDay, ID %in% valid_IDs)

data_clone_score_clonetagDay$Mean_Day_score <- as.numeric(data_clone_score_clonetagDay$Mean_Day_score)
data_clone_score_clonetagDay$Mean_Day_pseudotime <- as.numeric(data_clone_score_clonetagDay$Mean_Day_pseudotime)

data_clone_score_clonetagDay$Day <- factor(data_clone_score_clonetagDay$Day, levels = c("Day_0", "Day_3"))

# reprogramming Day
p_clone_scMutraceScore <- ggplot(data_clone_score_clonetagDay, mapping = aes(x = as.factor(Day), y = Mean_Day_score)) +
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = ID), color = "gray", alpha = 0.1) +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = "scMutraceScore") +
  theme_classic()

p_clone_pseudotime <- ggplot(data_clone_score_clonetagDay, aes(x = as.factor(Day), y = Mean_Day_pseudotime)) +
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = ID), color = "gray", alpha = 0.1) +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = "Pseudotime") +
  theme_classic()

p_cell_scMutraceScore <- ggplot(data = res_index_scMut_meta, mapping = aes(x = Day, y = scMutScore, fill = Outcome)) +
  geom_boxplot(outliers = FALSE) +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = "scMutraceScore") +
  theme_classic()

p_cell_pseudotime <- ggplot(data = res_index_scMut_meta[res_index_scMut_meta$pseudotime != "NotAva"], mapping = aes(x = Day, y = pseudotime, fill = Outcome)) +
  geom_boxplot(outliers = FALSE) +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = "pseudotime") +
  theme_classic()

res_index_scMut_meta_success <- res_index_scMut_meta[res_index_scMut_meta$Outcome == "Successful", , c("Day", "scMutScore", "pseudotime")]
res_index_scMut_meta_deadend <- res_index_scMut_meta[res_index_scMut_meta$Outcome == "DeadEnd", , c("Day", "scMutScore", "pseudotime")]
res_index_scMut_meta_success_deadend <- rbind(res_index_scMut_meta_success, res_index_scMut_meta_deadend)

# Outcome
p_outcome_scMutraceScore <- ggplot(res_index_scMut_meta[res_index_scMut_meta$Outcome %in% c("DeadEnd", "Successful")], mapping = aes(x = Day, y = scMutScore, fill = Outcome)) +
  labs(x = "", y = "scMutScore") +
  theme_classic()

```

```

geom_boxplot(outliers = FALSE) +
stat_compare_means(method = "t.test", method.args = list(alternative = "greater")) +
scale_fill_manual(values = c("#7876B1FF", "#FFDC91FF")) +
theme_classic()

p_outcome_pseudotime <- ggplot(res_index_scMut_meta[res_index_scMut_meta$Outcome %in% c("DeadEnd", "Successful")],
labs(x = "", y = "pseudotime") +
geom_boxplot(outliers = FALSE) +
stat_compare_means(method = "t.test", method.args = list(alternative = "greater")) +
scale_fill_manual(values = c("#7876B1FF", "#FFDC91FF")) +
theme_classic()

p_reprogramming_scMutraceScore <- ggplot(res_index_scMut_meta[res_index_scMut_meta$reprogramming %in% c("F_reprogramming", "T_reprogramming")],
labs(x = "", y = "scMutScore") +
geom_boxplot(outliers = FALSE) +
stat_compare_means(method = "t.test", method.args = list(alternative = "greater")) +
scale_fill_manual(values = c("#7876B1FF", "#FFDC91FF")) +
theme_classic()

p_reprogramming_pseudotime <- ggplot(res_index_scMut_meta[res_index_scMut_meta$reprogramming %in% c("F_reprogramming", "T_reprogramming")],
labs(x = "", y = "pseudotime") +
geom_boxplot(outliers = FALSE) +
stat_compare_means(method = "t.test", method.args = list(alternative = "greater")) +
scale_fill_manual(values = c("#7876B1FF", "#FFDC91FF")) +
theme_classic()

timepoint_color <- c(Day_6 = '#7D5188', Day_9 = '#7D71A5', Day_12 = '#738DAF', Day_15 = '#6EA5B0', Day_18 = '#6A8C9E')
cluster_color <- c(c0 = '#D5393E', c1 = '#5A6B93', c2 = '#519376', c3 = '#6E8371', c4 = '#A55F7F', c5 = '#A55F7F', c6 = '#A55F7F')
outcom_color <- c(BAB = "lightgray", DeadEnd = "#7876B1FF", Successful = "#FFDC91FF")
reprogramming_colors <- c(F_reprogramming = '#7876B1FF', T_reprogramming = '#FFDC91FF', NotAva = "#d9d9d9")

p_umap_reprogramming <- ggplot(data = res_index_scMut_meta[res_index_scMut_meta$reprogramming %in% c("F_reprogramming", "T_reprogramming")],
scale_color_manual(values = reprogramming_colors) +
xlab("") + ylab("") +
theme_classic()

p_umap_outcome <- ggplot(data = res_index_scMut_meta[res_index_scMut_meta$Outcome %in% c("DeadEnd", "Successful")],
scale_color_manual(values = outcom_color) +
xlab("") + ylab("") +
theme_classic()

p_umap_cell_day <- ggplot(data = res_index_scMut_meta) + geom_point(mapping = aes(x = TSNE_1, y = TSNE_2)) +
scale_color_manual(values = color_settings_2) +
xlab("") + ylab("") +
theme_classic()

res_index_scMut_meta$cluster <- paste0("c", res_index_scMut_meta$cluster)
p_umap_cell_cluster <- ggplot(data = res_index_scMut_meta) + geom_point(mapping = aes(x = TSNE_1, y = TSNE_2)) +
scale_color_manual(values = cluster_color) +
xlab("") + ylab("") +
theme_classic()

p_umap_cell_Pseudotime <- ggplot(res_index_scMut_meta[res_index_scMut_meta$pseudotime != "NotAva",], aes(x = TSNE_1, y = TSNE_2)) +

```

```

geom_point(shape = 21, size = 2, alpha = 0.3) +
scale_fill_gradientn(colours = c("#313695", "#4575b4", "#74add1", "#abd9e9", "#e0f3f8", "#ffffbf", "#f1b60d", "#d73027", "#a65628", "#81c784", "#4daf4a", "#1b9e77", "#4d9cfc", "#f781bf", "#999999")) +
theme_classic() +
labs(fill = "Pseudotime")

p_umap_cell_scMutScore <- ggplot(res_index_scMut_meta[res_index_scMut_meta$scMutScore != "NotAva",], aes(x = pseudotime, y = scMutScore)) +
geom_point(shape = 21, size = 2, alpha = 0.3) +
scale_fill_gradientn(colours = c("#313695", "#4575b4", "#74add1", "#abd9e9", "#e0f3f8", "#ffffbf", "#f1b60d", "#d73027", "#a65628", "#81c784", "#4daf4a", "#1b9e77", "#4d9cfc", "#f781bf", "#999999")) +
theme_classic() +
labs(fill = "scMutScore")

data_scMutScore_pseudotime <- res_index_scMut_meta[,c("scMutScore", "pseudotime")]
data_scMutScore_pseudotime <- data_scMutScore_pseudotime[data_scMutScore_pseudotime$scMutScore != "NotAva",]
data_scMutScore_pseudotime$scMutScore <- as.numeric(data_scMutScore_pseudotime$scMutScore)
data_scMutScore_pseudotime$pseudotime <- as.numeric(data_scMutScore_pseudotime$pseudotime)
data_scMutScore_pseudotime$pseudotime_scale <- (data_scMutScore_pseudotime$pseudotime - min(data_scMutScore_pseudotime$pseudotime)) / (max(data_scMutScore_pseudotime$pseudotime) - min(data_scMutScore_pseudotime$pseudotime))

p_cor_scMutScore_pseudotime <- ggplot(data_scMutScore_pseudotime, aes(x = scMutScore, y = pseudotime_scale)) +
geom_point() +
geom_smooth(method = "lm", se = TRUE, color = "blue") +
stat_cor(method = "pearson", label.x = 0.5, label.y = 1) +
labs(x = "scMutScore", y = "Pseudotime") +
theme_classic()

p_cor_scMutScore_sumMut <- ggplot(res_index, aes(x = scMutScore, y = log2(sumMut))) +
geom_point() +
geom_smooth(method = "lm", se = TRUE, color = "blue") +
stat_cor(method = "pearson", label.x = 0.5, label.y = 1) +
labs(x = "scMutScore", y = "sum of matrix") +
theme_classic()

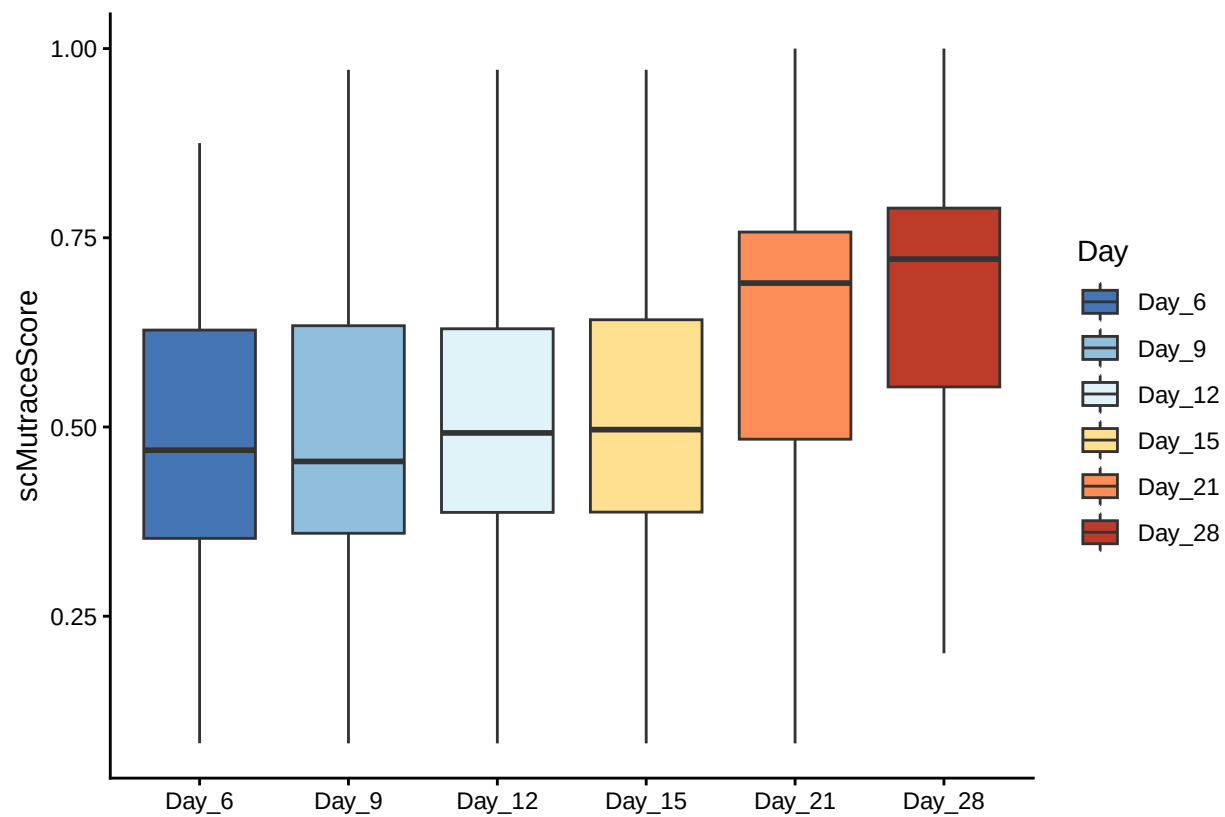
cor.test(data_scMutScore_pseudotime$scMutScore, data_scMutScore_pseudotime$pseudotime_scale)
##
## Pearson's product-moment correlation
##
## data: data_scMutScore_pseudotime$scMutScore and data_scMutScore_pseudotime$pseudotime_scale
## t = 24.639, df = 5007, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.3039094 0.3533129
## sample estimates:
## cor
## 0.3288361
cor.test(res_index$scMutScore, res_index$sumMut)
##
## Pearson's product-moment correlation
##
## data: res_index$scMutScore and res_index$sumMut
## t = 24.382, df = 6916, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.2594998 0.3029015
## sample estimates:

```

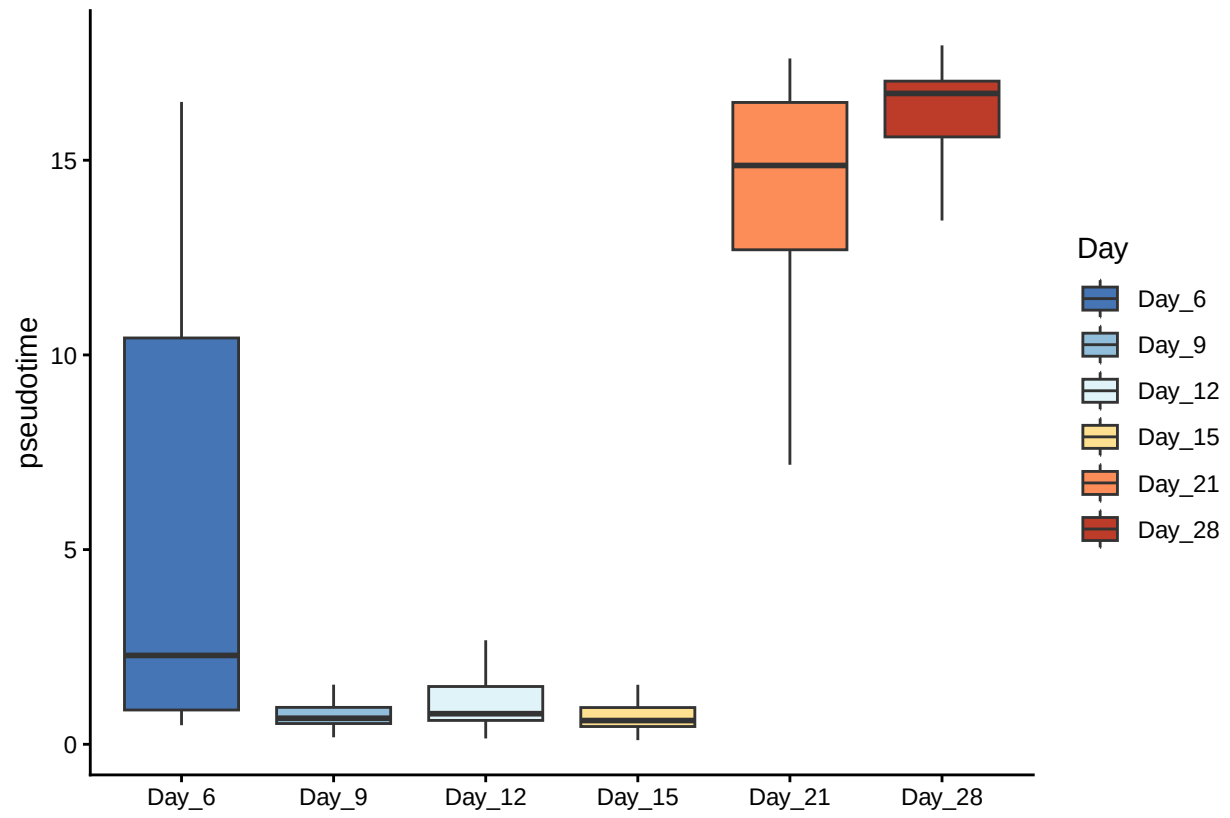
```
## cor
## 0.2813445
```

13 compareClonesScore figures

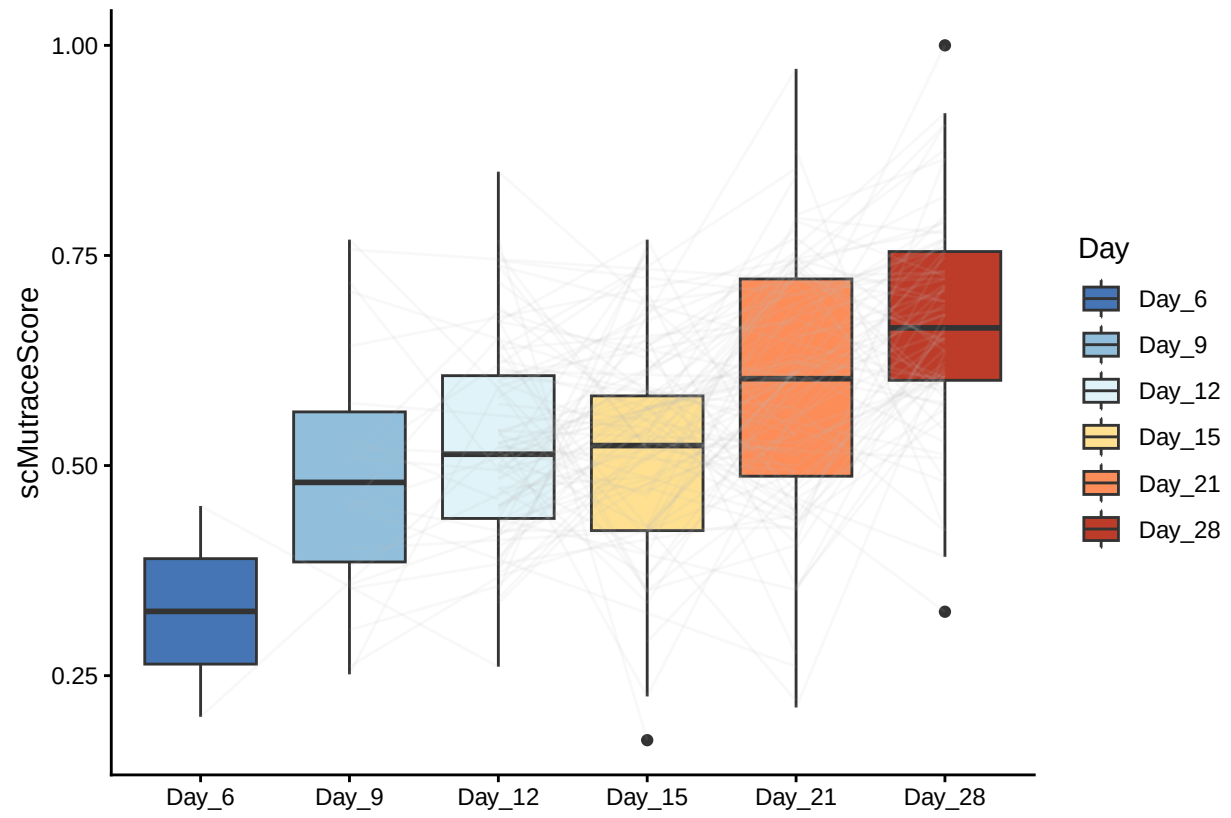
```
p_cell_scMutraceScore
```



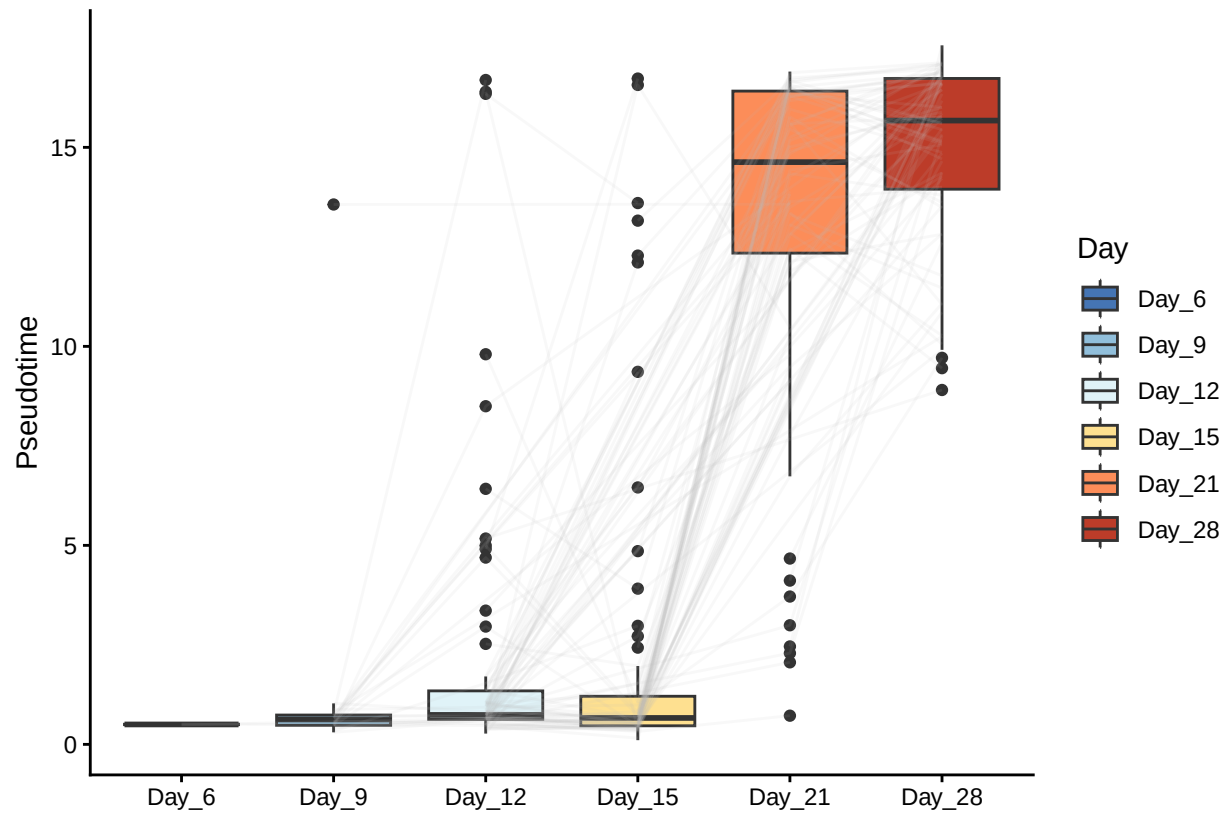
```
p_cell_pseudotime
```



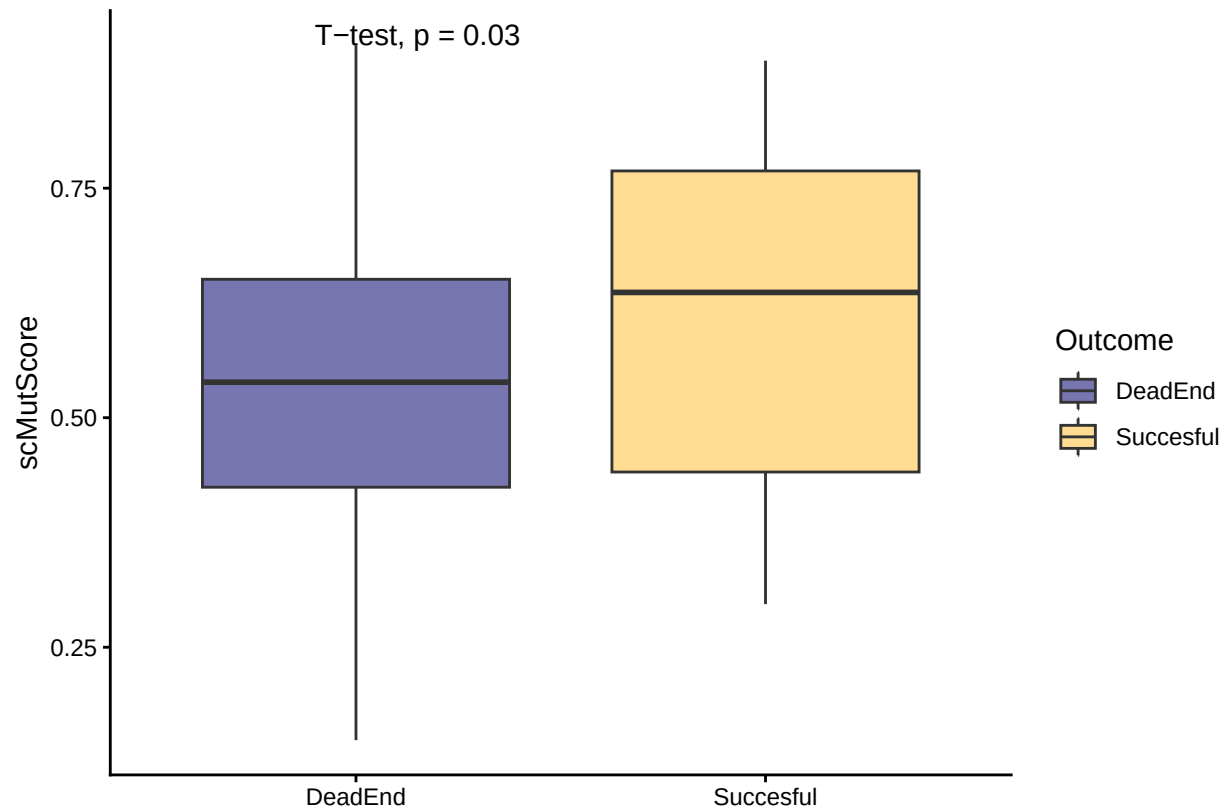
p_clone_scMutraceScore



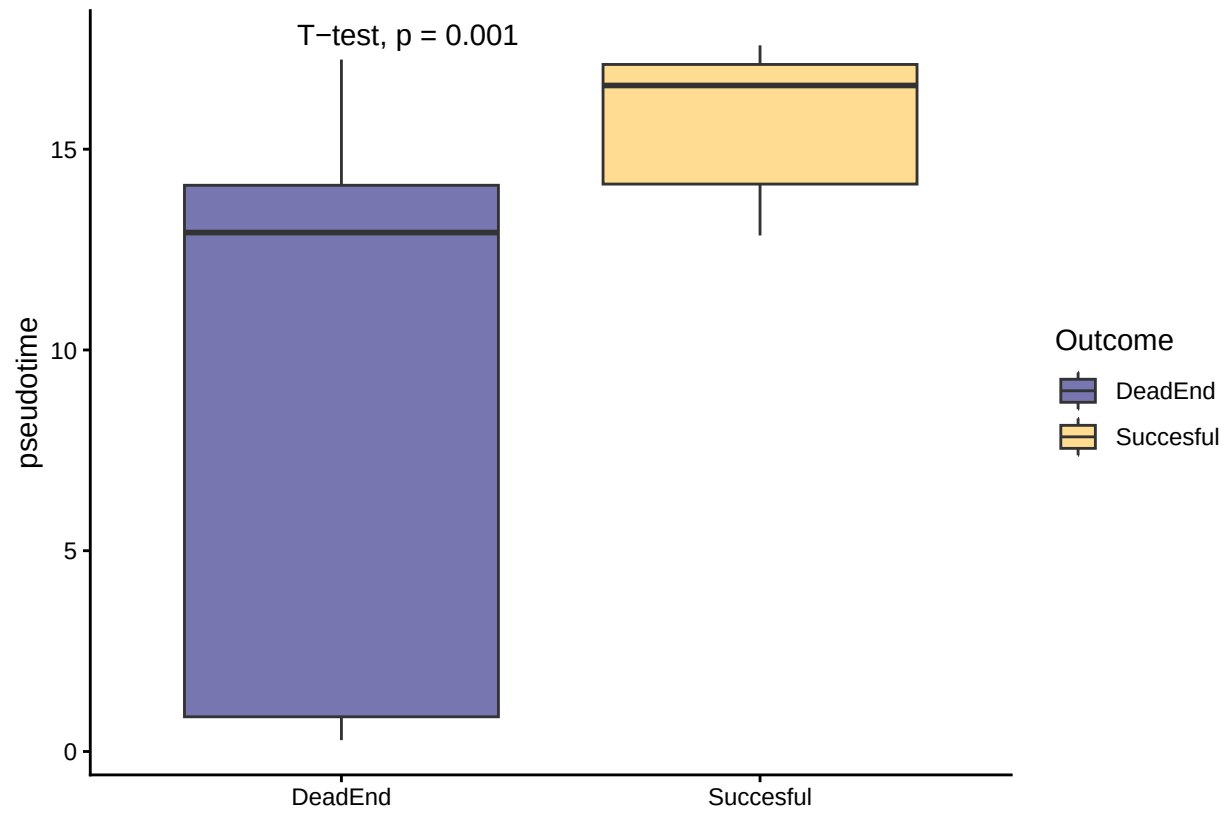
p_clone_pseudotime



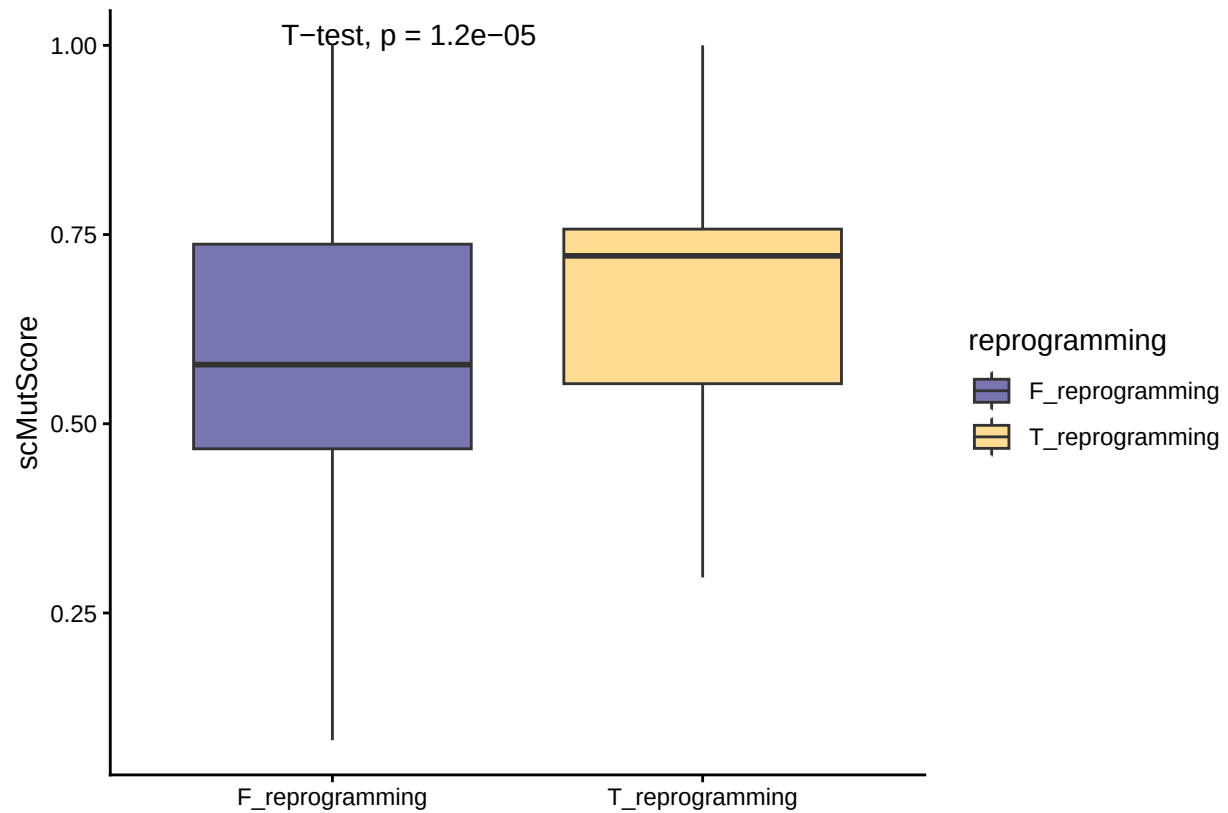
p_outcome_scMutraceScore



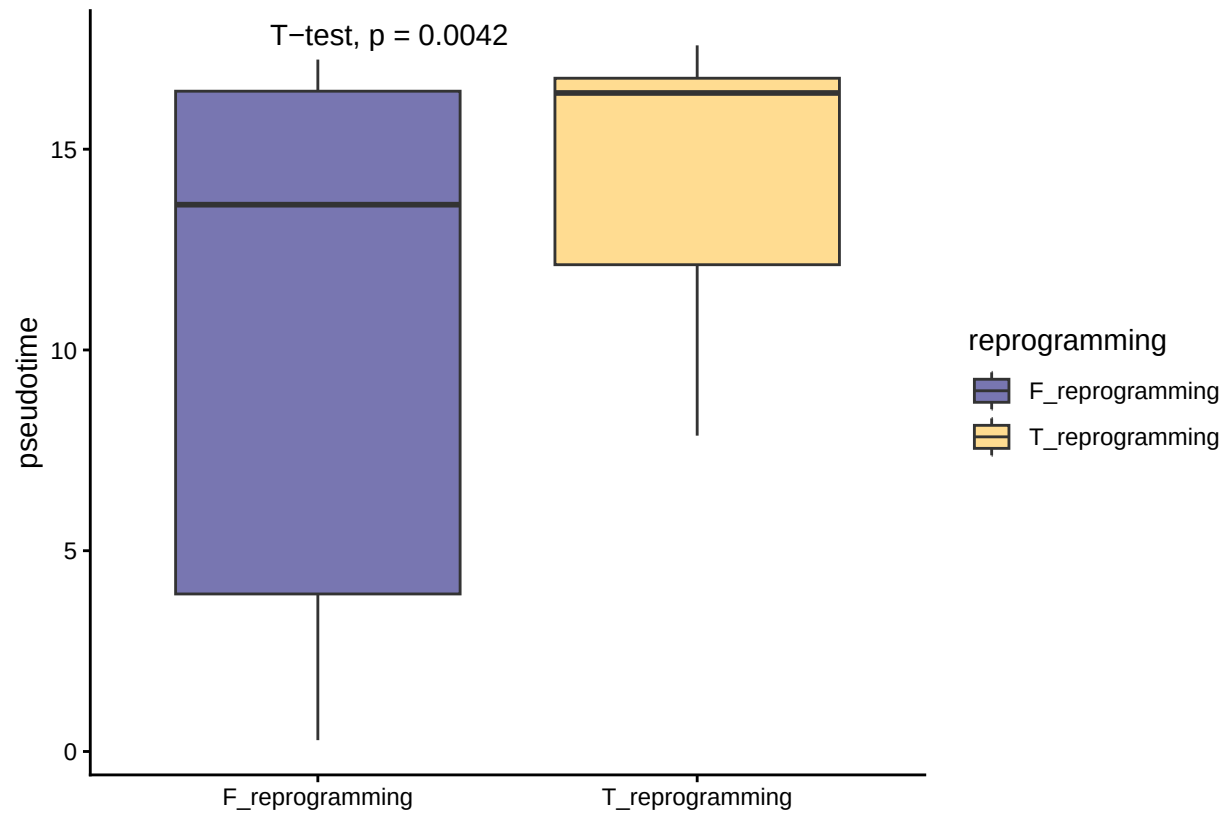
```
p_outcome_pseudotime
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning: Removed 102 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
## Warning: Removed 102 rows containing non-finite outside the scale range
## (`stat_compare_means()`).
```

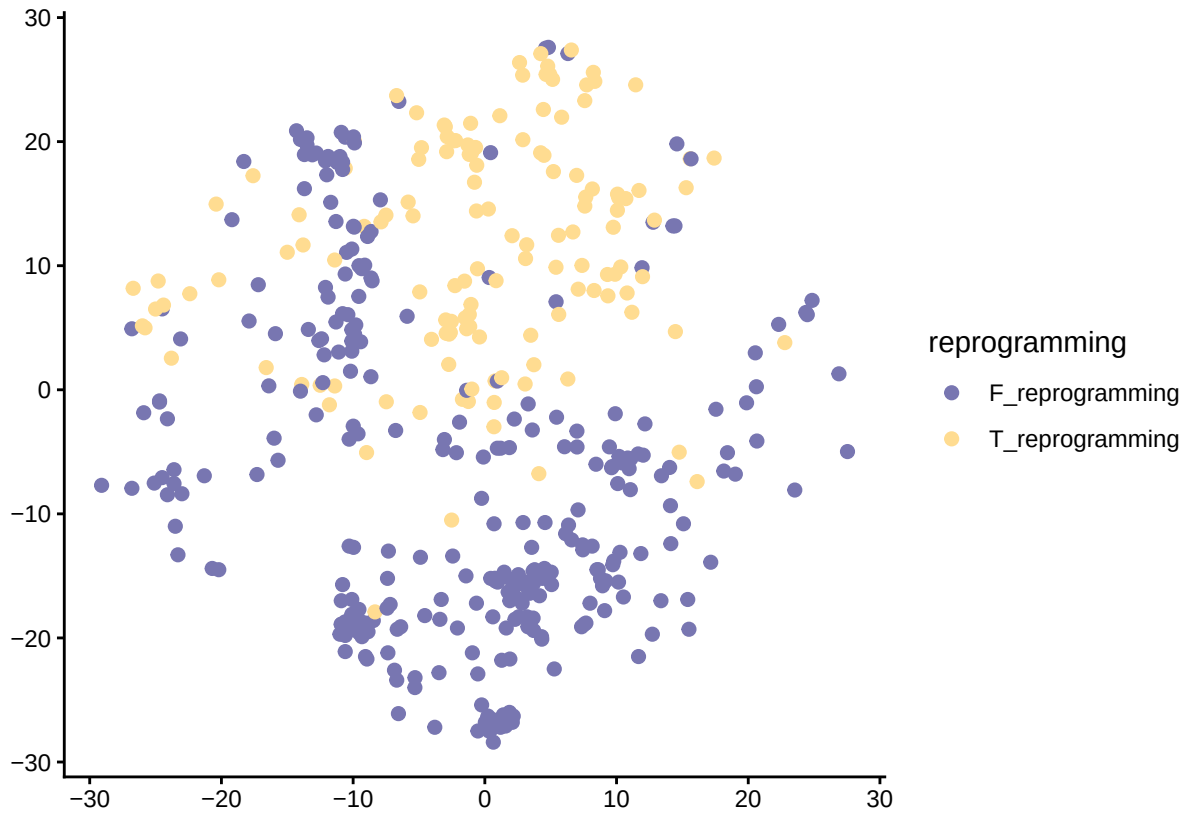
p_reprogramming_scMutraceScore



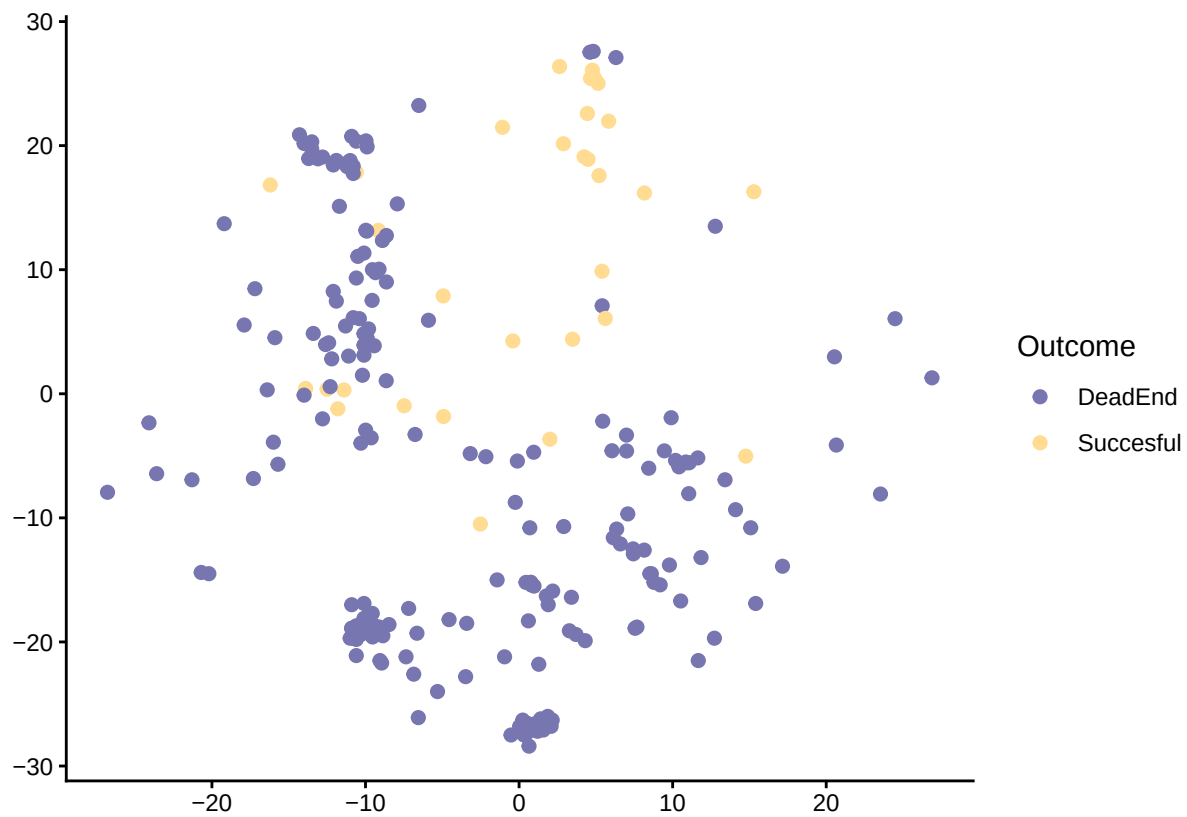
```
p_reprogramming_pseudotime
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning: Removed 135 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
## Warning: Removed 135 rows containing non-finite outside the scale range
## (`stat_compare_means()`).
```



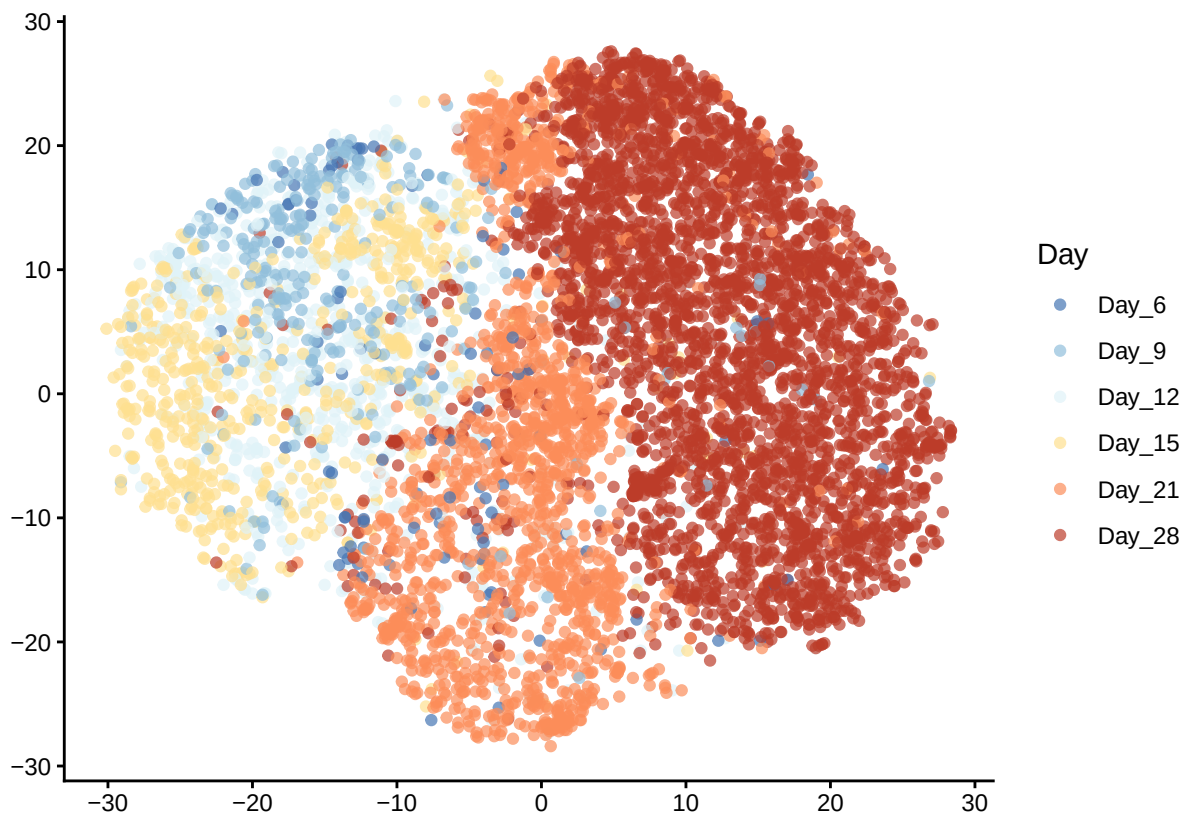
p_umap_reprogramming



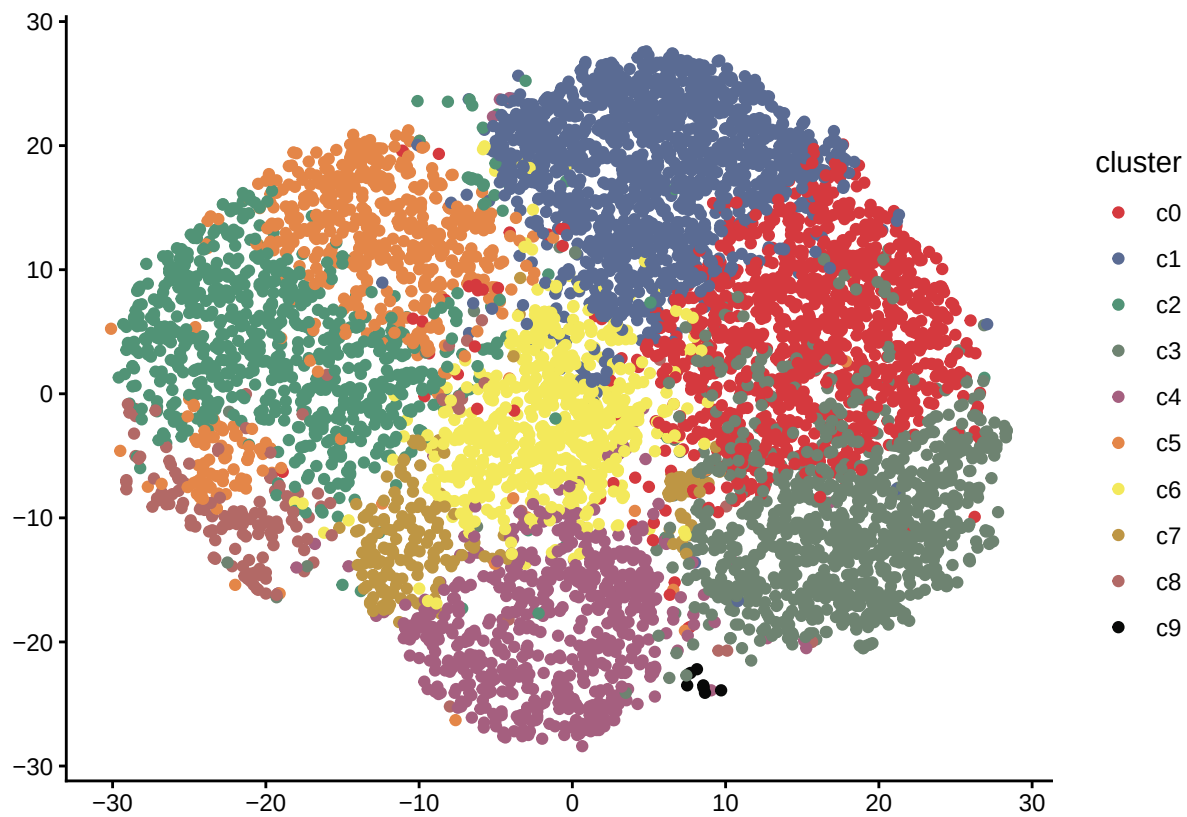
`p_umap_outcome`



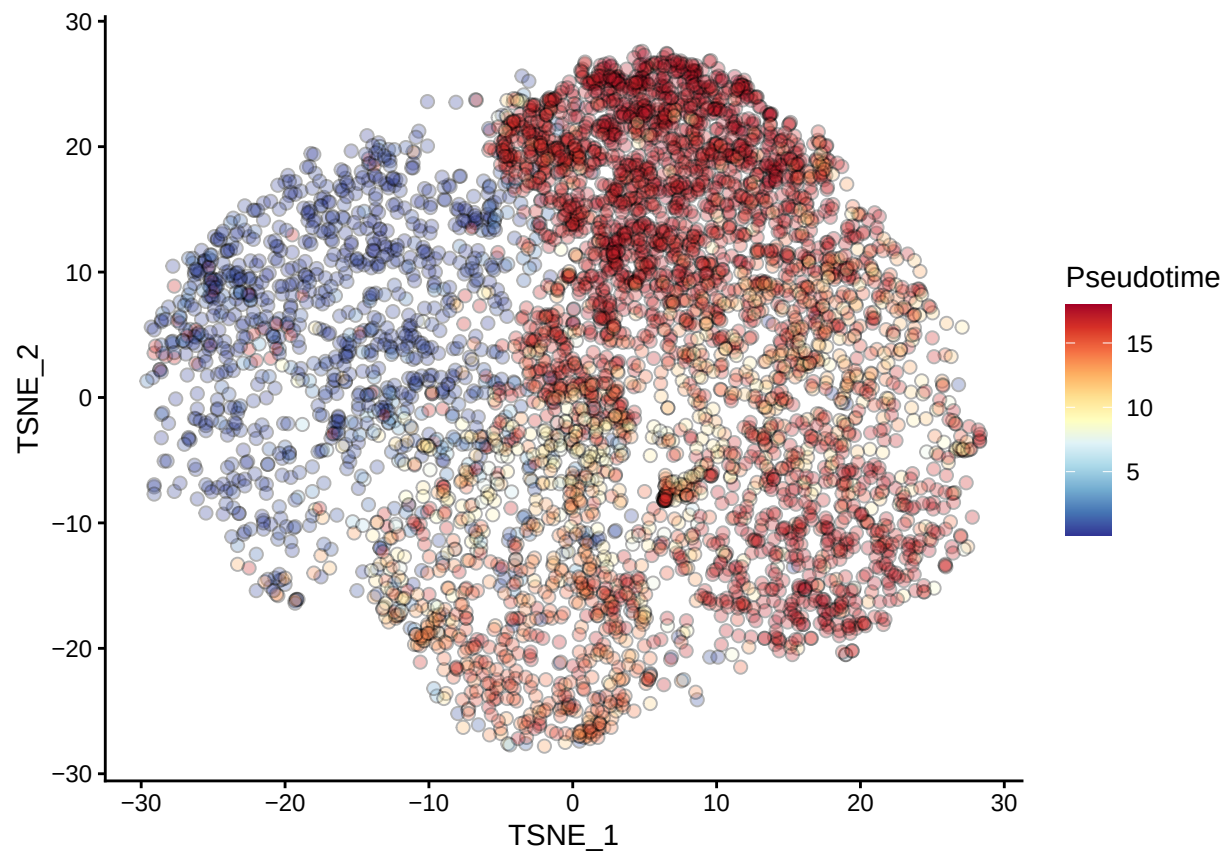
`p_umap_cell_day`



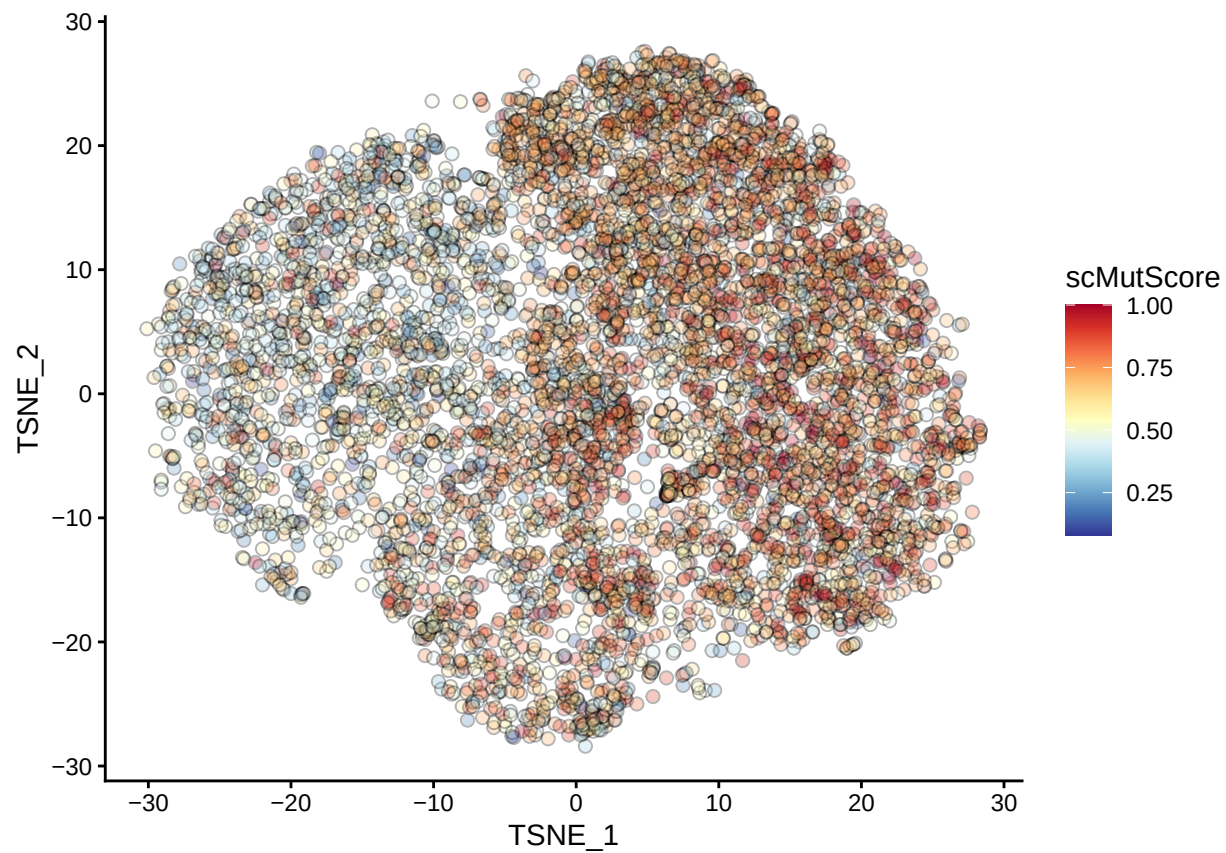
p_umap_cell_cluster



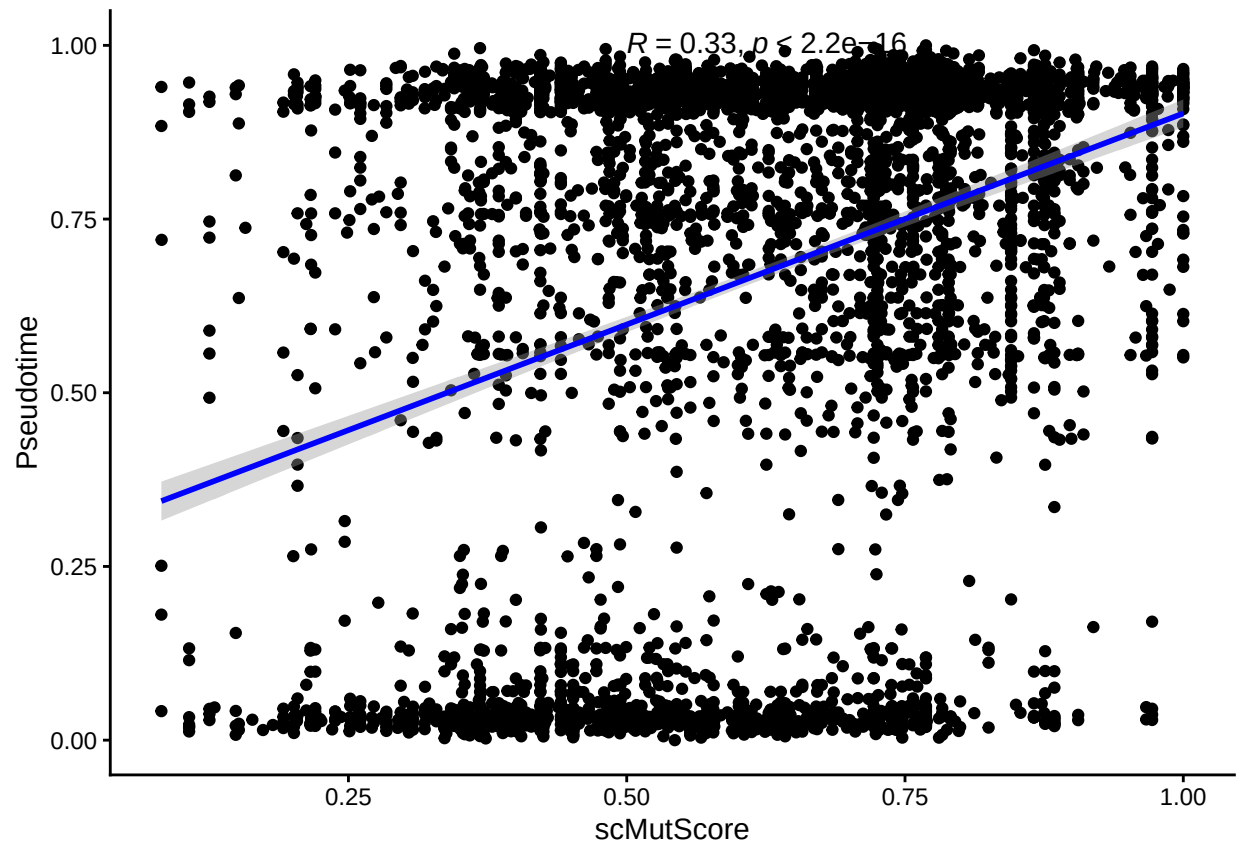
p_umap_cell_Pseudotime



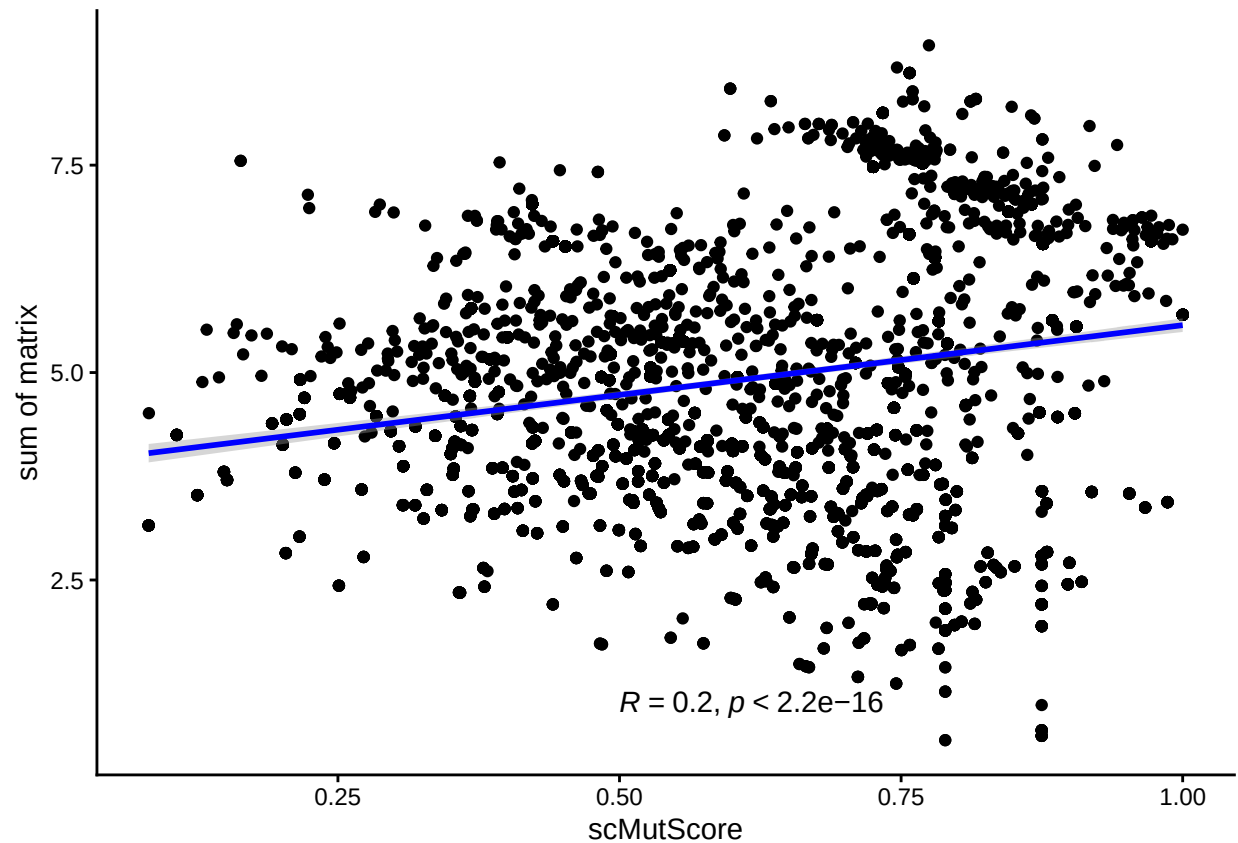
p_umap_cell_scMutScore



```
p_cor_scMutScore_pseudotime  
## `geom_smooth()` using formula = 'y ~ x'
```



```
p_cor_scMutScore_sumMut  
## `geom_smooth()` using formula = 'y ~ x'
```



14 save figures

```
# save data
pdf("../Figure/MEF/p_cell_scMutraceScore.pdf", width = 5.36, height = 3.42)
p_cell_scMutraceScore
dev.off()

pdf("../Figure/MEF/p_cell_pseudotime.pdf", width = 5.36, height = 3.42)
p_cell_pseudotime
dev.off()

pdf("../Figure/MEF/p_clone_scMutraceScore.pdf", width = 5.36, height = 3.42)
p_clone_scMutraceScore
dev.off()

pdf("../Figure/MEF/p_clone_pseudotime.pdf", width = 5.36, height = 3.42)
p_clone_pseudotime
dev.off()

pdf("../Figure/MEF/p_umap_cell_day.pdf", width = 5.36, height = 3.42)
p_umap_cell_day
dev.off()
```

```
pdf("../Figure/MEF/p_umap_cell_scMutScore.pdf", width = 5.36, height = 3.42)
p_umap_cell_scMutScore
dev.off()

pdf("../Figure/MEF/p_umap_cell_Pseudotime.pdf", width = 5.36, height = 3.42)
p_umap_cell_Pseudotime
dev.off()

pdf("../Figure/MEF/p_success_faile_scMutraceScore.pdf", width = 5.36, height = 3.42)
p_reprogramming_scMutraceScore
dev.off()
```

15 check randomly clone scMutScore

```
selected_Days <- c("Day_6", "Day_9", "Day_12", "Day_15", "Day_21", "Day_28")
set.seed(2025)
data_random_cells <- res_index_scMut_meta[sample(nrow(res_index_scMut_meta), 1000, replace = TRUE), c("scMutScore", "Day")]
data_random_cells$Day <- "Day_6"

for (i in 2:length(selected_Days)){
  random_rows_1 <- res_index_scMut_meta[sample(nrow(res_index_scMut_meta), 50), c("scMutScore", "Day")]
  random_rows_1$Day <- selected_Days[i]
  data_random_cells <- rbind(data_random_cells, random_rows_1)
}

data_random_cells$Day <- factor(data_random_cells$Day, levels = c("Day_6", "Day_9", "Day_12", "Day_15", "Day_21", "Day_28"))

p_random_cell_days <- ggplot(data = data_random_cells, mapping = aes(x = Day, y = scMutScore, fill = Day)) +
  #geom_violin(trim = TRUE) +
  geom_boxplot(outliers = TRUE) +
  #stat_summary(fun.data="mean_sdl", fun.args = list(mult=1), geom="pointrange", color = "black") +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = "scMutScore") +
  theme_classic()

data_random_clones <- data_clone_score_clonetagDay[sample(nrow(data_clone_score_clonetagDay), 400, replace = TRUE), c("Mean_Day_score", "Day")]
data_random_clones$Day <- "Day_6"

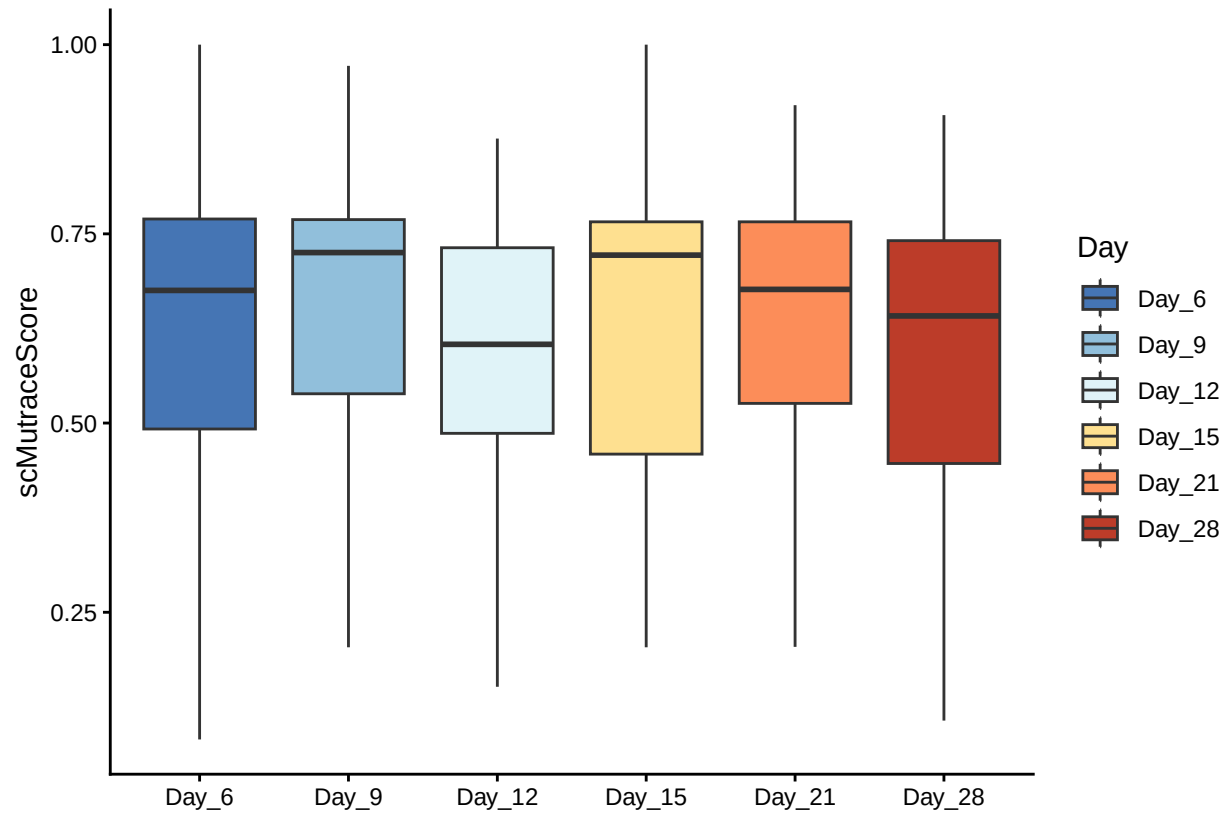
for (i in 2:length(selected_Days)){
  random_rows_1 <- data_clone_score_clonetagDay[sample(nrow(data_clone_score_clonetagDay), 50), c("Mean_Day_score", "Day")]
  random_rows_1$Day <- selected_Days[i]
  data_random_clones <- rbind(data_random_clones, random_rows_1)
}

data_random_clones$Day <- factor(data_random_clones$Day, levels = c("Day_6", "Day_9", "Day_12", "Day_15", "Day_21", "Day_28"))

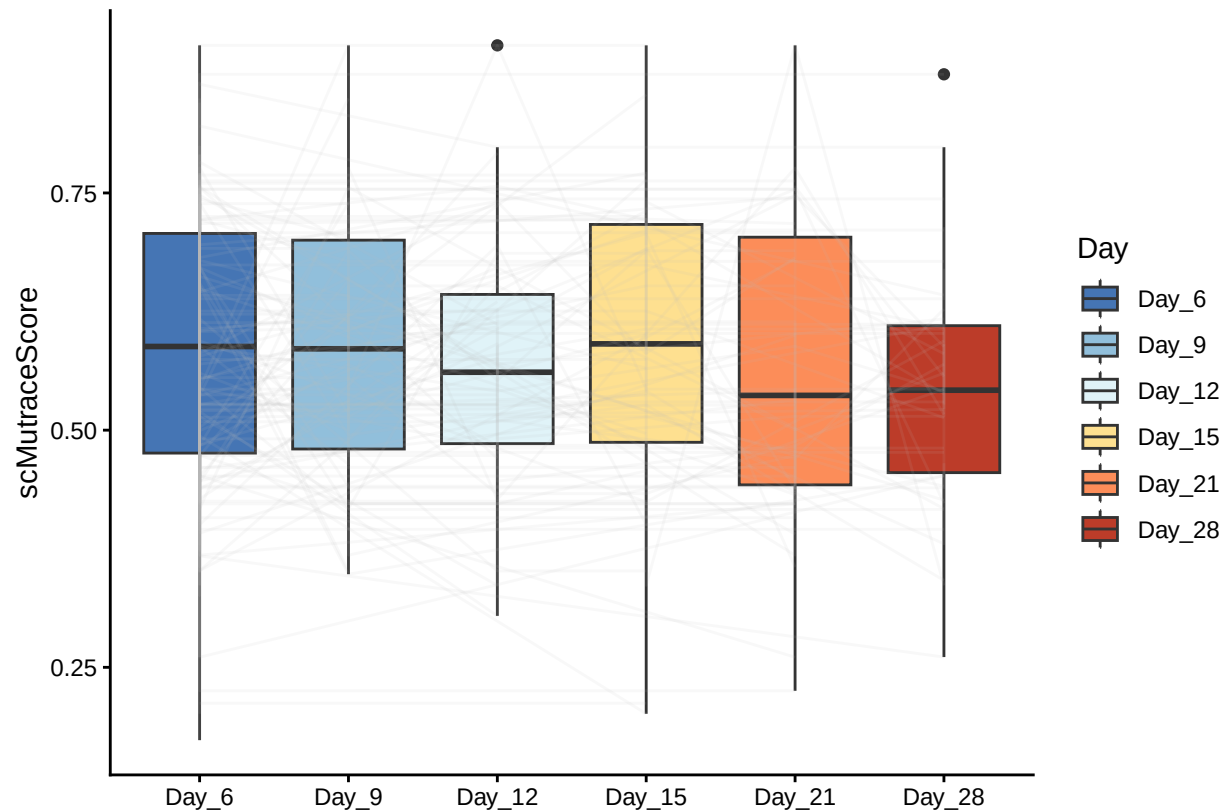
p_random_clone_days <- ggplot(data_random_clones, mapping = aes(x = as.factor(Day), y = Mean_Day_score, fill = Day)) +
  #geom_violin(trim = TRUE) +
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = ID), color = "gray", alpha = 0.1) +
  #geom_point(size = 1, shape = 21, color = "black") +
```

```
scale_fill_manual(values = color_settings_2) +
labs(x = "", y = "scMutraceScore") +
theme_classic()
```

p_random_cell_days



p_random_clone_days



```
library(clinfun)
# cells scMutScore
res_index_selected <- res_index_scMut_meta[res_index_scMut_meta$Day %in% c('Day_6', 'Day_9', 'Day_12',
res_index_selected$Day <- factor(res_index_selected$Day , levels = c('Day_6', 'Day_9', 'Day_12', 'Day_15',
jonckheere.test(x = res_index_selected$scMutScore, g = res_index_selected$Day, alternative = "increasing",
## Warning in jonckheere.test(x = res_index_selected$scMutScore, g = res_index_selected$Day, : Sample size > 100 or data
## p-value based on normal approximation. Specify nperm for permutation p-value
##
## Jonckheere-Terpstra test
##
## data:
## JT = 10520356, p-value < 2.2e-16
## alternative hypothesis: increasing
## JT = 10520356, p-value < 2.2e-16
## alternative hypothesis: increasing

res_index_selected <- res_index_scMut_meta[res_index_scMut_meta$Day %in% c('Day_6', 'Day_9', 'Day_12',
res_index_selected <- res_index_scMut_meta[res_index_scMut_meta$pseudotime != "NotAva",]
res_index_selected$Day <- factor(res_index_selected$Day , levels = c('Day_6', 'Day_9', 'Day_12', 'Day_15',
jonckheere.test(x = as.numeric(res_index_selected$pseudotime), g = res_index_selected$Day, alternative = "increasing",
## Warning in jonckheere.test(x = as.numeric(res_index_selected$pseudotime), : Sample size > 100 or data
## p-value based on normal approximation. Specify nperm for permutation p-value
##
## Jonckheere-Terpstra test
```

```

##
## data:
## JT = 6857494, p-value < 2.2e-16
## alternative hypothesis: increasing
## JT = 6857494, p-value < 2.2e-16
## alternative hypothesis: increasing

# clone scMutscore
data_clone_score_clonetagDay_selected <- data_clone_score_clonetagDay[data_clone_score_clonetagDay$Day %> 0]
data_clone_score_clonetagDay_selected$Day <- factor(data_clone_score_clonetagDay_selected$Day , levels = c('Day_6', 'Day_9', 'Day_12', 'Day_15'))
jonckheere.test(x = data_clone_score_clonetagDay_selected$Mean_Day_score, g = data_clone_score_clonetagDay_selected$Day, alternative = "increasing")
## Warning in jonckheere.test(x = data_clone_score_clonetagDay_selected$Mean_Day_score, : Sample size > 10000
## p-value based on normal approximation. Specify nperm for permutation p-value
##
## Jonckheere-Terpstra test
##
## data:
## JT = 31093, p-value = 8.216e-15
## alternative hypothesis: increasing
## JT = 31093, p-value = 8.216e-15
## alternative hypothesis: increasing
jonckheere.test(x = as.numeric(data_clone_score_clonetagDay_selected$Mean_Day_pseudotime), g = data_clone_score_clonetagDay_selected$Day, alternative = "increasing")
## Warning in jonckheere.test(x = as.numeric(data_clone_score_clonetagDay_selected$Mean_Day_pseudotime), : Sample size > 10000
## p-value based on normal approximation. Specify nperm for permutation p-value
##
## Jonckheere-Terpstra test
##
## data:
## JT = 37857, p-value < 2.2e-16
## alternative hypothesis: increasing
## JT = 37857, p-value < 2.2e-16
## alternative hypothesis: increasing

# data_random_cells
data_random_cells$Day <- factor(data_random_cells$Day , levels = c('Day_6', 'Day_9', 'Day_12', 'Day_15'))
jonckheere.test(x = data_random_cells$scMutScore, g = data_random_cells$Day, alternative = "increasing")
## Warning in jonckheere.test(x = data_random_cells$scMutScore, g = data_random_cells$Day, : Sample size > 10000
## p-value based on normal approximation. Specify nperm for permutation p-value
##
## Jonckheere-Terpstra test
##
## data:
## JT = 135420, p-value = 0.657
## alternative hypothesis: increasing
## JT = 135420, p-value = 0.657
## alternative hypothesis: increasing

#data_random_clones
data_random_clones$Day <- factor(data_random_clones$Day , levels = c('Day_6', 'Day_9', 'Day_12', 'Day_15'))
jonckheere.test(x = data_random_clones$Mean_Day_score, g = data_random_clones$Day, alternative = "increasing")
## Warning in jonckheere.test(x = data_random_clones$Mean_Day_score, g = data_random_clones$Day, : Sample size > 10000
## p-value based on normal approximation. Specify nperm for permutation p-value
##

```

```
## Jonckheere-Terpstra test
##
## data:
## JT = 58798, p-value = 0.9372
## alternative hypothesis: increasing
## JT = 58798, p-value = 0.9372
## alternative hypothesis: increasing
```

16 tracing allClones

```
res_index_scMut_meta_MEF <- res_index_scMut_meta[res_index_scMut_meta$CellTagD0_48k != "NotAva" & res_index_scMut_meta$CellTagD3_48k != "NotAva" & res_index_scMut_meta$CellTagD13_48k != "NotAva",]

res_index_scMut_meta_D3 <- res_index_scMut_meta[res_index_scMut_meta$CellTagD3_48k != "NotAva" & res_index_scMut_meta$CellTagD13_48k != "NotAva",]

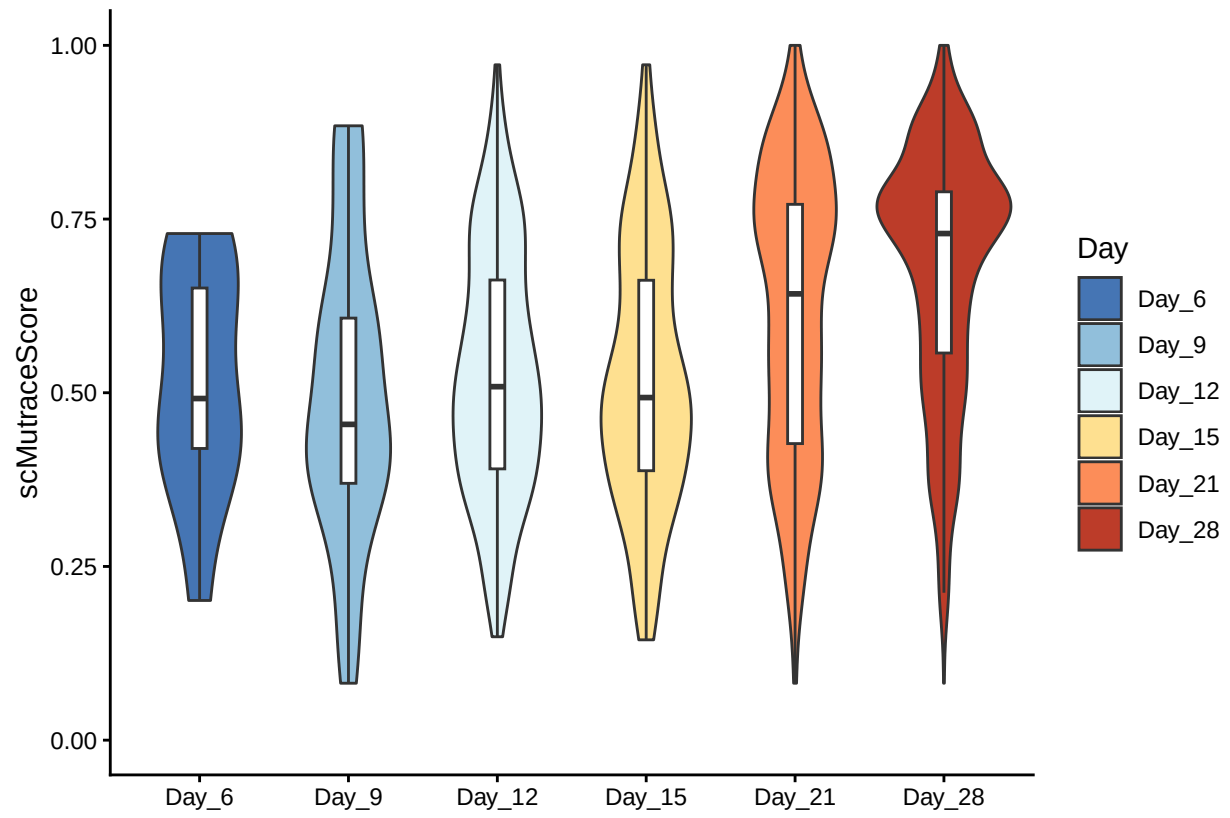
res_index_scMut_meta_D13 <- res_index_scMut_meta[res_index_scMut_meta$CellTagD13_48k != "NotAva",]

p_cell_MEF <- ggplot(data = res_index_scMut_meta_MEF, mapping = aes(x = Day, y = scMutScore, fill = Day)) +
  geom_violin(trim = TRUE) + coord_cartesian(ylim = c(0, 1)) +
  geom_boxplot(width = 0.1, fill = "white", outliers = FALSE) +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = "scMutraceScore") +
  theme_classic()

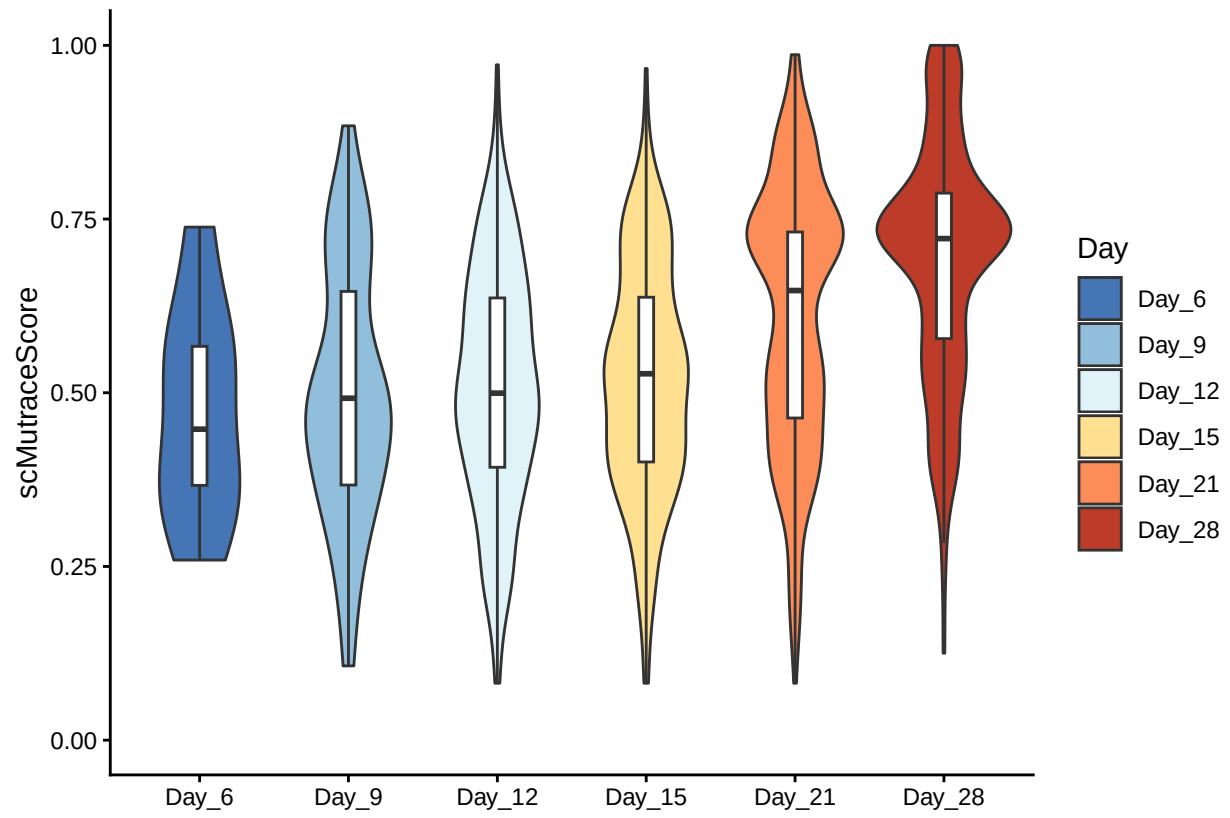
p_cell_D3 <- ggplot(data = res_index_scMut_meta_D3, mapping = aes(x = Day, y = scMutScore, fill = Day)) +
  geom_violin(trim = TRUE) + coord_cartesian(ylim = c(0, 1)) +
  geom_boxplot(width = 0.1, fill = "white", outliers = FALSE) +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = "scMutraceScore") +
  theme_classic()

p_cell_D13 <- ggplot(data = res_index_scMut_meta_D13, mapping = aes(x = Day, y = scMutScore, fill = Day)) +
  geom_violin(trim = TRUE) + coord_cartesian(ylim = c(0, 1)) +
  geom_boxplot(width = 0.1, fill = "white", outliers = FALSE) +
  #stat_summary(fun.data="mean_sdl", fun.args = list(mult=1),geom="pointrange", color = "black") +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = "scMutraceScore") +
  theme_classic()

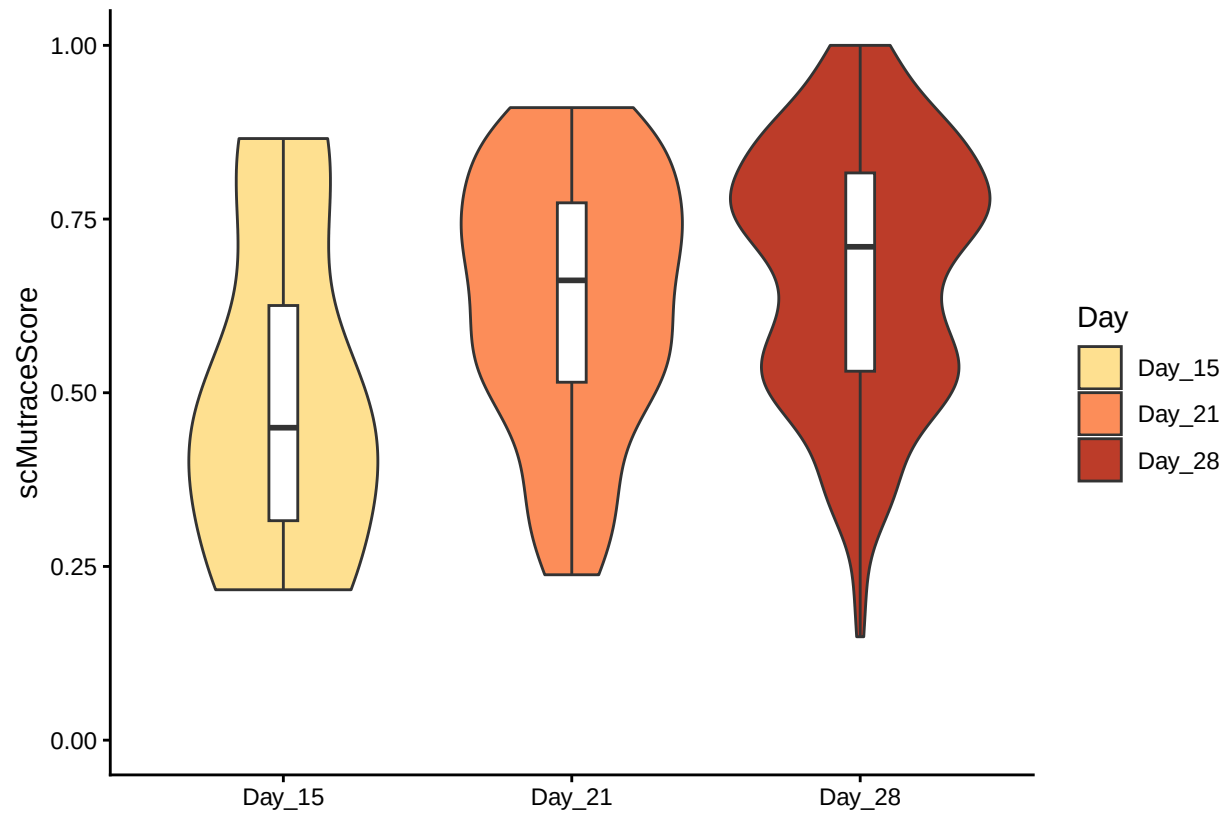
p_cell_MEF
```

p_cell1_D3



p_cell_D13



17 randomlySelectMut

```
color_settings <- c(d0 = "#FFDC91FF", d3 = "#7876B1FF", d6 = "#4575b4",
  d9 = "#91b9db", d12 = "#e0f3f8", d15 = "#fee090",
  d21 = "#fc8d59", d28 = "#BC3C29FF")

color_settings_2 <- c(Day_0 = "#FFDC91FF", Day_3 = "#7876B1FF", Day_6 = "#4575b4",
  Day_9 = "#91b9db", Day_12 = "#e0f3f8", Day_15 = "#fee090",
  Day_21 = "#fc8d59", Day_28 = "#BC3C29FF")

col_names_days <- c("Day_6", "Day_9", "Day_12", "Day_15", "Day_21", "Day_28", "sampleSize")
reps_random <- 100
row_names_reps <- paste("iter", c(1:reps_random), sep = "_")

sample_num <- c(10, 30, 50, 100, 150)

data_random_muts2cell_hf1 <- as.data.frame(matrix(nrow = reps_random * length(sample_num), ncol = length(sample_num)))
data_random_muts2cell_hf2 <- as.data.frame(matrix(nrow = reps_random * length(sample_num), ncol = length(sample_num)))

colnames(data_random_muts2cell_hf1) <- col_names_days
colnames(data_random_muts2cell_hf2) <- col_names_days

#rownames(data_random_muts2cell_hf1) <- row_names_reps
```

```

#rownames(data_random_muts2cell_hf2) <- row_names_reps

hf1_muts <- data_CellPerMut_hf1$Location
hf2_muts <- data_CellPerMut_hf2$Location

set.seed(2025)
line_count <- 0
for (i in 1:reps_random){

  for (j in sample_num){

    line_count <- line_count + 1
    hf1_muts_tmp <- sample(hf1_muts, j, replace = TRUE)
    hf2_muts_tmp <- sample(hf2_muts, j, replace = TRUE)

    hf1_cells <- retrieveCellsByLocation(MEF_CSM_hf1, hf1_muts_tmp)$Mut
    hf2_cells <- retrieveCellsByLocation(MEF_CSM_hf2, hf2_muts_tmp)$Mut

    data_random_muts2cell_hf1[line_count, "Day_6"] <- sum(str_count(hf1_cells, "hf1.d6"))
    data_random_muts2cell_hf1[line_count, "Day_9"] <- sum(str_count(hf1_cells, "hf1.d9"))
    data_random_muts2cell_hf1[line_count, "Day_12"] <- sum(str_count(hf1_cells, "hf1.d12"))
    data_random_muts2cell_hf1[line_count, "Day_15"] <- sum(str_count(hf1_cells, "hf1.d15"))
    data_random_muts2cell_hf1[line_count, "Day_21"] <- sum(str_count(hf1_cells, "hf1.d21"))
    data_random_muts2cell_hf1[line_count, "Day_28"] <- sum(str_count(hf1_cells, "hf1.d28"))
    data_random_muts2cell_hf1[line_count, "sampleSize"] <- j

    data_random_muts2cell_hf2[line_count, "Day_6"] <- sum(str_count(hf2_cells, "hf2.d6"))
    data_random_muts2cell_hf2[line_count, "Day_9"] <- sum(str_count(hf2_cells, "hf2.d9"))
    data_random_muts2cell_hf2[line_count, "Day_12"] <- sum(str_count(hf2_cells, "hf2.d12"))
    data_random_muts2cell_hf2[line_count, "Day_15"] <- sum(str_count(hf2_cells, "hf2.d15"))
    data_random_muts2cell_hf2[line_count, "Day_21"] <- sum(str_count(hf2_cells, "hf2.d21"))
    data_random_muts2cell_hf2[line_count, "Day_28"] <- sum(str_count(hf2_cells, "hf2.d28"))
    data_random_muts2cell_hf2[line_count, "sampleSize"] <- j
  }
}

data_random_muts2cell_hf1$iteration <- rownames(data_random_muts2cell_hf1)
data_random_muts2cell_hf2$iteration <- rownames(data_random_muts2cell_hf2)

data_random_muts2cell_hf1_long <- data_random_muts2cell_hf1 %>%
  pivot_longer(
    cols = starts_with("Day"),
    names_to = "Day",
    values_to = "CellNum"
  )

data_random_muts2cell_hf2_long <- data_random_muts2cell_hf2 %>%
  pivot_longer(
    cols = starts_with("Day"),
    names_to = "Day",
    values_to = "CellNum"
  )

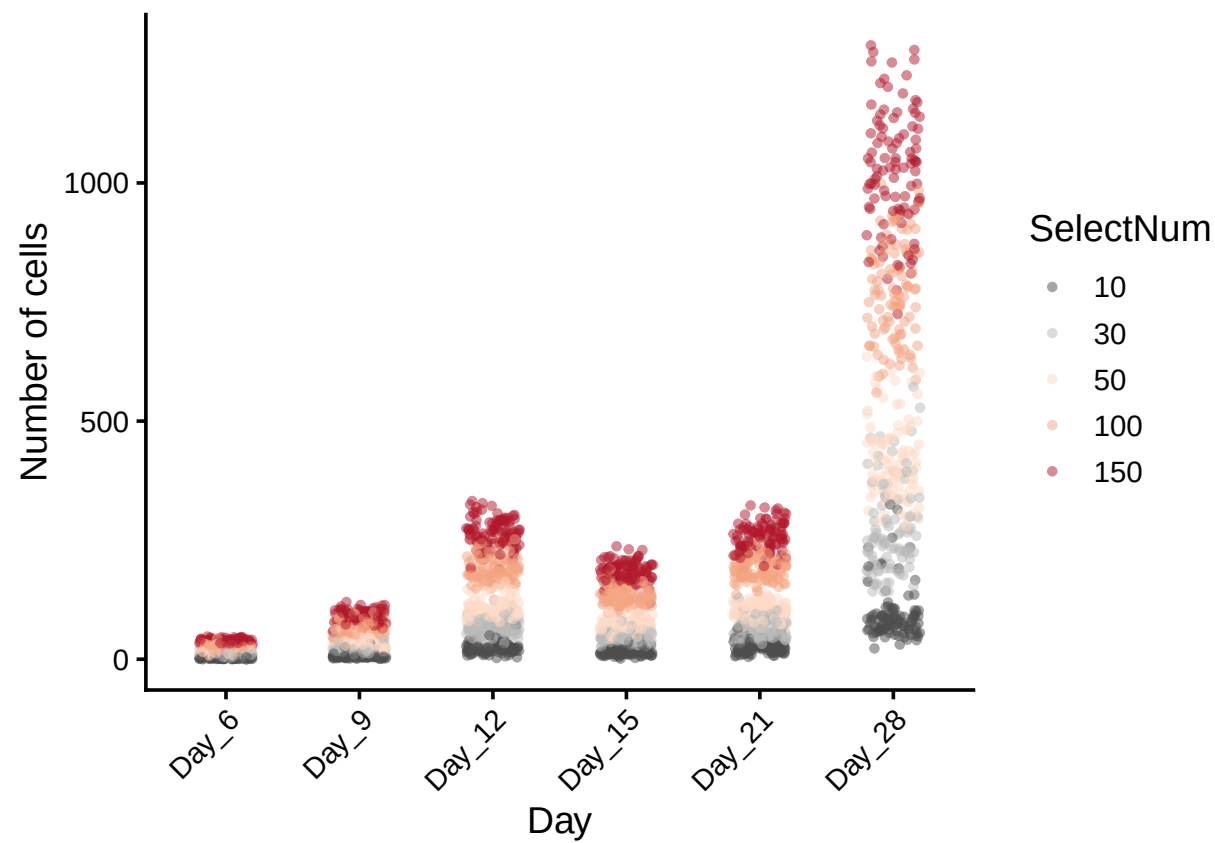
```

```
data_random_muts2cell_hf1_long$Day <- factor(data_random_muts2cell_hf1_long$Day, levels = c("Day_6", "D
data_random_muts2cell_hf2_long$Day <- factor(data_random_muts2cell_hf2_long$Day, levels = c("Day_6", "D
```

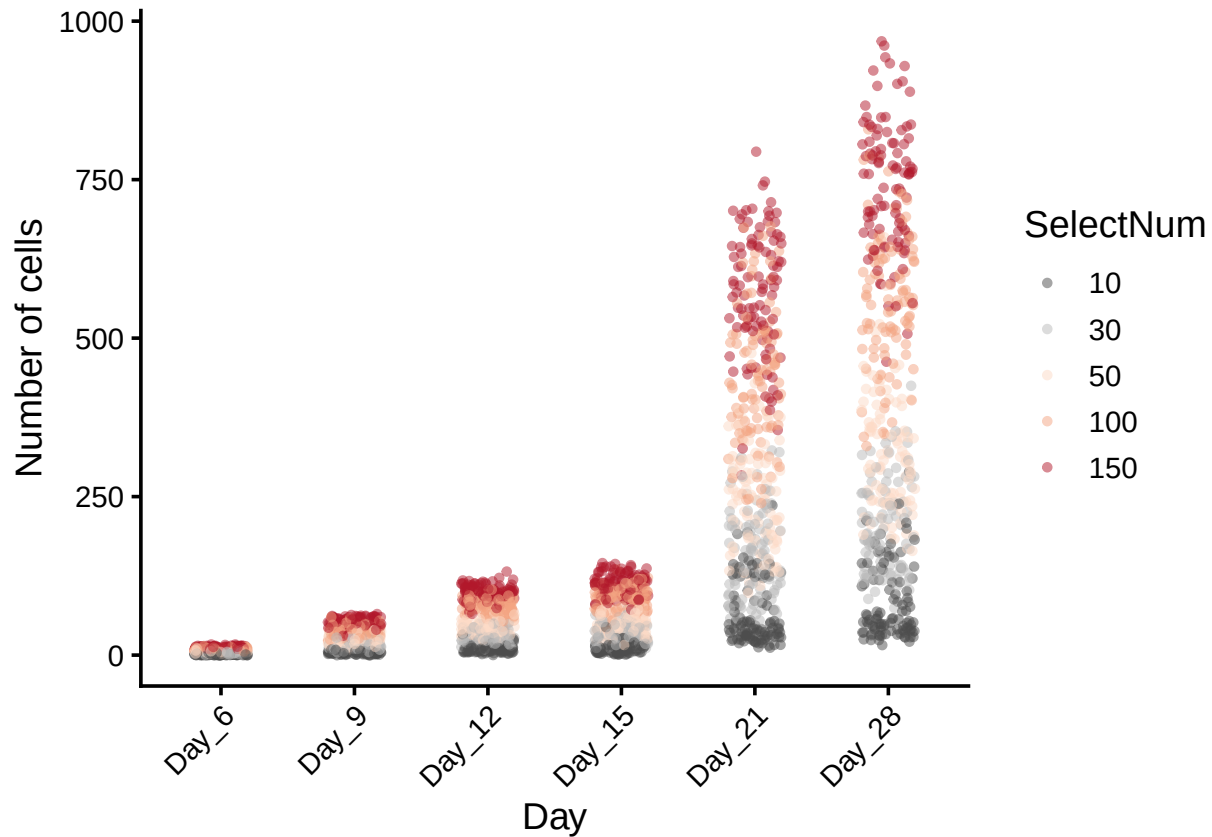
```
p_Mut2Cell_hf1 <- ggplot(data_random_muts2cell_hf1_long,
  aes(x = Day, y = CellNum, color = factor(sampleSize))) +
  geom_jitter(width = 0.2, alpha = 0.5, size = 1) +
  ylab("Number of cells") +
  xlab("Day") +
  labs(color = "SelectNum") +
  scale_color_manual(values = c(
    "10" = "#4d4d4d",
    "30" = "#bababa",
    "50" = "#fddbc7",
    "100" = "#f4a582",
    "150" = "#b2182b"
  )) +
  theme_classic(base_size = 14) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
p_Mut2Cell_hf2 <- ggplot(data_random_muts2cell_hf2_long,
  aes(x = Day, y = CellNum, color = factor(sampleSize))) +
  geom_jitter(width = 0.2, alpha = 0.5, size = 1) +
  ylab("Number of cells") +
  xlab("Day") +
  labs(color = "SelectNum") +
  scale_color_manual(values = c(
    "10" = "#4d4d4d",
    "30" = "#bababa",
    "50" = "#fddbc7",
    "100" = "#f4a582",
    "150" = "#b2182b"
  )) +
  theme_classic(base_size = 14) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
p_Mut2Cell_hf1
```



p_Mut2Cell1_hf2



18 lineageTracingExample

```
selected_start_cloneID <- 2352
data_clone_score_selected <- res_index_scMut_meta[res_index_scMut_meta$CellTagD0_48k == selected_start_cloneID, ]
colnames(data_clone_score_selected) <- c("Cellbarcode", "scMutScore", "D0_tag", "D3_tag", "D13_tag")

# Categorization of all situations
data_clone_score_selected_long <- bind_rows(
  # C1: D3 == NotAva & D13 != NotAva → D0 → D13
  data_clone_score_selected %>%
    dplyr::filter(D3_tag == "NotAva" & D13_tag != "NotAva") %>%
    dplyr::select(source = D0_tag, target = D13_tag),
  # C2: D3 == NotAva & D13 == NotAva → D0 → D0_self
  data_clone_score_selected %>%
    dplyr::filter(D3_tag == "NotAva" & D13_tag == "NotAva") %>%
    dplyr::mutate(target = paste0(D0_tag, "_self")) %>%
    dplyr::select(source = D0_tag, target = target),
  # C3: D3 != NotAva & D13 != NotAva → D0→D3
  data_clone_score_selected %>%
    dplyr::filter(D3_tag != "NotAva" & D13_tag != "NotAva") %>%
    dplyr::select(source = D0_tag, target = D13_tag)
```

```

    dplyr::select(source = D0_tag, target = D3_tag),

# C4: D3 != NotAva & D13 != NotAva → D3→D13
data_clone_score_selected %>%
  dplyr::filter(D3_tag != "NotAva" & D13_tag != "NotAva") %>%
  dplyr::select(source = D3_tag, target = D13_tag),

# C5: D3 != NotAva & D13 == NotAva → D0→D3
data_clone_score_selected %>%
  dplyr::filter(D3_tag != "NotAva" & D13_tag == "NotAva") %>%
  dplyr::select(source = D0_tag, target = D3_tag),

# C6: D3 != NotAva & D13 == NotAva → D3→D3_self
data_clone_score_selected %>%
  dplyr::filter(D3_tag != "NotAva" & D13_tag == "NotAva") %>%
  dplyr::mutate(target = paste0(D3_tag, "_self")) %>%
  dplyr::select(source = D3_tag, target)
)

# Counting the number of pairs
clone_links <- data_clone_score_selected_long %>%
  dplyr::group_by(source, target) %>%
  dplyr::summarise(value = n()) %>%
  dplyr::arrange(desc(value))
## `summarise()` has grouped output by 'source'. You can override using the
## `.groups` argument.

clone_links <- as.data.frame(clone_links)

clone_nodes <- data.frame(
  name=c(as.character(clone_links$source),
  as.character(clone_links$target)) %>% unique()
)

clone_nodes$group <- ""

for (i in 1:nrow(clone_nodes)){
  checkNode <- as.character(clone_nodes[i,"name"])
  if (checkNode %in% as.character(data_clone_score_selected$D0_tag)){
    clone_nodes[i,"group"] <- "MEF"
  }

  if (checkNode %in% as.character(data_clone_score_selected$D3_tag)){
    clone_nodes[i,"group"] <- "D3"
  }

  if (checkNode %in% as.character(data_clone_score_selected$D13_tag)){
    clone_nodes[i,"group"] <- "D13"
  }
}

clone_links$IDsource <- match(clone_links$source, clone_nodes$name)-1
clone_links$IDtarget <- match(clone_links$target, clone_nodes$name)-1

```

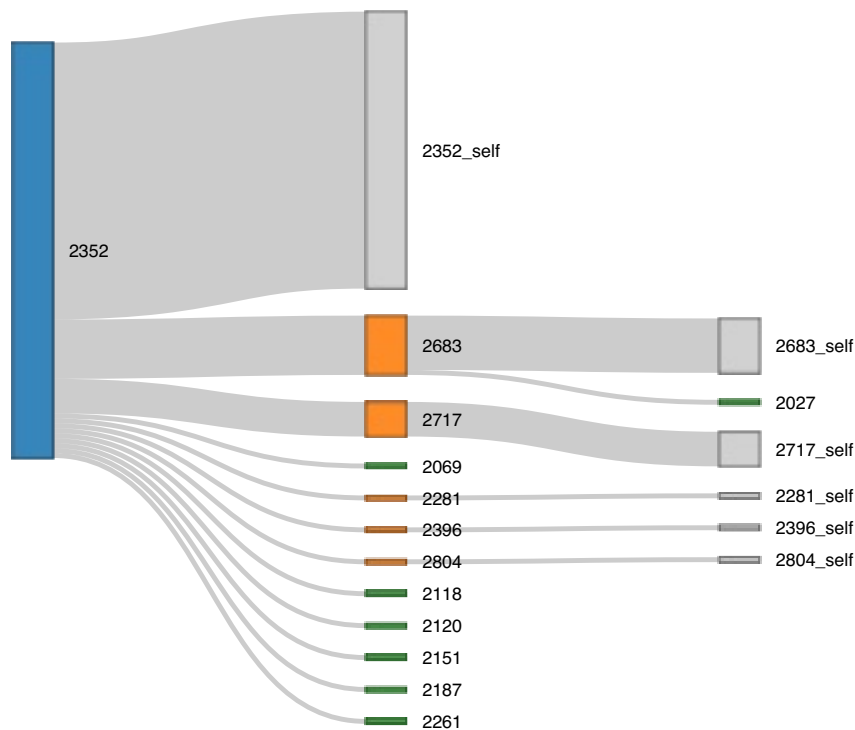


```

p_sankeyNetwork <- sankeyNetwork(Links = clone_links, Nodes = clone_nodes,
  Source = "IDsource", Target = "IDtarget",
  Value = "value", NodeID = "name",
  sinksRight=FALSE,
  colourScale = JS("d3.scaleOrdinal()
    .domain(['MEF', 'D3', 'D13'])
    .range(['#1f77b4', '#ff7f0e', '#2ca02c'])"),
  NodeGroup = "group")

p_sankeyNetwork

```



19 lineageTracingExample with scMutScore

```

selected_start_cloneID <- c(124, 487, 2352)

data_clone_score_selected <- res_index_scMut_meta[res_index_scMut_meta$CellTagD0_48k %in% selected_start_cloneID]
colnames(data_clone_score_selected) <- c("Cellbarcode", "scMutScore", "D0_tag", "D3_tag", "D13_tag")

# Categorization of all situations
data_clone_score_selected_long <- bind_rows(
  # C1: D3 == NotAva & D13 != NotAva → D0 → D13

```

```

data_clone_score_selected %>%
  dplyr::filter(D3_tag == "NotAva" & D13_tag != "NotAva") %>%
  dplyr::select(source = D0_tag, target = D13_tag),

# C2: D3 == NotAva & D13 == NotAva → D0 → D0_self
data_clone_score_selected %>%
  dplyr::filter(D3_tag == "NotAva" & D13_tag == "NotAva") %>%
  dplyr::mutate(target = paste0(D0_tag, "_self")) %>%
  dplyr::select(source = D0_tag, target),

# C3: D3 != NotAva & D13 != NotAva → D0→D3
data_clone_score_selected %>%
  dplyr::filter(D3_tag != "NotAva" & D13_tag != "NotAva") %>%
  dplyr::select(source = D0_tag, target = D3_tag),

# C4: D3 != NotAva & D13 != NotAva → D3→D13
data_clone_score_selected %>%
  dplyr::filter(D3_tag != "NotAva" & D13_tag != "NotAva") %>%
  dplyr::select(source = D3_tag, target = D13_tag),

# C5: D3 != NotAva & D13 == NotAva → D0→D3
data_clone_score_selected %>%
  dplyr::filter(D3_tag != "NotAva" & D13_tag == "NotAva") %>%
  dplyr::select(source = D0_tag, target = D3_tag),

# C6: D3 != NotAva & D13 == NotAva → D3→D3_self
data_clone_score_selected %>%
  dplyr::filter(D3_tag != "NotAva" & D13_tag == "NotAva") %>%
  dplyr::mutate(target = paste0(D3_tag, "_self")) %>%
  dplyr::select(source = D3_tag, target)
)

# Counting the number of pairs
clone_links <- data_clone_score_selected_long %>%
  dplyr::group_by(source, target) %>%
  dplyr::summarise(value = n()) %>%
  dplyr::arrange(desc(value))
## `summarise()` has grouped output by 'source'. You can override using the
## `.groups` argument.

clone_links <- as.data.frame(clone_links)

clone_links <- clone_links[!grepl("_self", clone_links$target), ]

#clone_links <- clone_links[clone_links$value > 1,]

clone_nodes <- data.frame(
  name=c(as.character(clone_links$source),
  as.character(clone_links$target)) %>% unique()
)

clone_nodes$group <- ""

```

```

for (i in 1:nrow(clone_nodes)){
  checkNode <- as.character(clone_nodes[i,"name"])
  if (checkNode %in% as.character(data_clone_score_selected$D0_tag)){
    clone_nodes[i,"group"] <- "MEF"
  }

  if (checkNode %in% as.character(data_clone_score_selected$D3_tag)){
    clone_nodes[i,"group"] <- "D3"
  }

  if (checkNode %in% as.character(data_clone_score_selected$D13_tag)){
    clone_nodes[i,"group"] <- "D13"
  }
}

# cal node scMutScore
clone_nodes$scMutScore <- 0

data_MEF_D3_D13 <- data.frame()

for (i in 1:nrow(clone_nodes)){
  group_tmp <- clone_nodes[i, "group"]
  node_tmp <- clone_nodes[i, "name"]
  print(node_tmp)
  print(group_tmp)
  if (group_tmp == "MEF"){
    data_mef_tmp <- res_index_scMut_meta[res_index_scMut_meta$CellTagD0_48k %in% selected_start_cloneID &
    if(nrow(data_mef_tmp) == 0){
      data_mef_tmp_tmp <- res_index_scMut_meta[res_index_scMut_meta$CellTagD0_48k == selected_start_cloneID &
      data_mef_tmp_tmp$Tag <- "MEF"
      clone_nodes[i, "scMutScore"] <- min(data_mef_tmp_tmp$scMutScore)
      data_MEF_D3_D13 <- rbind(data_MEF_D3_D13, data_mef_tmp_tmp)
    }else{
      data_mef_tmp$Tag <- "MEF"
      clone_nodes[i, "scMutScore"] <- min(data_mef_tmp$scMutScore)
      data_MEF_D3_D13 <- rbind(data_MEF_D3_D13, data_mef_tmp)
    }
  }
  if (group_tmp == "D3"){
    data_d3_tmp <- res_index_scMut_meta[res_index_scMut_meta$CellTagD0_48k %in% selected_start_cloneID &
    if(nrow(data_d3_tmp) == 0){
      data_d3_tmp_tmp <- res_index_scMut_meta[res_index_scMut_meta$CellTagD0_48k == selected_start_cloneID &
      data_d3_tmp_tmp$Tag <- "D3"
      clone_nodes[i, "scMutScore"] <- min(data_d3_tmp_tmp$scMutScore)
      data_MEF_D3_D13 <- rbind(data_MEF_D3_D13, data_d3_tmp_tmp)
    }else{
      data_d3_tmp$Tag <- "D3"
      clone_nodes[i, "scMutScore"] <- min(data_d3_tmp$scMutScore)
      data_MEF_D3_D13 <- rbind(data_MEF_D3_D13, data_d3_tmp)
    }
  }
  if (group_tmp == "D13"){
    data_d13_tmp <- res_index_scMut_meta[res_index_scMut_meta$CellTagD0_48k %in% selected_start_cloneID &

```

```

data_d13_tmp$Tag <- "D13"
clone_nodes[i, "scMutScore"] <- min(data_d13_tmp$scMutScore)
data_MEF_D3_D13 <- rbind(data_MEF_D3_D13, data_d13_tmp)
}
}
## [1] "487"
## [1] "MEF"
## [1] "124"
## [1] "MEF"
## [1] "2352"
## [1] "MEF"
## [1] "204"
## [1] "D3"
## [1] "240"
## [1] "D3"
## [1] "212"
## [1] "D3"
## [1] "2683"
## [1] "D3"
## [1] "767"
## [1] "D3"
## [1] "488"
## [1] "D3"
## [1] "2717"
## [1] "D3"
## [1] "543"
## [1] "D3"
## [1] "779"
## [1] "D3"
## [1] "83"
## [1] "D13"
## [1] "93"
## [1] "D13"
## [1] "770"
## [1] "D3"
## [1] "784"
## [1] "D3"
## [1] "104"
## [1] "D13"
## [1] "43"
## [1] "D13"
## [1] "47"
## [1] "D13"
## [1] "99"
## [1] "D13"
## [1] "2069"
## [1] "D13"
## [1] "2118"
## [1] "D13"
## [1] "2120"
## [1] "D13"
## [1] "2151"
## [1] "D13"

```

```

## [1] "2187"
## [1] "D13"
## [1] "2261"
## [1] "D13"
## [1] "2281"
## [1] "D3"
## [1] "2396"
## [1] "D3"
## [1] "2804"
## [1] "D3"
## [1] "178"
## [1] "D13"
## [1] "2027"
## [1] "D13"
## [1] "110"
## [1] "D13"
## [1] "522"
## [1] "D3"
## [1] "128"
## [1] "D13"

data_MEF_D3_D13$Tag <- factor(data_MEF_D3_D13$Tag, levels = c("MEF", "D3", "D13"))
clone_nodes$group <- factor(clone_nodes$group, levels = c("MEF", "D3", "D13"))
data_MEF_D3_D13$Day <- factor(data_MEF_D3_D13$Day, levels = c("Day_0", "Day_3", "Day_6", "Day_9", "Day_12"))

network <- graph_from_data_frame(d=clone_links, vertices=clone_nodes, directed=T)

color_settings_network_1 <- c( D13 = "#e41a1c", D3 = "#377eb8", MEF = "#4daf4a")

color_settings_network_2 <- colorRampPalette(c("#fee8c8", "#fdbb84", "#e34a33"))

colors <- color_settings_network_2(5)

breaks <- unique(quantile(V(network)$scMutScore, probs = seq(0, 1, length.out = 6)))

value_group <- cut(V(network)$scMutScore, breaks = breaks, include.lowest = TRUE, labels = FALSE)

my_color_2 <- colors[value_group]

edge_width <- log2(E(network)$value + 0.1)

vertex_frame_color <- color_settings_network_1[V(network)$group]

p_selected_clone_MEF <- ggplot(data = data_MEF_D3_D13[data_MEF_D3_D13$Tag == "MEF",], mapping = aes(x = clone_ID, y = scMutScore)) +
  geom_violin(trim = FALSE) + coord_cartesian(ylim = c(0, 1)) +
  geom_boxplot(width = 0.05, fill = "white", outliers = FALSE) +
  #stat_summary(fun.data="mean_sdl", fun.args = list(mult=1),geom="pointrange", color = "black") +
  scale_fill_manual(values = color_settings_2) +
  labs(x = "", y = paste0("scMutScore_", selected_start_cloneID)) +
  theme_classic()

p_selected_clone_D3 <- ggplot(data = data_MEF_D3_D13[data_MEF_D3_D13$Tag == "D3",], mapping = aes(x = clone_ID, y = scMutScore)) +
  geom_violin(trim = FALSE) + coord_cartesian(ylim = c(0, 1)) +

```

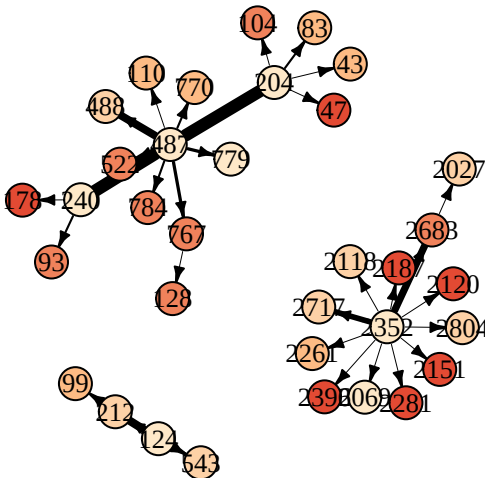
```

geom_boxplot(width = 0.05, fill = "white", outliers = FALSE) +
#stat_summary(fun.data="mean_sdl", fun.args = list(mult=1),geom="pointrange", color = "black") +
scale_fill_manual(values = color_settings_2) +
labs(x = "", y = paste0("scMutraceScore_", selected_start_cloneID)) +
theme_classic()

p_selected_clone_D13 <- ggplot(data = data_MEF_D3_D13[data_MEF_D3_D13$Tag == "D13",], mapping = aes(x =
geom_violin(trim = FALSE) + coord_cartesian(ylim = c(0, 1)) +
geom_boxplot(width = 0.05, fill = "white", outliers = FALSE) +
#stat_summary(fun.data="mean_sdl", fun.args = list(mult=1),geom="pointrange", color = "black") +
scale_fill_manual(values = color_settings_2) +
labs(x = "", y = paste0("scMutraceScore_", selected_start_cloneID)) +
theme_classic()

plot(network,
      vertex.color=my_color_2,
      vertex.label.color = "black",
      vertex.label.cex = 0.8,
      vertex.size = 15,
      #vertex.frame.width = 3,
      #vertex.frame.color = vertex_frame_color,
      edge.color = "black",
      edge.arrow.size = 0.4,
      edge.width = edge_width)

```



```
col_fun <- colorRampPalette(c("#fee8c8", "#fdbb84", "#e34a33"))
col_seq <- col_fun(length(my_color_2))
image(z = matrix(1:length(my_color_2), ncol = 1), col = col_seq,
      axes = FALSE, main = "Gradient Legend")
axis(4, at = seq(0, 1, length.out = 5), labels = round(seq(0, 1, length.out = 5), 2))
```

Gradient Legend



20 ModelFigure

```
m1 <- c(1, 1, 1, 1, 1, 1, 1, 1)
m2 <- c(0, 1, 1, 1, 1, 1, 1, 1)
m3 <- c(0, 0, 1, 0, 0, 0, 0, 0)
m4 <- c(0, 0, 0, 1, 1, 1, 1, 1)
m5 <- c(0, 0, 0, 0, 1, 1, 1, 1)
m6 <- c(0, 0, 0, 0, 0, 1, 0, 0)
m7 <- c(0, 0, 0, 0, 0, 0, 1, 0)
m8 <- c(0, 0, 0, 0, 0, 0, 0, 1)

t1 <- c(1, 0, 0, 0, 0, 0, 0, 0)
t2 <- c(1, 1, 0, 0, 0, 0, 0, 0)
t3 <- c(1, 1, 1, 1, 0, 0, 0, 0)
t4 <- c(1, 1, 1, 1, 0, 0, 0, 0)
```

```

t5 <- c(1, 1, 0, 1, 1, 0, 0, 0)
t6 <- c(1, 1, 0, 1, 1, 1, 1, 1)

test_Matrix_cm <- data.frame(m1 = m1, m2 = m2, m3 = m3, m4 = m4, m5 = m5, m6 = m6, m7 = m7, m8 = m8)
test_Matrix_mt <- data.frame(t1 = t1, t2 = t2, t3 = t3, t4 = t4, t5 = t5, t6 = t6)
rownames(test_Matrix_cm) <- c("c1", "c2", "c3", "c4", "c5", "c6", "c7", "c8")

test_Matrix_ct <- as.matrix(test_Matrix_cm) %*% as.matrix(test_Matrix_mt)
rownames(test_Matrix_ct) <- c("c1", "c2", "c3", "c4", "c5", "c6", "c7", "c8")

res <- cal_MultiSpe(test_Matrix_ct)

cal_scMutScore(test_Matrix_cm, test_Matrix_mt, NA)
## [1] 8 8
## [1] 8 6
## $diversity
##      t1      t2      t3      t4      t5      t6 Diversity
## c1  0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
## c2  0 0.1666667 0.1666667 0.1666667 0.1666667 0.1666667 0.8333333
## c3  0 0.1666667 0.1666667 0.1666667 0.1666667 0.1666667 0.8333333
## c4  0 0.1666667 0.1666667 0.1666667 0.1666667 0.1666667 0.8333333
## c5  0 0.1666667 0.1666667 0.1666667 0.1666667 0.1666667 0.8333333
## c6  0 0.1666667 0.1666667 0.1666667 0.1666667 0.1666667 0.8333333
## c7  0 0.1666667 0.1666667 0.1666667 0.1666667 0.1666667 0.8333333
## c8  0 0.1666667 0.1666667 0.1666667 0.1666667 0.1666667 0.8333333
##
## $score
##      Shannon   Simpson      Gini sumMut      Tau      GiniTau scMutScore
## c1  1.791759 0.8333333 0.0000000      6 0.0000000 0.000000000 0.00000000
## c2  1.767761 0.8264463 0.07575716     11 0.1000000 0.007575716 0.06691216
## c3  1.737983 0.8165680 0.16666590     13 0.3333333 0.055555299 0.48481273
## c4  1.736741 0.8177778 0.14444387     15 0.2000000 0.028888773 0.25320566
## c5  1.711547 0.8096886 0.20588163     17 0.3500000 0.072058569 0.62733159
## c6  1.692020 0.8024691 0.24073994     18 0.4800000 0.115555170 1.00000000
## c7  1.692020 0.8024691 0.24073994     18 0.4800000 0.115555170 1.00000000
## c8  1.692020 0.8024691 0.24073994     18 0.4800000 0.115555170 1.00000000
##
##      Diversity sumMut_norm GiniTau_norm scMutScoreQun
## c1  0.0000000 0.0000000 0.0000000 0.0000000
## c2  0.8333333 0.4166667 0.0655593 0.4166667
## c3  0.8333333 0.5833333 0.4807686 0.5833333
## c4  0.8333333 0.7500000 0.2499998 0.7500000
## c5  0.8333333 0.9166667 0.6235859 0.9166667
## c6  0.8333333 1.0000000 1.0000000 1.0000000
## c7  0.8333333 1.0000000 1.0000000 1.0000000
## c8  0.8333333 1.0000000 1.0000000 1.0000000
##
## $matrixD
##      t1 t2 t3 t4 t5 t6
## c1  1  1  1  1  1  1
## c2  1  2  2  2  2  2
## c3  1  2  3  3  2  2
## c4  1  2  3  3  3  3
## c5  1  2  3  3  4  4

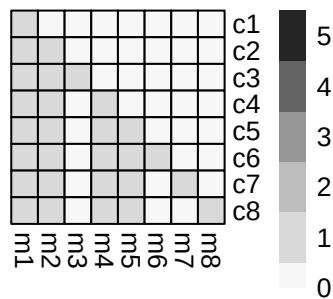
```



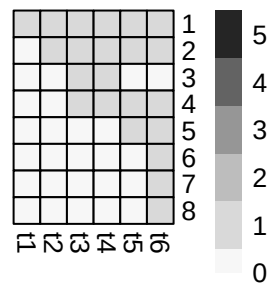
```
## c6 1 2 3 3 4 5
## c7 1 2 3 3 4 5
## c8 1 2 3 3 4 5
```

```
my_colors <- c("#f7f7f7", "#d9d9d9", "#bdbdbd", "#969696", "#636363", "#252525")
breaks <- c(0, 0.5, 1.5, 2.5, 3.5, 4.5, 5.5)
```

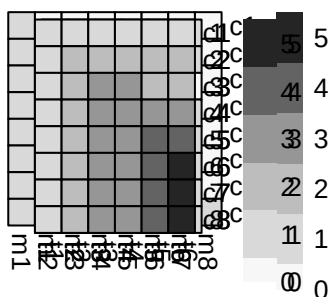
```
p_Matrix_cm <- pheatmap(test_Matrix_cm, cluster_rows = F, cluster_cols = F, color = my_colors, breaks =
```



```
p_Matrix_mt <- pheatmap(test_Matrix_mt, cluster_rows = F, cluster_cols = F, color = my_colors, breaks =
```



```
p_Matrix_ct <- pheatmap(test_Matrix_ct, cluster_rows = F, cluster_cols = F, color = my_colors, breaks =
p_Matrix_cm
p_Matrix_mt
p_Matrix_ct
```



21 sessionInfo

```
sessionInfo()
## R version 4.5.1 (2025-06-13)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.4.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Stockholm
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] clinfun_1.1.5      patchwork_1.3.2    lubridate_1.9.4    forcats_1.0.0
## [5] purrr_1.1.0        readr_2.1.5        tibble_3.3.0       tidyverse_2.0.0
```

```

## [9] networkD3_0.4.1      tidygraph_1.3.1      reshape2_1.4.4      pheatmap_1.0.13
## [13] corrplot_0.95        stringr_1.5.2        ggpmisc_0.6.2        ggpp_0.5.9
## [17] webshot_0.5.5        ggraph_2.2.2         igraph_2.1.4         Seurat_5.3.0
## [21] SeuratObject_5.2.0   sp_2.2-0             ggpubr_0.6.1         ggplot2_4.0.0
## [25] tidyr_1.3.1          dplyr_1.1.4          vegan_2.7-1          permute_0.9-8
## [29] ineq_0.2-13          FSA_0.10.0
##
## loaded via a namespace (and not attached):
## [1] RcppAnnoy_0.0.22      splines_4.5.1         later_1.4.4
## [4] polyclip_1.10-7       fastDummies_1.7.5     lifecycle_1.0.4
## [7] rstatix_0.7.2         processx_3.8.6        globals_0.18.0
## [10] lattice_0.22-7        MASS_7.3-65           backports_1.5.0
## [13] magrittr_2.0.4        plotly_4.11.0         rmarkdown_2.30
## [16] yaml_2.3.10           httpuv_1.6.16         sctransform_0.4.2
## [19] spam_2.11-1           spatstat.sparse_3.1-0 reticulate_1.43.0
## [22] cowplot_1.2.0         pbapply_1.7-4         RColorBrewer_1.1-3
## [25] abind_1.4-8           Rtsne_0.17            tweenr_2.0.3
## [28] data.tree_1.2.0       ggrepel_0.9.6         irlba_2.3.5.1
## [31] listenv_0.9.1         spatstat.utils_3.2-0  MatrixModels_0.5-4
## [34] goftest_1.2-3         RSpectra_0.16-2       spatstat.random_3.4-2
## [37] fitdistrplus_1.2-4    parallelly_1.45.1     codetools_0.2-20
## [40] ggforce_0.5.0         tidymodels_1.2.1      farver_2.1.2
## [43] viridis_0.6.5         matrixStats_1.5.0     spatstat.explore_3.5-3
## [46] jsonlite_2.0.0        progressr_0.16.0      Formula_1.2-5
## [49] ggribes_0.5.7         survival_3.8-3        tools_4.5.1
## [52] ica_1.0-3             Rcpp_1.1.0            glue_1.8.0
## [55] gridExtra_2.3         xfun_0.53             mgcv_1.9-3
## [58] withr_3.0.2           fastmap_1.2.0         SparseM_1.84-2
## [61] callr_3.7.6           digest_0.6.37         timechange_0.3.0
## [64] R6_2.6.1             mime_0.13             colorspace_2.1-2
## [67] scattermore_1.2       tensor_1.5.1          dichromat_2.0-0.1
## [70] spatstat.data_3.1-8   generics_0.1.4        data.table_1.17.8
## [73] graphlayouts_1.2.2    httr_1.4.7            htmlwidgets_1.6.4
## [76] uwot_0.2.3           pkgconfig_2.0.3       gtable_0.3.6
## [79] lmtest_0.9-40         S7_0.2.0              htmltools_0.5.8.1
## [82] carData_3.0-5         dotCall64_1.2          scales_1.4.0
## [85] png_0.1-8            spatstat.univar_3.1-4 knitr_1.50
## [88] rstudioapi_0.17.1     tzdb_0.5.0            nlme_3.1-168
## [91] zoo_1.8-14           cachem_1.1.0          KernSmooth_2.23-26
## [94] parallel_4.5.1        miniUI_0.1.2          pillar_1.11.1
## [97] grid_4.5.1           vctrs_0.6.5           RANN_2.6.2
## [100] promises_1.3.3        car_3.1-3             xtable_1.8-4
## [103] cluster_2.1.8.1       evaluate_1.0.5         mvtnorm_1.3-3
## [106] cli_3.6.5            compiler_4.5.1         rlang_1.1.6
## [109] future.apply_1.20.0    ggsignif_0.6.4        labeling_0.4.3
## [112] ps_1.9.1             plyr_1.8.9            stringi_1.8.7
## [115] viridisLite_0.4.2     deldir_2.0-4          lazyeval_0.2.2
## [118] spatstat.geom_3.6-0    quantreg_6.1          Matrix_1.7-4
## [121] RcppHNSW_0.6.0        hms_1.1.3            future_1.67.0
## [124] shiny_1.11.1          ROCR_1.0-11           broom_1.0.10
## [127] memoise_2.0.1         polynom_1.4-1

```